

Hierarchical Dynamical Spatio-Temporal Models

As described in Chapter 6, it is often natural to model spatio-temporal processes dynamically. In this chapter, we consider statistical models for such processes and refer to these models as *dynamical spatio-temporal models* (DSTMs). We shall consider both linear and nonlinear DSTMs, as well as data, process, and parameter models that could be Gaussian or non-Gaussian.

The essence of a DSTM is the hierarchical state-space framework described in Section 3.6. Specifically, we recognize that it is impossible to observe any process perfectly, and thus we require a mapping that relates a set of observations to the true process of interest. As mentioned previously, this is just the data-model level in a hierarchical framework. At the next level, we specify a model for this true (hidden/latent/state) process. In the context of DSTMs, this model typically assumes some sort of Markovian-dependence structure in time. As in the case of purely spatial (Chapter 4) or purely temporal (Chapter 3) processes, a prior distribution might be put on the collection of parameters used in the model specification at these two levels, to create a third level (BHM), or it may be decided to estimate them (EHM).

We can write the general DSTM schematically in terms of a data model, process model, and possibly parameter (prior) model as follows.

Data Models

The distribution of data can be generally written as

$$[\{ Z(\mathbf{x}; r) : \mathbf{x} \in D_s, r \in D_t \} | \{ Y(\mathbf{s}; t) : \mathbf{s} \in \mathcal{N}_s, t \in \mathcal{N}_t \}, \boldsymbol{\theta}_D], \quad (7.1)$$

where $Z(\mathbf{x}; r)$ is an (actual or potential) observation at spatial location $\mathbf{x}$ and time r , $Y(\mathbf{s}; t)$ represents the (hidden or latent) process of interest at spatial location $\mathbf{s}$ and time t , and $\boldsymbol{\theta}_D$ refers to data-model parameters that may also

vary in space and/or time. Note that the observation locations (in space and time) and their relative supports and/or alignments may be different than those for the process. In addition, we represent the “spatial neighborhood” of $\mathbf{x}$ as $\mathcal{N}_{\mathbf{x}}$ and, similarly, the “past neighborhood” of r as $\mathcal{N}_r$. Thus, equation (7.1) simply gives the distribution of observations at possibly different supports, conditional on values of the true process of interest at some other locations/levels of support and some parameters. Usually, the data are conditionally independent, yielding a data model that is a product of univariate conditional distributions. In the absence of conditional independence, the data model involves the joint (conditional) distribution of all observations given by (7.1).

The form of the DSTM data model given in (7.1) is really too general to be of much use. One can imagine an infinite number of formulations for data models that could be used in various spatio-temporal applications. Rather than continue to generalize, we find it more useful to discuss specific types of models that have proven useful for spatio-temporal models and that illustrate the power of the conditional framework. Thus, in Section 7.1, we present a discussion of the data model for a number of specific cases. Critical in the formulation of such models in practice is the parameterization of change of support, as well as dimension reduction. In the case of the latter, this corresponds to the notion that the true underlying (hidden) dynamical process is operating on a lower-dimensional manifold than physical space. Other issues of importance have to do with the distributional nature of the observations—for example, deciding whether the data, conditional on the process, are Gaussian or non-Gaussian, and deciding whether the observations are not conditionally independent (given the hidden process).

Process Model

The DSTM process distribution (i.e., process model) that we shall use can generally be written as

$$[Y(\mathbf{s}; t) | \{Y(\mathbf{w}; t - \tau_1); \mathbf{w} \in \mathcal{N}_{\mathbf{s}}^{(1)}\}, \dots, \{Y(\mathbf{w}; t - \tau_p); \mathbf{w} \in \mathcal{N}_{\mathbf{s}}^{(p)}\}, \boldsymbol{\theta}_P], \quad (7.2)$$

where $\mathcal{N}_{\mathbf{s}}^{(1)}, \dots, \mathcal{N}_{\mathbf{s}}^{(p)}$ are neighborhoods of the spatial location $\mathbf{s}$, corresponding to the time lags $0 < \tau_1 < \dots < \tau_p$, and $\boldsymbol{\theta}_P$ refers to the process-model parameters that may vary with space and/or time. Going from conditional distributions (7.2) to a joint distribution for the whole process $Y(\cdot; \cdot)$ requires the same type of theory developed in the spatial-only case in Section 4.2.3. This form of the DSTM process model is too general to be of much use. In Sections 7.2 and 7.3, we shall discuss specific formulations of linear and non-linear dynamical-process specifications, respectively. Then, in Section 7.4, we discuss multivariate processes in the context of the DSTM framework.

One of the critical aspects of using DSTMs is the curse of dimensionality. Specifically, we recognize that in many disciplines where spatio-temporal data are available, the number of time points at which there are data can be relatively

few. In such situations, specifying interactions of the process across space and time is problematic, unless one can efficiently reduce dimensionality of either the process or the parameters. These issues are even more pronounced in the case of nonlinear-evolution operators.

Parameter Models

In general, the parameter-model distribution is represented as

$$[\theta_D, \theta_P | \theta_h], \quad (7.3)$$

where θ_h refers to “hyperparameters.” Again, this framework is quite general, and ultimately it is more useful to consider specific examples; these will obviously depend on choices made at higher levels of the hierarchy. In many cases, one assumes (conditional) independence in the parameter distributions; for example, $[\theta_D, \theta_P | \theta_h] = [\theta_D | \theta_h][\theta_P | \theta_h]$. It is important to reiterate that in certain instances, it may be sufficient to estimate or specify either the data parameters or the process parameters (or hyperparameters), rather than model them at this third level of the hierarchy. Furthermore, we recognize that we might need even lower levels in complicated modeling scenarios. It is often the case that the choice of these prior distributions is driven by the underlying scientific problem.

It is important to note that it is not always possible to distinguish clearly between process and parameters. Indeed, just as one person’s mean structure is another’s correlation structure, so it is with processes and parameters. In many cases, the underlying Science makes the distinction clear. However, it really doesn’t matter, as long as one has plausible conditional probability structures. The general hierarchical model is not intended to be a rigid structure, but rather a framework for developing conditional-probability models to model complex problems.

7.1 DATA MODELS FOR THE DSTM

As with all hierarchical formulations, the power of the data model for DSTMs comes from conditioning. That is, the specification of the spatio-temporal data distribution given the true process of interest is typically much simpler than the unconditional distribution of the data. Specifically, if the data are $\mathbf{Z}_1, \dots, \mathbf{Z}_T$, it is exceedingly difficult to specify the joint distribution $[\mathbf{Z}_1, \dots, \mathbf{Z}_T]$. However, in most cases, the dependence in the data comes from the hidden process $\{\mathbf{Y}_t\}$, and not typically the measurement process. Thus, the conditional distribution of data given the process is often simplified greatly. For example, the data model might be

$$[\mathbf{Z}_1, \dots, \mathbf{Z}_T | \mathbf{Y}_1, \dots, \mathbf{Y}_T, \theta_D] = \prod_{t=1}^T [\mathbf{Z}_t | \mathbf{Y}_t, \theta_D], \quad (7.4)$$

where we have included dependence on parameters θ_D . The main point here is the assumption that the data are conditionally independent in time, given the hidden process.

Conditioning also plays a very important role in how multiple sources of information are handled. Indeed, one of the most powerful features of hierarchical models is their ability to accommodate multiple datasets (e.g., Section 2.1.5). For example, say we have observations $\mathbf{Z}_t^{(1)}$ and $\mathbf{Z}_t^{(2)}$ (perhaps at different supports or from different instruments) of some process $\mathbf{Y}_t$. In this case, we seek a model for the joint distribution of both data vectors, conditional upon the process $\mathbf{Y}_t$. It is often the case that, conditional on the hidden process, the two datasets are independent. That is, the errors in measurement (or aggregation, etc.) in the two datasets are not related, given one knows the true process. This implies the following simplifying assumption in the data model:

$$[\mathbf{Z}_t^{(1)}, \mathbf{Z}_t^{(2)} | \mathbf{Y}_t, \theta_D] = [\mathbf{Z}_t^{(1)} | \mathbf{Y}_t, \theta_D^{(1)}][\mathbf{Z}_t^{(2)} | \mathbf{Y}_t, \theta_D^{(2)}], \quad (7.5)$$

where $\theta_D \equiv \{\theta_D^{(1)}, \theta_D^{(2)}\}$ are parameters associated with the two data models. Clearly, the benefit of such an assumption is that for each dataset we can use one of the data models presented below, and we do not have to specify the joint dependencies in the datasets directly. Of course, the common dependence of $\mathbf{Z}_t^{(1)}$ and $\mathbf{Z}_t^{(2)}$ on $\mathbf{Y}_t$ would induce marginal dependence between $\mathbf{Z}_t^{(1)}$ and $\mathbf{Z}_t^{(2)}$.

Data models of this sort arise when one has multiple observation sources for a given process, including remotely sensed observations (e.g., from satellites or radars), deterministic model output (such as from numerical weather/climate models), and *in situ* observations. For example, Wikle et al. (2001) consider observations for tropical winds, as given by satellite-based scatterometers, as well as “observations” from numerical-weather-forecast models. This latter source may seem somewhat unusual since the “observations” from such models are not direct observations. However, such models assimilate observations from other sources and produce output that can be thought of as interpolated or smoothed observations of the hidden process. As such, it is information regarding the true state and can certainly be included as “data” in a hierarchical framework.

The hierarchical framework can be used with multiple datasets and multiple processes as well (e.g., Section 2.1.5). That is, assume that associated with each dataset $\mathbf{Z}_t^{(1)}$ and $\mathbf{Z}_t^{(2)}$ are (hidden) processes $\mathbf{Y}_t^{(1)}$ and $\mathbf{Y}_t^{(2)}$, respectively. Again, it is often reasonable to assume that the two datasets are conditionally independent, conditional on knowing the true states of the processes:

$$[\mathbf{Z}_t^{(1)}, \mathbf{Z}_t^{(2)} | \mathbf{Y}_t^{(1)}, \mathbf{Y}_t^{(2)}, \theta_D] = [\mathbf{Z}_t^{(1)} | \mathbf{Y}_t^{(1)}, \theta_D^{(1)}][\mathbf{Z}_t^{(2)} | \mathbf{Y}_t^{(2)}, \theta_D^{(2)}]. \quad (7.6)$$

In addition to characterizing the distributional nature of measurement error in such data models, differences in support/alignment and other transformations between the observations and the hidden process can also be accounted for. In the remainder of this section, we describe a variety of different parameterizations of the data model that account for such transformations.

7.1.1 Linear Mappings with Equal Dimensions

The simplest data model occurs when observation locations and process locations coincide (i.e., there are no missing data), and measurement errors are additive and conditionally independent in space and time. That is,

$$Z(\mathbf{s}; t) = Y(\mathbf{s}; t) + \epsilon(\mathbf{s}; t), \quad (7.7)$$

where the measurement errors $\epsilon(\mathbf{s}; t)$ have zero mean, common variance σ_ϵ^2 , are independent of $Y(\cdot; \cdot)$, and are independent across time and space. Although not necessary, it is often the case that a reasonable distributional assumption on the measurement errors is that they are Gaussian. Thus, for spatial locations $\mathbf{s}_1, \dots, \mathbf{s}_n$, the data model implies that

$$\mathbf{Z}_t = \mathbf{Y}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim iid \text{Gau}(\mathbf{0}, \sigma_\epsilon^2 \mathbf{I}), \quad (7.8)$$

where $\mathbf{Z}_t \equiv (Z(\mathbf{s}_1; t), \dots, Z(\mathbf{s}_n; t))'$, $\mathbf{Y}_t \equiv (Y(\mathbf{s}_1; t), \dots, Y(\mathbf{s}_n; t))'$, and $\boldsymbol{\varepsilon}_t \equiv (\epsilon(\mathbf{s}_1; t), \dots, \epsilon(\mathbf{s}_n; t))'$ is independent of $\mathbf{Y}_t$. The obvious advantage of this model is that it only contains one parameter, σ_ϵ^2 . Despite its simplicity, such a model can be quite useful, since there is often interest in modeling at the same locations/resolutions as the observations, and the conditionally independent error structure is reasonable in many real-world applications. If the measuring instruments are properly calibrated and the calibration results are available, measurement-error variances might be assumed known.

Perhaps the next level of sophistication is to make the same assumptions as above, with the further possibility of additive and multiplicative bias in the observations. For example, consider

$$Z(\mathbf{s}; t) = a + hY(\mathbf{s}; t) + \epsilon(\mathbf{s}; t), \quad (7.9)$$

where a and h represent the additive and multiplicative bias terms, respectively. Notice that in this formulation, any bias has been taken out of the error term $\epsilon(\mathbf{s}; t)$, leaving it with zero mean. In principle, it might be the case that the biases in (7.9) could vary with space and time; for spatial locations $\mathbf{s}_1, \dots, \mathbf{s}_n$, this would imply that

$$\mathbf{Z}_t = \mathbf{a}_t + \text{diag}(\mathbf{h}_t)\mathbf{Y}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim iid \text{Gau}(\mathbf{0}, \sigma_\epsilon^2 \mathbf{I}), \quad (7.10)$$

where $\mathbf{a}_t \equiv (a(\mathbf{s}_1; t), \dots, a(\mathbf{s}_n; t))'$, $\mathbf{h}_t \equiv (h(\mathbf{s}_1; t), \dots, h(\mathbf{s}_n; t))'$, and recall that $\text{diag}(\mathbf{h})$ is a matrix with the elements of $\mathbf{h}$ down the diagonal and zeros elsewhere. One can see that in this scenario it could be very difficult to estimate these spatially (and potentially temporally) varying parameters. In cases where there is prior knowledge about this bias structure, $\{\mathbf{a}_t\}$ and $\{\mathbf{h}_t\}$ could be further modeled (e.g., parameterized in terms of covariates or modeled in terms of some correlated random process). This could be captured through a parameter model for $\{\mathbf{a}_t\}$ and $\{\mathbf{h}_t\}$ (see Section 7.5).

A logical extension to (7.10) is to consider mappings that relate the observations to linear combinations of the complete process:

$$\mathbf{Z}_t = \mathbf{a}_t + \mathbf{H}_t \mathbf{Y}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim iid \text{Gau}(\mathbf{0}, \sigma_\varepsilon^2 \mathbf{I}), \quad (7.11)$$

where $\mathbf{H}_t$ is an $n \times n$ matrix. Obviously, it would be very difficult to estimate these time-varying coefficients without further assumptions. In fact, such a general framework is most often considered in the case where the dimension of the data and process vectors are not equal, as discussed in Section 7.1.2.

Perhaps more important is the case where the observations, after conditioning on the process (and bias/transformation corrections), have heterogeneous measurement errors in time and/or space. For example, satellite observations of an atmospheric quantity might be affected by the presence of clouds, so that measurements in cloudy regions have a different measurement-error variance than those in clear regions. In this case, the model is written as in (7.10) or (7.11), except we assume that $\boldsymbol{\varepsilon}_t \sim ind. \text{Gau}(\mathbf{0}, \text{diag}(\mathbf{v}_t))$, where $\mathbf{v}_t \equiv (v(\mathbf{s}_1; t), \dots, v(\mathbf{s}_n; t))'$ and $v(\mathbf{s}_i; t)$ corresponds to the variance associated with the i -th location at time t . Again, these spatially and temporally explicit variances could be difficult to estimate without further assumptions (that could be captured through a parameter model). For example, these might be modeled as spatially varying GARCH or stochastic volatility processes (e.g., Section 3.4.8).

The final extension to the basic scenario, with observations at the same locations as the process, is to relax the assumption of (conditional) spatial independence for the observations. In this case, the model is again as written in (7.10) or (7.11), except that now, $\boldsymbol{\varepsilon}_t \sim ind. \text{Gau}(\mathbf{0}, \mathbf{R}_t)$, again with the possibility that the covariance matrix is changing in time. Even in the case where $\mathbf{R}_t$ does not vary with time, estimation is typically problematic without further assumptions on the structure of the covariance matrix. In fact, for fixed t , $\mathbf{R}_t$ can be determined from a typical spatial model of dependence as discussed in Chapter 4. Clearly, the assumption of temporal independence may be relaxed, and all manner of time series models (see Chapter 3) could “in principle” be applied. Or, if one has both spatially and temporally correlated measurement errors, one might use a spatio-temporal covariance model such as discussed in Chapter 6.

7.1.2 Linear Mappings with Unequal Dimensions

Assume now a data model for DSTMs, where the observation dimension is different from the state-process dimension, n . For example, assume

$$\mathbf{Z}_t = \mathbf{H}_t \mathbf{Y}_t + \boldsymbol{\varepsilon}_t, \quad (7.12)$$

where $\mathbf{Z}_t$ is m_t -dimensional, indicating that there are observations at m_t spatial locations; $\mathbf{Y}_t$ is n -dimensional, representing the n -dimensional state process of interest; $\boldsymbol{\varepsilon}_t$ is m_t -dimensional with zero mean and covariance matrix $\mathbf{R}_t$; $\boldsymbol{\varepsilon}_t$ and $\mathbf{Y}_t$ are independent; and $\mathbf{H}_t$ is of dimension $m_t \times n$. It is usual to assume that $\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2, \dots$ are mutually independent. This form is quite general

and encompasses many different scenarios in spatio-temporal modeling, as described below. Of course, we could also include bias corrections as discussed in Section 7.1.1, but to keep the presentation simple, we leave it as an exercise for the reader.

In many cases we “know” the form of $\mathbf{H}_t$. Often this is because we have a good scientific sense of the measurement process. For example, in the case of satellite retrievals, it is often the case that this observation matrix corresponds to some representation of radiative transfer that maps the radiance measurements to the process. We consider below some special cases where the form of $\mathbf{H}_t$ is assumed known (in the sense that it is a deterministic function of where and when we have observations).

Incidence Matrix

Often, the observations at certain locations are associated with the process at a different set of locations. In Section 7.1.1, each observation is associated with one state-process element; then $\mathbf{H}_t$ is simply an *incidence matrix*, which is a matrix of zeros with the exception of a single one in each row (corresponding to the element that matches the observation with the process location).

The same 0–1 structure holds when $m_t > n$ or $m_t < n$, and it is certainly reasonable to allow an incidence matrix to vary temporally to account for different observation locations through time. As an example of how an incidence matrix works in this context, consider the case where we have three observation locations $\{\mathbf{x}_i : i = 1, 2, 3\}$ and two process locations $\{\mathbf{s}_1, \mathbf{s}_2\}$, such that observations at $\mathbf{x}_1$ and $\mathbf{x}_2$ are associated with the process at location $\mathbf{s}_1$ and the observation at $\mathbf{x}_3$ is associated with the process at location $\mathbf{s}_2$. That is, $m_t > n$.

In this case, the incidence matrix is given by

$$\mathbf{H}_t \equiv \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (7.13)$$

The implication of using (7.13) for $\mathbf{H}_t$ in (7.12) is that $Z(\mathbf{x}_1; t)$ and $Z(\mathbf{x}_2; t)$ are replicate observations with mean $Y(\mathbf{s}_1; t)$ and variance σ_ϵ^2 . An example of when we might consider such a data model is if we are predicting $Y(\cdot; t)$ at spatial locations on a grid or lattice, and we have (possibly) multiple observations at smaller spatial resolution than the process of interest. For example, there might be two point-support observations associated with a process’ gridbox value (e.g., Wikle, Berliner, and Cressie, 1998).

The case where m_t is less than n occurs when there are missing data. For example, say we have observations $\mathbf{Z}_t = (Z(\mathbf{s}_1; t), Z(\mathbf{s}_3; t))'$ and the process $\mathbf{Y}_t = (Y(\mathbf{s}_1; t), Y(\mathbf{s}_2; t), Y(\mathbf{s}_3; t))'$. Then the incidence matrix is given by

$$\mathbf{H}_t \equiv \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (7.14)$$

which accommodates the missing observation for $Y(\mathbf{s}_2; t)$.

Change-of-Support

The incidence-matrix form for $\mathbf{H}_t$ suggests more general possibilities by which one might assign observations to process locations. For example, if $\mathbf{Z}_t = (Z(\mathbf{x}_1; t), Z(\mathbf{x}_2; t), Z(\mathbf{x}_3; t))'$ and $\mathbf{Y}_t = (Y(\mathbf{s}_1; t), Y(\mathbf{s}_2; t))'$, then consider

$$\mathbf{H}_t = \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \\ h_{31} & h_{32} \end{pmatrix}, \quad (7.15)$$

where $h_{ij} > 0$ ($i = 1, \dots, m_t$; $j = 1, \dots, n$) is the weight associated with the observation at location $\mathbf{x}_i$ and the process at location $\mathbf{s}_j$, and $\sum_j h_{ij} = 1$. The obvious question is how should these weights be selected.

A situation where one might know $\{h_{ij}\}$ is with the change-of-support problem. With a slight abuse of notation, assume that locations $\mathbf{x}_i$ and $\mathbf{s}_j$ correspond to areal support, where the areas are given by $|\mathbf{x}_i|$ and $|\mathbf{s}_j|$, respectively. Furthermore, consider the possibility that $|\mathbf{x}_i| \neq |\mathbf{s}_j|$. That is, perhaps the observations are at larger or smaller levels of spatial support than the process of interest (e.g., if observations are from satellites with a resolution smaller than the prediction grid, then $|\mathbf{x}_i| < |\mathbf{s}_j|$). As described briefly in Section 4.1.3, Wikle and Berliner (2005) show that, with minor assumptions, the weights in this case are

$$h_{ij} = |\mathbf{x}_i \cap \mathbf{s}_j| / |\mathbf{x}_i|, \quad (7.16)$$

for $i = 1, \dots, m_t$; $j = 1, \dots, n$. These weights simply reflect the intersection (area of overlap) between the observation and process support locations. In particular, they account directly for change-of-support, as well as change in alignment between observation and process locations. Note that many of the weights will be zero, since the intersection of most data-support and process-support locations is null.

In addition, change-of-support has implications for the measurement covariance matrix ($\mathbf{R}_t$, defined above) as well. Again, this is described briefly in Section 4.1.3. Suppressing the temporal index t , assume

$$Y(\mathbf{r}) = Y(\mathbf{s}_j) + \gamma(\mathbf{r}), \quad (7.17)$$

for any (point-support) location $\mathbf{r} \in \mathbf{s}_j$ in the spatial domain of interest, where $\{\gamma(\mathbf{r})\}$ is a correlated process with spatial covariance function $C_\gamma(\mathbf{r}, \mathbf{r}')$, independent of $\{Y(\mathbf{s}_j): j = 1, \dots, n\}$.

In this illustration, assume that the data supports, $\{|\mathbf{x}_i|: i = 1, \dots, m\}$, involve smaller areas than the desired prediction supports, $\{|\mathbf{s}_j|: j = 1, \dots, n\}$, and recall that block support for some arbitrary block R is defined as

$$Y(R) = \frac{1}{|R|} \int_R Y(\mathbf{r}) d\mathbf{r}, \quad |R| > 0, \quad (7.18)$$

where $|R| \equiv \int_R d\mathbf{r}$. Then, using (7.17) and (7.18), we obtain

$$\begin{aligned} Y(\mathbf{x}_i) &= \frac{1}{|\mathbf{x}_i|} \sum_{j=1}^n \left(\int_{\mathbf{x}_i} Y(\mathbf{s}_j) d\mathbf{r} \right) + \frac{1}{|\mathbf{x}_i|} \int_{\mathbf{x}_i} \gamma(\mathbf{r}) d\mathbf{r} \\ &= \mathbf{h}'_i \mathbf{Y} + \frac{1}{|\mathbf{x}_i|} \int_{\mathbf{x}_i} \gamma(\mathbf{r}) d\mathbf{r}, \end{aligned} \quad (7.19)$$

where $\mathbf{Y} \equiv (Y(\mathbf{s}_1), \dots, Y(\mathbf{s}_n))'$, $\mathbf{h}_i \equiv (h_{i1}, \dots, h_{in})'$, and h_{ij} is defined by (7.16). Now, for $\mathbf{Y}^{(x)} \equiv (Y(\mathbf{x}_1), \dots, Y(\mathbf{x}_m))'$, which is the process corresponding to the data supports, we have

$$\mathbf{Y}^{(x)} | \mathbf{Y}, \Sigma_Y \sim \text{Gau}(\mathbf{H}\mathbf{Y}, \Sigma_Y), \quad (7.20)$$

where the matrix $\mathbf{H}$ has (i, j) th element h_{ij} , and the (p, q) th element of Σ_Y is given by

$$\frac{1}{|\mathbf{x}_q| |\mathbf{x}_p|} \int_{\mathbf{x}_p} \int_{\mathbf{x}_q} C_Y(\mathbf{r}, \mathbf{r}') d\mathbf{r} d\mathbf{r}'. \quad (7.21)$$

Given the measurement-error model,

$$\mathbf{Z} | \mathbf{Y}^{(x)}, \Sigma_Z \sim \text{Gau}(\mathbf{Y}^{(x)}, \Sigma_Z), \quad (7.22)$$

one can use (7.20) and (7.22) to show that

$$\mathbf{Z} | \mathbf{Y}, \Sigma_Y, \Sigma_Z \sim \text{Gau}(\mathbf{H}\mathbf{Y}, \Sigma_Y + \Sigma_Z). \quad (7.23)$$

Thus, as noted in Section 4.1.3, change-of-support has induced a correlated structure, $\Sigma_Y + \Sigma_Z$, in the data model. Note that similar structures for this covariance matrix and $\mathbf{H}$ arise if the support of the data is larger than the desired prediction support. For a more complete discussion, see Wikle and Berliner (2005).

Finally, we mention that in many problems where this approach to change of spatial support is used in a spatio-temporal framework, it is possible that the weights $\mathbf{H}_t$ will change with time, due to differing observation locations or missing observations. Furthermore, the change-of-support can be given in terms of temporal aggregation as well.

7.1.3 Dimension Reduction

Characterizing the dependence structures in the hidden (state) process can be problematic, due to the curse of dimensionality. It is often necessary in such cases to reduce the effective dimensionality of the (spatial) state vector. In many cases, this is reasonable, since the true dynamical process exists on

a lower-dimensional manifold, or at least it is reasonably approximated by the assumption that it does (e.g., see the discussion of dynamical systems in Section 3.2).

Recall, in the linear, additive-error data model (with no additive bias or transformation), we might assume, $\mathbf{Z}_t = \mathbf{H}_t \mathbf{Y}_t + \boldsymbol{\varepsilon}_t$, where $\mathbf{Z}_t$ and $\boldsymbol{\varepsilon}_t$ are both m_t -dimensional vectors, $\mathbf{Y}_t$ is n -dimensional, and $\boldsymbol{\varepsilon}_t$ is independent of $\mathbf{Y}_t$. Now, let us assume that the process $\mathbf{Y}_t$ can be decomposed as

$$\mathbf{Y}_t = \Phi \boldsymbol{\alpha}_t + \mathbf{v}_t, \quad (7.24)$$

where the lower-dimensional process is p_α -dimensional, $\boldsymbol{\alpha}_t \equiv (\alpha_t(1), \dots, \alpha_t(p_\alpha))'$, Φ is an $n \times p_\alpha$ matrix with $p_\alpha \ll n$, and $\mathbf{v}_t$ is an n -dimensional vector accounting for the residual process not included in $\Phi \boldsymbol{\alpha}_t$. Note that $\boldsymbol{\alpha}_t$ may not be “spatially referenced” but $\Phi \boldsymbol{\alpha}_t$ is. The model (7.24) is a “process model” in the usual HM nomenclature (see Sections 7.2 and 7.3). However, if the modeling focus is on $\boldsymbol{\alpha}_t$ rather than $\mathbf{Y}_t$, it can be convenient to rewrite the data model in terms of $\boldsymbol{\alpha}_t$. In this case,

$$\mathbf{Z}_t = \mathbf{H}_t \Phi \boldsymbol{\alpha}_t + \mathbf{H}_t \mathbf{v}_t + \boldsymbol{\varepsilon}_t, \quad (7.25)$$

where $\mathbf{H}_t$ can take any of the forms described previously, and one may find it useful to combine the error processes into a single (more complicated) process; for example, write $\boldsymbol{\gamma}_t = \mathbf{H}_t \mathbf{v}_t + \boldsymbol{\varepsilon}_t$. If one does this combining, one loses the ease of predicting $\mathbf{Y}_t$ in order to concentrate on predicting $\boldsymbol{\alpha}_t$ or, equivalently, $\Phi \boldsymbol{\alpha}_t$.

The key here is the dimension-reduction matrix Φ . We often refer to this as a matrix of basis functions (e.g., see Section 4.1.4) or the “spectral transformation matrix” (e.g., see Section 3.5). Now Φ is (often) assumed known, does not have to have orthogonal columns, and serves to relate the process vector $\mathbf{Y}_t$ to the reduced-dimensional vector $\boldsymbol{\alpha}_t$. As was seen in Section 6.6, we refer to the decomposition (7.24) as a *spectral representation*, where we use the word “spectral” rather broadly to refer to general basis function expansions. In that sense, $\boldsymbol{\alpha}_t$ can be called a vector of spectral coefficients.

There are many possible choices for the spectral transformation matrix Φ , and “novel” formulations of spatio-temporal dynamical models have often (and continue to be) proposed by suggesting a different choice for Φ . Such novelty is perhaps overstated, but the concept is powerful. Some specific choices for Φ are described in the following subsections.

Finally, it is certainly the case that $\mathbf{v}_t$ could be further decomposed in an additional expansion; for example, write

$$\mathbf{v}_t = \Psi \boldsymbol{\xi}_t + \boldsymbol{\omega}_t,$$

where $\boldsymbol{\omega}_t$ is an error process, $\boldsymbol{\xi}_t$ is a p_ξ -dimensional coefficient vector associated with the $n \times p_\xi$ matrix of basis functions Ψ , which typically are not

of the same form as Φ . The vector ξ_t may be allowed to vary dynamically in the process model, similarly to α_t . Why would we need to have two different spectral components for the $\{Y_t\}$ process? One reason would be if we were attempting to capture the variability associated with spatio-temporal processes on different spatial scales (and with different dynamical properties). For example, Wikle et al. (2001) consider such a case, where Φ corresponds to normal modes of a deterministic dynamical system for atmospheric winds, and Ψ are wavelet basis functions that accommodate the multiresolutional nature of turbulent scaling relationships in atmospheric winds. In this case, the associated spectral coefficients, α_t and ξ_t , follow first-order dynamical process models (although, with substantially different parameterizations). Overall, the model, $Y_t = \Phi\alpha_t + \Psi\xi_t + \omega_t$, provides a flexible and powerful decomposition of a spatio-temporal process. Of course, additional decompositions (e.g., of ω_t) can be specified if there is sufficient scientific justification.

Pure Spectral Decomposition

For motivation, we discuss first the case where we have spectral operators for Φ , and $p_\alpha = n$. In this case, $Y_t \equiv \Phi\alpha_t$, and α_t is n -dimensional. That is, there is a simple linear transformation of α_t that gives Y_t , with no explicit dimension reduction of the process (i.e., Φ is an $n \times n$ matrix and $v_t = \mathbf{0}$ for all t). One may ask why this is considered “dimension reduction,” and what the advantage is of such an assumption. One advantage is that the parameterization of the process at the next level of the HM is often simpler. That is, for certain basis functions such as the Fourier, wavelet, and EOF bases, the implied process α_t is simpler, since the spectral transformation acts as a *decorrelator* (see Section 3.5). Thus, in this case, “dimension reduction” refers to the implied reduction in the number of process-model *parameters*, even if the dimension of the process itself has not been reduced.

As an example, consider the simple case where $Y_t \sim \text{Gau}(\mathbf{0}, \mathbf{R})$. Then, it follows that $\alpha_t \sim \text{Gau}(\mathbf{0}, \mathbf{R}_\alpha)$, where $\mathbf{R}_\alpha \equiv (\Phi'\Phi)^{-1}\Phi'\mathbf{R}\Phi(\Phi'\Phi)^{-1}$, for $\Phi'\Phi$ nonsingular. Notice that if Φ is orthogonal such that $\Phi'\Phi = \mathbf{I}$, then $\mathbf{R}_\alpha = \Phi'\mathbf{R}\Phi$. Recall that, for second-order stationary processes, it is often the case that $\mathbf{R}_\alpha$ is reasonably well approximated by a diagonal matrix, $\mathbf{R}_\alpha = \text{diag}(\mathbf{d})$, where $\mathbf{d}$ is a p_α -dimensional vector of variances (see Section 3.5). Furthermore, for known classes of valid spatial covariance functions, $\mathbf{R}(\theta)$ depends on parameters θ (e.g., the Matérn class considered in Section 4.1.1). If there is an analytical form for the spectral transform (e.g., Fourier), one can parameterize the elements of $\mathbf{d}$; we write it here as $\mathbf{d}(\theta)$. In addition to the likely simplification in the dynamics at the process level, this suggests powerful computational advantages for spatial prediction over large domains with large numbers of spatial grid points, even in the hierarchical-Bayesian-model case where θ is given a prior distribution (see Wikle, 2002b; Royle and Wikle, 2005; Paciorek, 2007). Similar computational savings can be shown when multiresolution wavelet basis functions are used (e.g., Nychka, Wikle, and Royle, 2002).

These have the additional advantage of easily accounting for nonstationarity, but they are limited in that, typically, analytical forms for $\mathbf{R}_\alpha$ are not available.

A criticism of these spectral-transformation approaches is that the process has to be modeled on some regular grid in order to take advantage of the computational efficiencies of the spectral operator. However, in all but the rarest cases, we can deal with this by specifying a very-high-resolution grid such that each observation location corresponds to a unique process location. Alternatively, we can use one of the formulations of $\mathbf{H}_t$ given above to account for a regular process-model grid.

It should also be mentioned that many processes, especially those in the atmospheric sciences, are studied theoretically and via data analysis, in the spectral domain. In such cases, process modeling in the spectral domain is not only simpler in terms of model structures and computation, but conceptually as well.

Empirical Orthogonal Functions

Empirical orthogonal functions (EOFs; see Sections 4.1.4 and 5.3) have often been used as the spectral basis functions in Φ for dimension reduction. That is, if we had access to the first p_α eigenvectors of a stationary covariance matrix for the process $\{\mathbf{Y}_t\}$ (typically an estimated or “empirical” covariance matrix calculated from $\{\mathbf{Z}_t\}$, or from some other source of information such as a deterministic numerical model), then the process model, $\mathbf{Y}_t = \Phi \alpha_t + \mathbf{v}_t$ [see (7.24)], is somewhat simplified, in that α_t has a diagonal covariance matrix with diagonal elements given by the eigenvalues of the original covariance matrix used to generate the EOFs. However, the dynamics of α_t may suggest more complicated structures than implied by the EOF decomposition.

Typically, one only considers the first p_α EOFs that account for some significant portion of $\text{var}(\mathbf{Y}_t)$. Since EOFs typically decrease in spatial scale with decreasing variance explained, it is often reasonable in such cases to assume that $\mathbf{v}_t \sim \text{Gau}(\mathbf{0}, \sigma_v^2 \mathbf{I})$. Consequently, σ_v^2 and σ_ϵ^2 (the measurement-error variance) are not identifiable without additional (prior) information.

In cases where p_α is selected such that a reasonable amount of spatial structure is left in $\mathbf{v}_t$, one can model it by either choosing a valid spatial covariance function (see Chapter 4) or using a portion of the remaining EOFs to characterize the covariance matrix of $\mathbf{v}_t$, say Σ_v . That is, one could simply assume

$$\Sigma_v = c\mathbf{I} + \sum_{k=p_\alpha+1}^{p_\alpha+p_v} \lambda_k \Phi_k \Phi_k', \quad (7.26)$$

where c is an unknown constant, and λ_k and Φ_k , for $k = p_\alpha + 1, \dots, p_\alpha + p_v$, are the next p_v (after p_α) eigenvalues and eigenvectors, respectively, of the EOF decomposition. The constant c must be estimated (EHM) or given a prior distribution (BHM) to account for the extra variability lost in the EOF

truncation and to ensure that Σ_y is positive-definite. It may be the case that $\{\lambda_k\}$ in (7.26) are estimated or given prior distributions as well.

Despite their prevalence (especially in atmospheric/ocean science applications) and their success as a means of dimension reduction, there are several issues that can make the EOF basis function less appealing in some situations. First, if we have no missing data and many more time replicates (T) than spatial locations (n), then from the empirical covariance matrix based on Z_t , EOF basis functions can be used to approximate $\text{var}(Y_t)$. However, when $T < n$, as is often the case in practice, we can only get T EOFs, and they are sometimes of dubious quality due to the sampling variability caused by small T . Furthermore, when data are missing, or interest is in predicting at locations where there are no observations, we cannot get EOFs directly without some sort of data smoothing (e.g., Wikle and Cressie, 1999). This pre-smoothing is less than appealing, given that smoothing is often the goal of the spatio-temporal analysis in the first place, and it is not always clear how much the final predictions are influenced by the pre-smoothing operation. A more rigorous framework for building EOFs in this case makes use of localized basis functions (e.g., Obled and Creutin, 1986; Wikle, 1996), but this can be a computational and practical burden. Alternatively, one can shrink the empirical covariance estimates to some parametric-based covariance matrix (Nott and Dunsmuir, 2002).

Other issues that influence the EOF basis functions concern the sensitivity of EOFs to the geometry of the spatial domain upon which the data are defined. For example, as discussed in Sections 4.1.4 and 5.3, it can be shown that EOFs can be thought of as a discretization of the Karhunen–Loève integral equation. Such a discretization must account for the area of influence of each observation (around its spatial location) if the locations of observations are not on a regular grid (Cohen and Jones, 1969; Buell, 1972; Karl, Koscielny, and Diaz, 1982). In addition, Buell (1975) showed that the EOF basis functions (patterns) are heavily influenced by the shape of the spatial domain where the observations were taken (see Jolliffe, 2002, p. 297, for an in-depth discussion of this issue).

Perhaps a more critical concern is that, although it is the case that the truncated set of EOF basis functions are optimal in the sense of describing the most variance among any orthogonal set of p_α basis functions, they are not necessarily optimal in terms of representing the spatio-temporal dynamics of the process. This was clearly demonstrated in Aubry, Lian, and Titi (1993), who gave an example of an EOF decomposition that accounted for 99.9995% of the variation in the process, but it could not adequately model the dynamics of a particular process. In that case, the process was nonlinear; from a “chaos” perspective (Section 3.2.4), it is perhaps not surprising that a component that represents such a small part of the overall variation can play an important role in the dynamics. This is particularly problematic when the process includes abrupt transitions between different regimes. A statistical approach that can help with this problem is to select the reduced set of EOF basis functions stochastically, rather than by the amount of variation explained. For example, Wikle and Holan (2011) used the stochastic search variable selection methodology of George and McCulloch (1993) to choose the appropriate basis functions for a

hierarchical DSTM. However, it is also the case that, in general, the EOF basis may not provide the best low-order reduction for *representing the dynamics* of even linear systems.

Despite these concerns, our experience is that EOFs are one of the better sets of basis functions to use, when the degree of dimension reduction and the computational simplicity are factored in. If nothing else, the EOF set of basis functions should be the standard by which other sets of basis functions are judged.

Alternative System-Reduction Approaches

There is a substantial literature in Control Theory and Fluid Dynamics that addresses alternative sets of basis functions for dimension reduction in dynamical systems [e.g., see Antoulas (2005) for an overview]. As discussed in Sections 7.2 and 7.3, the spatio-temporal behavior of a process is governed by its dynamical operator and its noise-forcing covariance matrix. Thus, it is not surprising that optimal reduction strategies for spatio-temporal systems typically consider one or both of these components. This is one of the motivations for principle oscillation patterns (POPs) as discussed in Section 5.5.

As an example of a different approach to POPs, consider the notion of stochastic optimal analysis (SOA) as described, for example, in Farrell and Ioannou (2001). In order to perform SOA, one must know the form of the governing dynamics. For example, as discussed in Sections 3.4.7 and 6.4.2, consider the vector autoregressive (VAR) process of order one:

$$\mathbf{Y}_t = \mathbf{M}\mathbf{Y}_{t-1} + \boldsymbol{\eta}_t; \quad t = 1, 2, \dots,$$

where $E(\boldsymbol{\eta}_t) = \mathbf{0}$, $\text{var}(\boldsymbol{\eta}_t) \equiv \mathbf{Q}_\eta$, and $\boldsymbol{\eta}_t$ is independent of $\mathbf{Y}_{t-1}$, for $t = 1, 2, \dots$. Furthermore, assume that $E(\mathbf{Y}_t) = \mathbf{0}$ and $\text{var}(\mathbf{Y}_t) \equiv \boldsymbol{\Sigma}_Y$, for all t . Recall from Section 3.4.7 that we can write this VAR(1) process as an infinite-order MA process:

$$\mathbf{Y}_t = \sum_{i \geq 0} \mathbf{M}^i \boldsymbol{\eta}_{t-i},$$

which implies that

$$\boldsymbol{\Sigma}_Y = \sum_{i \geq 0} \mathbf{M}^i \mathbf{Q}_\eta (\mathbf{M}')^i.$$

These series converge if all eigenvalues of $\mathbf{M}$ are less than 1 in absolute value. Now, consider the expected total variation in $\mathbf{Y}_t$:

$$\begin{aligned} E(\mathbf{Y}_t' \mathbf{Y}_t) &= E(\text{tr}(\mathbf{Y}_t \mathbf{Y}_t')) \\ &= \text{tr}(\boldsymbol{\Sigma}_Y) \\ &= \text{tr} \left(\sum_{i \geq 0} \mathbf{M}^i \mathbf{Q}_\eta (\mathbf{M}')^i \right). \end{aligned}$$

Let $\mathbf{Q}_\eta = \mathbf{F}\mathbf{F}'$, so that

$$\begin{aligned} E(\mathbf{Y}_t' \mathbf{Y}_t) &= tr \left(\sum_{i \geq 0} \mathbf{M}^i \mathbf{F} \mathbf{F}' (\mathbf{M}')^i \right) \\ &= \sum_{i \geq 0} tr(\mathbf{F}' (\mathbf{M}')^i \mathbf{M}^i \mathbf{F}) \\ &= tr \left(\mathbf{F}' \left\{ \sum_{i \geq 0} (\mathbf{M}')^i \mathbf{M}^i \right\} \mathbf{F} \right). \end{aligned}$$

Now, SOA is concerned with the eigenvectors and eigenvalues in the diagonalization of

$$\mathbf{B} \equiv \sum_{i \geq 0} (\mathbf{M}')^i \mathbf{M}^i.$$

The resulting eigenvectors are (spatial) basis functions that correspond to the dominant patterns of *forcing* that maximize the total variation of the system *response*. Compare this to EOFs, which are the basis functions that diagonalize $\Sigma_Y = \sum_{i \geq 0} \mathbf{M}^i \mathbf{Q} (\mathbf{M}')^i$ and correspond to the spatial structures that exhibit the most variation (over time). Obviously, when the noise forcing is “unitary” (i.e., $\mathbf{Q}_\eta = \mathbf{I}$) and the transition matrix is “normal” (i.e., $\mathbf{M}\mathbf{M}' = \mathbf{M}'\mathbf{M}$), then the EOFs and *stochastic optimals* (SOs) are identical. However, these assumptions would rarely hold in practice.

Analogous to the solution for the *observability Gramian* in discrete-time linear systems theory (e.g., Kailath, 1980), $\mathbf{B}$ can be found by solving the Lyapunov equation:

$$\mathbf{M}'\mathbf{B}\mathbf{M} - \mathbf{B} + \mathbf{I} = \mathbf{0}, \quad (7.27)$$

which can be obtained numerically, for example, by the Schur decomposition method of Kitagawa (1977). As an example, say we have

$$\mathbf{M} = \begin{pmatrix} 0.4 & 0.01 \\ 4.0 & 0.1 \end{pmatrix}. \quad (7.28)$$

In this case, the Schur decomposition method gives

$$\mathbf{B} = \begin{pmatrix} 22.5467 & 0.5387 \\ 0.5387 & 1.0135 \end{pmatrix}, \quad (7.29)$$

and the eigenvectors (SOs) and eigenvalues of the decomposition of $\mathbf{B}$ are, respectively,

$$\mathbf{E}_B = \begin{pmatrix} 0.9997 & -0.0250 \\ -0.0250 & 0.9997 \end{pmatrix} \quad \text{and} \quad \lambda_B = \begin{pmatrix} 22.56 \\ 1.00 \end{pmatrix}. \quad (7.30)$$

Thus, the first SO is the eigenvector associated with the largest eigenvalue, 22.56, and is given by $(0.9997, -0.0250)'$. This SO is responsible for *producing* most (its eigenvalue is 95.8% of the sum of eigenvalues) of the variation in the dynamical system. The second SO, which is given by $(-0.0250, 0.9997)'$, is orthogonal to the first and is only responsible for *producing* about 4.2% of the dynamical system's variability. In the context of dimension reduction, it seems reasonable to keep only the first SO and to project the full-dimensional process onto this basis function.

Now, consider the EOFs associated with the dynamical system defined by $\mathbf{M}$ as in (7.28) above, and let $\mathbf{Q}_\eta = \mathbf{I}$. Note that we can obtain the marginal covariance matrix Σ_Y from (6.73), or by recognizing that this can be obtained by solving the discrete Lyapunov equation:

$$\mathbf{M}\Sigma_Y\mathbf{M}' - \Sigma_Y + \mathbf{Q}_\eta = \mathbf{0}. \quad (7.31)$$

In this case,

$$\Sigma_Y = \begin{pmatrix} 1.2135 & 2.1347 \\ 2.1347 & 22.3467 \end{pmatrix}, \quad (7.32)$$

and the associated eigenvectors (EOFs) and eigenvalues are, respectively:

$$\mathbf{E}_Y = \begin{pmatrix} -0.9950 & 0.0995 \\ 0.0995 & 0.9950 \end{pmatrix} \quad \text{and} \quad \lambda_Y = \begin{pmatrix} 1.00 \\ 22.56 \end{pmatrix}. \quad (7.33)$$

Here, the EOF, $(0.0995, 0.9950)'$, corresponding to the largest eigenvalue, 22.56, *accounts for* most (its eigenvalue is 95.8% of the sum of eigenvalues) of the variance in the dynamical system. If we truncated the expansion at this first EOF, we might expect that it would represent the dynamics reasonably well. However, this EOF is quite similar in structure (first element near zero, second element near one) to the second SO in (7.30), and recall that the second SO is responsible for *producing* only a small portion (4.2%) of the variation in the system response. Correspondingly, the first SO is responsible for *producing* most of the variation in the system response, and it is very similar in structure to the second EOF, which only *accounts for* a relatively small amount of the variance of $\mathbf{Y}_t$. This suggests that the system would not be well approximated by projecting on *either* the first EOF *or* the first SO. Note the seemingly subtle distinction between “accounting for” variation and “producing” variation; from a dynamical-systems perspective, they can be quite different. Recall from the discussion just above about how the SO provides structures that, when forced, lead to the system response. Thus, they are much more related to the mechanism of dynamical evolution. On the other hand, EOFs simply correspond to the spatial (not dynamical) structures that exhibit variation in the system.

A natural development is to find basis functions that simultaneously diagonalize Σ_Y and $\mathbf{B}$. In the control literature, this is related to the notion of finding a *balanced truncation* (e.g., Moore, 1981; Farrell and Ioannou, 2001). We note that there are many additional approaches to model reduction, and

this is an active area of research [e.g., see the introduction to Crommelin and Majda (2004); also see Antoulas (2005)]. Most of these approaches have not made their way into the Statistics literature. The most likely impediment to this lack of technology transfer is the “jargon differential” between researchers in Control Theory and Fluid Dynamics and researchers in Statistics.

We note that these dimension-reduction approaches do assume specific dynamical models. This may be unappealing to some practitioners, and it partially explains why (dynamically) nonoptimal basis functions such as EOFs and smoothing kernels (see just below) are so popular. In addition, dynamically based optimal dimension-reduction procedures can be much more difficult to compute.

Smoothing-Kernel Parameterizations

Although more difficult to justify from a dynamical perspective, a useful, statistically motivated choice for Φ is based on smoothing kernels. That is, the process $\{\mathbf{Y}_t\}$ (or often, the observations $\{\mathbf{Z}_t\}$) is approximated by some hidden low-dimensional dynamical process $\{\alpha_t\}$, such that $\mathbf{Y}_t$ is just a smoothed version of the individual elements of α_t . For example, consider the process model,

$$Y_t(s_i) = \int_{D_s} \phi(s_i, \mathbf{x}; \theta) \alpha_t(\mathbf{x}) d\mathbf{x} + \nu_t(s_i), \quad (7.34)$$

where $\phi(s_i, \mathbf{x}; \theta)$ is a smoothing kernel for location s_i , $i = 1, \dots, n$, applied to locations $\mathbf{x}$ in spatial domain D_s and depends on kernel parameters θ (that could also depend on spatial location s_i). Note that we have included an error term $\nu_t(s_i)$ in (7.34) to account for model error associated with the smoothing assumption. This error might exhibit spatial dependence, but typically we assume that it is independent across time.

Practically, it is more common to select some reference locations $\{\mathbf{x}_j: j = 1, \dots, p_\alpha\}$, and thus we discretize (7.34) to obtain

$$Y_t(s_i) = \sum_{j=1}^{p_\alpha} \phi_{ij}(\theta) \alpha_t(\mathbf{x}_j) + \nu_t(s_i), \quad (7.35)$$

for $i = 1, \dots, n$ and $j = 1, \dots, p_\alpha$ (where θ may depend on i or j as well). Note, as suggested in Section 7.2.6, one can also effectively perform a discretization of (7.34) by expanding the kernel and process in terms of some basis functions. In vector notation, we can write (7.35) as

$$\mathbf{Y}_t = \Phi(\theta) \alpha_t + \nu_t, \quad (7.36)$$

where $\Phi(\theta)$ is an $n \times p_\alpha$ matrix depending on kernel parameters θ . Often, θ is assumed fixed and known. However, depending on the assumed structure of the process $\{\alpha_t\}$, these parameters might be estimated (EHM) or given prior

distributions (BHM). As discussed below, in general one has to be careful about identifiability related to inference on θ .

One can imagine complicated modeling situations, where θ is allowed to vary in space (e.g., Higdon, 1998) and/or time. However, we have to be careful in characterizing just where the dynamics are to be modeled in this case. For example, say that the smoothing kernels are spatially varying and represented by relatively high-dimensional vectors $\{\theta_t\}$. Even in the case where the kernels are described by only one (spatially varying) parameter, θ_t is n -dimensional. If we allowed this to vary with time, for example, in a Markovian fashion, then we would have to model the conditional distributions $\{[\theta_t|\theta_{t-1}]: t = 1, 2, \dots\}$, which depend on their own set of parameters. This might be a viable model if $\{\alpha_t\}$ were assumed to be white noise. However, we would effectively be replacing a relatively low-dimensional spatio-temporal process $\{\alpha_t\}$ with a high-dimensional one, $\{\theta_t\}$, of (most likely) equal or greater complexity.

General Parameterization of Dimension Reduction

It is intuitively appealing simply to estimate Φ in (7.24) (or the product $\Phi_H \equiv \mathbf{H}\Phi$) without the basis function or kernel assumptions described in previous sections. However, in the high-dimensional spatio-temporal context, it is difficult to imagine enough data to adequately estimate Φ (or Φ_H). In addition, there are well known identifiability issues with trying to estimate these matrices in the presence of an unknown dynamical model propagator and covariance matrix. These are, in many respects, just the typical identifiability concerns that one experiences with traditional factor analysis (e.g., Johnson and Wichern, 2007). Without strong assumptions on the structure of Φ (or Φ_H) or the parameters associated with the process $\{\alpha_t\}$, it is not possible to identify both the data-model parameters and the process-model parameters.

A property of the Bayesian paradigm is that strong prior information mitigates this lack of identifiability. Thus, one can imagine putting an informative prior on the matrix Φ (or Φ_H), say $[\Phi]$. In fact, this issue has received substantial attention in recent years. In the problem of “blind source separation” found in the engineering literature, one has several different source signals, but the observations of the system are typically unknown mixtures of these sources. In the linear case, understanding these mixtures is just the problem of doing inference on Φ . Specific algorithms have been developed, based on process assumptions [e.g., Independent Component Analysis (Hyvärinen, Karhunen, and Oja, 2001)], and it is known that some of these algorithms can be derived from the Bayesian perspective by the appropriate choice of a matrix prior, $[\Phi]$. For example, Knuth (1999) discusses the prior,

$$[\Phi|\sigma] = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left\{-\frac{\|\mathbf{I} - \Phi'\Phi\|_F^2}{2\sigma^2}\right\}, \quad (7.37)$$

where $\mathbf{I}$ is the identity matrix, $\|\cdot\|_F$ represents the Frobenius norm, and σ accounts for uncertainty regarding the orthogonality of Φ . Alternative prior formulations for the “mixing matrix” Φ are formulated by Knuth (2005).

As an example of prior structure in the DSTM data model, Lopes, Salazar, and Gamerman (2008) put a Gaussian (spatial) random field prior on each column of Φ (or Φ_H). This is motivated by the dynamical factor models for multiple time series discussed by Aguilar and West (2000). In this case, vague priors for these fields require strong priors for the hidden dynamical process $\{\alpha_t\}$, to ensure identifiability [see also Calder (2007) for a similar implementation in the context of multivariate DSTMs; and see Section 7.4.2]. Similarly, Sansó, Schmidt, and Nobre (2008) consider a dynamic specification of the linear model of coregionalization (see Section 4.1.5 for a discussion in the context of spatial processes), which models Φ as the lower-triangular matrix decomposition of a covariance matrix. The parameters of this matrix can then be inferred, with appropriate limitations on the process dynamics.

7.1.4 Nonlinear Mappings

In many problems, it may be more appropriate for observations to be thought of as

$$Z(s_i; t) = h_{i,t}(\mathbf{Y}_t) + \varepsilon(s_i; t), \quad (7.38)$$

where $h_{i,t}(\cdot)$ is a *nonlinear* mapping of the process $\{\mathbf{Y}_t\}$ and $\varepsilon(s_i; t)$ is an additive error process (typically, assumed to be Gaussian), for $i = 1, \dots, m_t$. In most cases, the functional form for $h_{i,t}(\cdot)$ is known, at least up to some parameters. When the parameters are not known, the nonlinearity complicates process prediction in both the EHM (e.g., Kalman filter) and BHM (e.g., particle filter or MCMC) settings (see Chapter 8).

If the functional form of $h_{i,t}(\cdot)$ is known, it is sometimes the case that a Taylor-series-based linearization of the measurement function is considered. In this approach, a first-order expansion about some estimate of the process, $\hat{\mathbf{Y}}_t$, gives

$$h_{i,t}(\mathbf{Y}_t) = h_{i,t}(\hat{\mathbf{Y}}_t) + (\mathbf{h}'_{i,t}) \cdot (\mathbf{Y}_t - \hat{\mathbf{Y}}_t) + \text{higher-order terms},$$

where

$$\mathbf{h}_{i,t} \equiv \left(\frac{\partial h_{i,t}(\mathbf{Y}_t)}{\partial \mathbf{Y}_t} \right) \bigg|_{\mathbf{Y}_t = \hat{\mathbf{Y}}_t}$$

is an n -dimensional vector. Then, assuming the first-order truncation is reasonable, the data model is given by

$$\mathbf{Z}_t = \boldsymbol{\mu}_t(\hat{\mathbf{Y}}_t) + \mathbf{H}_t \mathbf{Y}_t + \boldsymbol{\varepsilon}_t,$$

where

$$\mathbf{H}_t \equiv \begin{pmatrix} \mathbf{h}'_{1,t} \\ \vdots \\ \mathbf{h}'_{m_t,t} \end{pmatrix}$$

is an $m_t \times n$ matrix, and

$$\mu_t(\hat{\mathbf{Y}}_t) \equiv \begin{pmatrix} h_{1,t}(\hat{\mathbf{Y}}_t) \\ \vdots \\ h_{m_t,t}(\hat{\mathbf{Y}}_t) \end{pmatrix} - \mathbf{H}_t \hat{\mathbf{Y}}_t.$$

In this case, the error process, $\{\epsilon_t\}$, is assumed to account for effects due to measurement error, as well as Taylor-series truncation error. Clearly, higher-order approximations could be considered if necessary.

There are numerous choices for $\hat{\mathbf{Y}}_t$ in such a parameterization. For example, these might include forecasts (e.g., the mean of $[\mathbf{Y}_t | \mathbf{Z}_{t-1}, \mathbf{Z}_{t-2}, \dots]$), the prior mean $E(\mathbf{Y}_t)$, estimates from some other model (e.g., a deterministic model of the process), or some average based on an extended dataset (e.g., climatology in the case of a meteorological process).

Perhaps the simplest form of a nonlinear data model extends (7.9) to include a power transformation. For instance, if $\{\mathbf{Y}_t\}$ were positive and transformed in the sense of a Box–Cox transformation (Box and Cox, 1964), the general data model could be written as:

$$Z(\mathbf{s}; t) = a(\mathbf{s}; t) + h(\mathbf{s}; t)Y(\mathbf{s}; t)^{b(\mathbf{s}; t)} + \epsilon(\mathbf{s}; t), \quad (7.39)$$

where $-\infty < b(\mathbf{s}; t) < \infty$, and $\epsilon(\cdot; \cdot)$ is measurement error independent of the process $Y(\cdot; \cdot)$. For example, Sansó and Guenni (1999) used such a data model in the context of modeling a precipitation process. As is the case for a and h , if one allows b to vary with space and/or time, then inference on $Y(\cdot; \cdot)$ might be problematic without strong prior information on b in the parameter-model level of the hierarchical model.

Clearly, the discussion above focuses on the case where at least the functional form of the nonlinear measurement function is known. Alternatively, one might consider semiparametric or nonparametric functions in this context, as is often done in the time-series literature (e.g., Fan and Yao, 2005).

7.1.5 Non-Gaussian Data Models

One of the advantages of the increase in computational power over the last several decades, and the associated development of efficient numerical estimation procedures, is that non-Gaussian data models can now be accounted for readily in most problems encountered in practice. As mentioned previously, the key benefit from the hierarchical formulation of the data model comes if we can justify the *conditional-independence* assumption. Such a formulation is not surprising in data models, particularly in the linear, Gaussian setting where we simply have additive errors. In non-Gaussian settings, this assumption is absolutely critical. It is exceedingly difficult to work with multivariate non-Gaussian distributions, but it is not so difficult to work with a product of a large number of univariate non-Gaussian (conditional) distributions.

In general, we consider the exponential class of data models, conditional on the true process. Specifically, let $\mathbf{Z}_t$ denote the m_t -dimensional vector of data at time t and assume

$$[\mathbf{Z}_t | \boldsymbol{\gamma}_t] \propto \exp\{\boldsymbol{\gamma}_t' \mathbf{Z}_t - b_t(\boldsymbol{\gamma}_t) - c_t(\mathbf{Z}_t)\}, \quad (7.40)$$

where $E(\mathbf{Z}_t | \boldsymbol{\gamma}_t) \equiv \boldsymbol{\mu}_t$ and $\boldsymbol{\gamma}_t$ are m_t -dimensional natural parameters that will depend on the process $\{\mathbf{Y}_t\}$ and the parameters $\boldsymbol{\theta}_D$.

Given the usual regularity conditions for exponential families (e.g., McCulloch and Searle, 2001), the link function $g(\cdot)$ is given in its general form by

$$g(\boldsymbol{\mu}_t) = \mathbf{X}_t \boldsymbol{\beta} + \mathbf{H}_t(\boldsymbol{\theta}_h) \mathbf{Y}_t + \boldsymbol{\nu}_t, \quad (7.41)$$

where $\mathbf{H}_t(\boldsymbol{\theta}_h)$ is the $m_t \times n$ observation matrix, which is often assumed known but may be parameterized in terms of hyperparameters $\boldsymbol{\theta}_h$; $\mathbf{X}_t$ is a matrix of potential covariates (assumed known), with associated (unknown) parameters $\boldsymbol{\beta}$; and $\mathbf{Y}_t$ and $\boldsymbol{\nu}_t$ are independent for all t . As before, there are many options regarding assumptions on the components of variation in (7.41). For example, we may neither need, nor be able to justify, the additive error term $\boldsymbol{\nu}_t$, nor will we always have covariates $\mathbf{X}_t$. Similarly, $\boldsymbol{\beta}$ and/or $\boldsymbol{\theta}_h$ may be time-varying, and we may relate the process $\{\mathbf{Y}_t\}$ to a lower-dimensional dynamical process, as described in (7.24). As an example, a typical model for spatio-temporal counts might assume

$$\mathbf{Z}_t | \boldsymbol{\gamma}_t \sim \text{ind. Poi}(\boldsymbol{\gamma}_t), \quad (7.42)$$

$$\log(\boldsymbol{\gamma}_t) = \mathbf{X}_t \boldsymbol{\beta} + \boldsymbol{\Phi} \boldsymbol{\alpha}_t + \boldsymbol{\nu}_t, \quad (7.43)$$

where $\boldsymbol{\Phi}$ could represent one of the types of dimension-reduction basis functions described above, $\{\boldsymbol{\nu}_t\}$ are independent Gaussian with mean zero, and $\boldsymbol{\nu}_t$ is independent of $\boldsymbol{\alpha}_t$, for all t . In this case, note that $\boldsymbol{\gamma}_t$ has a non-Gaussian distribution. Typically, in this setting the random components would be assigned distributions at the next level of the hierarchical model; for example, $\{\boldsymbol{\alpha}_t\}$ would be a DSTM. The general framework in (7.41) could be referred to as a generalized linear dynamical mixed model (GLDMM). However, we prefer to indicate specifically that the process is *spatio-temporal* and refer to this model as a generalized linear dynamical spatio-temporal model (GLDSTM). In this case, the process $\{\boldsymbol{\nu}_t\}$ is assumed to have no temporal structure, but it could be modeled as having (independent across time) spatial structure.

In problems with missing data, or different supports, (7.42) can be modified to explicitly account for it, as follows:

$$\mathbf{Z}_t | \boldsymbol{\gamma}_t \sim \text{ind. Poi}(\mathbf{H}_t \boldsymbol{\gamma}_t), \quad (7.44)$$

where $\mathbf{H}_t$ is an $m_t \times n$ measurement matrix, and $\boldsymbol{\gamma}_t$ is now n -dimensional. In this case, the matrix $\mathbf{H}_t$ accounts for the mapping of the mean process to observation locations as described in (7.11) for Gaussian models. Here,

a Gaussian assumption for $\{\gamma_t\}$ would be inappropriate, since they are non-negative intensities, although a log Gaussian assumption like (7.43) would be appropriate.

Another common choice within this class of models is the binomial sampling distribution. Now,

$$\mathbf{Z}_t | \gamma_t, \mathbf{p}_t \sim \text{ind. Bin}(\gamma_t, \mathbf{p}_t), \quad (7.45)$$

where the observations (e.g., counts) are distributed independently as binomial distributions conditioned on the true counts, $\{\gamma_t\}$, and probabilities, $\{\mathbf{p}_t\}$. In the case where neither γ_t nor $\mathbf{p}_t$ are known, we cannot identify both components without additional information (e.g., replicate observations or strong priors). This is common in the ecological sciences, where one has observations $\{\mathbf{Z}_t\}$ of some ecological process (e.g., birds) conditional upon the true abundance $\{\gamma_t\}$ and the probability of detection $\{\mathbf{p}_t\}$ at the spatial locations (and times) at which observations were taken (e.g., see Royle and Dorazio, 2008).

7.2 PROCESS MODELS FOR THE DSTM: LINEAR MODELS

As discussed in Section 3.2, a key issue with dynamical processes is specifying how the process transitions from one time to the next. In the spatio-temporal statistical setting, this is relatively complicated due to the curse of dimensionality. As a simple example, assume we have just three spatial locations of interest for the process, $Y(s_1; t)$, $Y(s_2; t)$, and $Y(s_3; t)$. An ideal spatio-temporal model would completely characterize the joint distribution of these process locations for all times, namely $[Y(s_1; t), Y(s_2; t), Y(s_3; t) : t \in D_t]$. That is, we would need to characterize the distributional aspects of all possible interactions of $Y(s_i; t)$ and $Y(s_j; r)$ for any and all times t and r and spatial locations $i = 1, 2, 3$ and $j = 1, 2, 3$.

The joint specification of such spatio-temporal distributions is difficult, as described in Chapter 6, although there has been much recent work done regarding the development of valid joint spatio-temporal covariance functions (Section 6.1). However, these formulations are often unrealistic for real-world processes, and inference is still burdened by the curse of dimensionality. Recall that many processes in the environmental sciences exhibit high dimensionality (either in prediction locations or data locations), nonstationarity (typically in space), nonseparability, and complicated dynamics with features such as advection, diffusion, repulsion, growth, and so on. In addition, the process may evolve nonlinearly. It can be very difficult to incorporate what is known about such processes into the joint-dependence structure.

As has been suggested throughout this book, most models of real-world spatio-temporal processes are dynamical, in that the state of the process at the current time can be characterized, conditional on the process in the recent past. Incorporating this recognition into the modeling framework leads to dramatic simplifications through conditional-probability specifications about the

process. Consider n finite spatial locations $\{s_1, \dots, s_n\}$ and discrete times $t \in \{0, 1, 2, \dots, T\}$. Let $\mathbf{Y}_t \equiv (Y_t(s_1), \dots, Y_t(s_n))'$ be an arbitrary vector representation of the process at the n spatial locations at time t . The joint distribution of the spatio-temporal process can be factored as follows:

$$[\mathbf{Y}_0, \dots, \mathbf{Y}_T] = [\mathbf{Y}_T | \mathbf{Y}_{T-1}, \dots, \mathbf{Y}_0][\mathbf{Y}_{T-1} | \mathbf{Y}_{T-2}, \dots, \mathbf{Y}_0] \\ \dots [\mathbf{Y}_2 | \mathbf{Y}_1, \mathbf{Y}_0][\mathbf{Y}_1 | \mathbf{Y}_0][\mathbf{Y}_0].$$

We can simplify this greatly with Markov-type assumptions. For example, the first-order Markov assumption for $\{\mathbf{Y}_t\}$ is

$$[\mathbf{Y}_t | \mathbf{Y}_{t-1}, \mathbf{Y}_{t-2}, \dots] = [\mathbf{Y}_t | \mathbf{Y}_{t-1}].$$

Thus, under this assumption, the joint distribution can be written as

$$[\mathbf{Y}_0, \dots, \mathbf{Y}_T] = [\mathbf{Y}_0] \prod_{t=1}^T [\mathbf{Y}_t | \mathbf{Y}_{t-1}].$$

Critical to the specification of this distribution is a model for $[\mathbf{Y}_t | \mathbf{Y}_{t-1}]$. That is, we need some stochastic model that describes the evolution of the process from time $t - 1$ to time t . In general, we can write such a model as

$$\mathbf{Y}_t = \mathcal{M}(\mathbf{Y}_{t-1}, \boldsymbol{\eta}_t; \boldsymbol{\theta}_P),$$

where $\mathcal{M}(\cdot, \cdot; \cdot)$ is some function of the process at the previous time, an error term, and process parameters $\boldsymbol{\theta}_P$ that describe the dynamical transition. This function can be linear or nonlinear, and the associated distribution can be Gaussian or non-Gaussian, with additive or multiplicative errors. More generally, higher-order Markov assumptions could be considered as well, but many of the issues regarding parameterization of dynamical processes can be illustrated with the first-order model; hence, that will be our focus here.

Consider the simple example above, where we were interested in the process at just three locations $\{s_1, s_2, s_3\}$, so $n = 3$. In this case, we need to specify the relationship between $Y_t(s_i)$ and $Y_{t-1}(s_1), Y_{t-1}(s_2), Y_{t-1}(s_3)$, for $i = 1, 2, 3$. For now, just consider the linear relationship

$$\begin{bmatrix} Y_t(s_1) \\ Y_t(s_2) \\ Y_t(s_3) \end{bmatrix} = \begin{bmatrix} m_{11}Y_{t-1}(s_1) + m_{12}Y_{t-1}(s_2) + m_{13}Y_{t-1}(s_3) \\ m_{21}Y_{t-1}(s_1) + m_{22}Y_{t-1}(s_2) + m_{23}Y_{t-1}(s_3) \\ m_{31}Y_{t-1}(s_1) + m_{32}Y_{t-1}(s_2) + m_{33}Y_{t-1}(s_3) \end{bmatrix} + \begin{bmatrix} \eta_t(s_1) \\ \eta_t(s_2) \\ \eta_t(s_3) \end{bmatrix},$$

or

$$\begin{bmatrix} Y_t(s_1) \\ Y_t(s_2) \\ Y_t(s_3) \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix} \begin{bmatrix} Y_{t-1}(s_1) \\ Y_{t-1}(s_2) \\ Y_{t-1}(s_3) \end{bmatrix} + \begin{bmatrix} \eta_t(s_1) \\ \eta_t(s_2) \\ \eta_t(s_3) \end{bmatrix},$$

where the *redistribution weights* $\{m_{ij}\}$ describe how the process at location j at the previous time influences the process at location i at the current time (i.e., how the process *propagates* or *transitions* from one time to the next). We have also added a contemporaneous noise process $\{\eta_t(\mathbf{s}_i)\}$ to “force” the system, or to account for model uncertainty. In general, the linear-evolution equation with Gaussian errors can be written in matrix form,

$$\mathbf{Y}_t = \mathbf{M}\mathbf{Y}_{t-1} + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim iid \text{Gau}(\mathbf{0}, \mathbf{Q}_\eta), \quad (7.46)$$

where the propagator matrix (or transition matrix, or redistribution matrix) $\mathbf{M}$ is an $n \times n$ matrix of typically unknown parameters $\{m_{ij}\}$; $\{\boldsymbol{\eta}_t\}$ is independent over time and Gaussian with mean zero and covariance matrix $\mathbf{Q}_\eta$ (although other error processes could be considered); and $\boldsymbol{\eta}_t$ is independent of $\mathbf{Y}_{t-1}$. Note, we have assumed that $\{\mathbf{Y}_t\}$ has mean zero in this example. If this is not the case, we can simply rewrite the evolution equation accounting for the mean vector, $\boldsymbol{\mu}_t$:

$$\mathbf{Y}_t - \boldsymbol{\mu}_t = \mathbf{M}(\mathbf{Y}_{t-1} - \boldsymbol{\mu}_{t-1}) + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim iid \text{Gau}(\mathbf{0}, \mathbf{Q}_\eta). \quad (7.47)$$

Unless specified otherwise, we shall assume $\{\mathbf{Y}_t\}$ has mean zero throughout this discussion.

Consider the i th element of $\mathbf{Y}_t$ and the associated evolution equation implied by the model (7.46),

$$Y_t(\mathbf{s}_i) = \sum_{j=1}^n m_{ij} Y_{t-1}(\mathbf{s}_j) + \eta_t(\mathbf{s}_i),$$

where the redistribution weight m_{ij} corresponds to the i th row and j th column of $\mathbf{M}$. It is seldom the case in real-world problems that one knows the elements of $\mathbf{M}$ with certainty. When n is relatively small, we can estimate $\mathbf{M}$ outright (e.g., see the discussion of DSTM estimation in Section 8.3). But, in realistic spatio-temporal problems with high-dimensional state spaces, this is not possible. Generally speaking, obtaining reliable estimates of the $n \times n$ propagator matrix $\mathbf{M}$ is a challenge, unless $T \gg n$, which is seldom the case. Thus, the key issue becomes how $\mathbf{M}$ can be parameterized to capture adequately the dynamics using relatively few parameters. This is the fundamental problem in spatio-temporal dynamical modeling! Furthermore, as difficult as this is in the linear case, it is much more complicated in the nonlinear case, where one must have a strong sense of the underlying dynamics to make efficient and useful parameterizations of the transition operator (see Section 7.3, below).

In the remainder of this section, we consider various approaches to parameterizing the propagator matrix for first-order, linear DSTMs. Similar approaches will apply to higher-order models (i.e., models with higher lags). Note that higher-order VAR models can always be reparameterized and written as first-order models [e.g., see equations (3.38) and (3.39) in Section 3.1.1; and see

Lütkepohl (2005, p. 15)]. However, this does not reduce dimensionality of the parameter space.

Ultimately, we must parameterize the propagator (or transition) matrix $\mathbf{M}$ in terms of some parameters θ_P that have dimensionality significantly less than the n^2 required to estimate $\mathbf{M}$ directly. There are several strategies that one can use, many of which are motivated by the underlying dynamical process. The following subsections describe various types of model specifications for first-order linear models, starting from the very simple and increasing in complexity.

7.2.1 Spatio-Temporal Random Walk

Perhaps the simplest parameterization of the DSTM process model is the multivariate extension of the random walk (see Section 3.4.2). That is,

$$\mathbf{Y}_t = \mathbf{Y}_{t-1} + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim iid \text{Gau}(\mathbf{0}, \mathbf{Q}_\eta).$$

In this case, the process value at location $\mathbf{s}_i$ and time t is just the value of the process at the same location $\mathbf{s}_i$ at the previous time $(t - 1)$, plus some noise. Clearly, in this case, $\mathbf{M} = \mathbf{I}$. However, there is the possibility that the forcing term $\boldsymbol{\eta}_t$ is spatially dependent. The obvious advantage of this parameterization is that there are very few parameters (there are none in the transition matrix, since it is just the identity matrix). The disadvantage is that it is nonstationary in time (leading to increasing variance with time) and cannot directly accommodate dynamical interactions across space and time. In this sense, the parameterization is not very realistic for most spatio-temporal processes (at least those on relatively large spatial and temporal scales).

An obvious way to introduce stationarity in this model is to allow a single autoregressive parameter in the transition operator:

$$\mathbf{Y}_t = m\mathbf{Y}_{t-1} + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim iid \text{Gau}(\mathbf{0}, \mathbf{Q}_\eta),$$

where $|m| < 1$. That is, for each spatial location, the transition between the past state and the current state is determined by a parameter m , which is homogeneous across space. Of course, this is still a very-low-dimensional parameterization, but it is probably not all that realistic for most spatio-temporal processes. In both of these models, spatial dependence in $\mathbf{Q}_\eta$ does imply quite interesting marginal spatio-temporal covariance structure for the process $\{\mathbf{Y}_t\}$, but it does not allow dynamical interaction, which is needed for, say, forecasting.

7.2.2 Spatial Autoregressive Model

An obvious extension of the simple autoregressive model given above, is to allow the AR parameters to vary heterogeneously across space. For example,

in the three-location example,

$$\begin{bmatrix} Y_t(\mathbf{s}_1) \\ Y_t(\mathbf{s}_2) \\ Y_t(\mathbf{s}_3) \end{bmatrix} = \begin{bmatrix} m_{11}Y_{t-1}(\mathbf{s}_1) \\ m_{22}Y_{t-1}(\mathbf{s}_2) \\ m_{33}Y_{t-1}(\mathbf{s}_3) \end{bmatrix} + \begin{bmatrix} \eta_t(\mathbf{s}_1) \\ \eta_t(\mathbf{s}_2) \\ \eta_t(\mathbf{s}_3) \end{bmatrix},$$

or, for general n ,

$$\mathbf{Y}_t = \text{diag}(\mathbf{m})\mathbf{Y}_{t-1} + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \text{iid } \text{Gau}(\mathbf{0}, \mathbf{Q}_\eta), \quad (7.48)$$

where $\mathbf{m} \equiv (m(\mathbf{s}_1), \dots, m(\mathbf{s}_n))'$. Again, the advantages of this model are that there are relatively few parameters, that it is nonseparable (provided $\mathbf{Q}_\eta$ is not diagonal), and that it is fairly easy to estimate (or model) the parameters. That is, one might have strong prior assumptions for $\mathbf{m}$, and their distribution could be specified at the parameter-model level of the hierarchy (e.g., $\mathbf{m}$ could be represented as a spatial random field). The complicated spatial dependencies in $\mathbf{m}$ will be reflected in the marginal distribution of $\{\mathbf{Y}_t\}$, and hence the spatio-temporal dependencies can be quite rich in this setting. However, this model obviously cannot handle direct interaction between space and time for multiple spatial locations. As such, many processes that are of interest to physical and biological scientists, such as advection and diffusion, cannot be accommodated directly in (7.48).

7.2.3 “Lagged Nearest-Neighbor” Model

Now consider the case where we let $\mathbf{M}$ be parameterized such that only the m_{ij} corresponding to the location $\mathbf{s}_i$ and its nearest neighbors $\{\mathbf{s}_j; j \in \mathcal{N}_i\}$ at the previous time are important (e.g., Wikle, Berliner, and Cressie, 1998). This model assumes that we have a reasonable way to characterize the nearest neighbors. This might be in the context of a process with areal support (say county-level data), or it might be for processes on a regular grid, with point or areal support. This model is written as

$$Y_t(\mathbf{s}_i) = \sum_{j \in \mathcal{N}_i} m_{ij}Y_{t-1}(\mathbf{s}_j) + \eta_t(\mathbf{s}_i), \quad (7.49)$$

where $\mathcal{N}_i$ refers to a prespecified “neighborhood” of $\mathbf{s}_i$, $i = 1, \dots, n$. We typically assume that the errors are Gaussian: $\eta_t \sim \text{iid } \text{Gau}(\mathbf{0}, \mathbf{Q}_\eta)$. These models are relatively sparse, in the sense that many of the elements of $\mathbf{M}$ are zero, yet lagged-neighbor interaction can accommodate a very large class of dynamical processes (as will be shown below). However, even in this case, with relatively large numbers of process locations, the estimation of the nonzero m_{ij} can be difficult, if not impossible, given typical datasets. This suggests that further parameterization is needed. The specification for parameterized neighbor interactions is not necessarily simple, but it can often be facilitated by using scientific theory related to the process of interest. An example of this, based on partial-differential-equation dynamics, is given in the next subsection.

We note that one is tempted to consider this model a special case of the space–time autoregressive (STAR) models outlined in Section 6.4.2. Recall that, as shown in (6.94), $\mathbf{M} \equiv (\mathbf{I} - \mathbf{B}_0)^{-1} \mathbf{B}_1$ for STAR models, where the matrix $\mathbf{B}_0$ has zeros down the main diagonal (representing “instantaneous” spatial dependence). Thus, even if $\mathbf{B}_0$ and $\mathbf{B}_1$ are formulated in terms of the first-order (nearest) neighbors, $\mathbf{M}$ does not have such a diagonal structure due to the typically unbanded nature of $(\mathbf{I} - \mathbf{B}_0)^{-1}$. However, this is an interesting approach for *parameterizing* $\mathbf{M}$ in terms of $\mathbf{B}_0$ and $\mathbf{B}_1$, since each is parameterized in terms of only a few parameters. Obviously, if $\mathbf{B}_0 = \mathbf{0}$, then $\mathbf{M}$ and $\mathbf{B}_1$ are equivalent.

Of course, as in all of the dynamical models presented in this section, higher-order lags in time can be considered if desired.

7.2.4 PDE-Based Parameterizations

For many environmental processes, the underlying spatio-temporal process may follow partial differential equations (PDEs) or ordinary differential equations (ODEs). In the case of linear PDEs, standard finite differencing implies linear Markov spatio-temporal models and suggests parameterizations of the propagator matrix $\mathbf{M}$ as described above (e.g., Section 6.3; Wikle, 2003b; Wikle and Hooten, 2010). For example, consider the general PDE,

$$\frac{\partial Y}{\partial t} = \mathcal{M}(Y, w, \theta),$$

where $\mathcal{M}$ is some function of the variable of interest, Y , other potential variables, w , and process-model parameters, θ . As a process model, it is deterministic.

Simple finite-difference representations (e.g., see Haberman, 1987) suggest an approximate difference-equation model,

$$\mathbf{Y}_t = m(\mathbf{Y}_{t-\Delta t}, \mathbf{w}_t, \theta) + \eta_t,$$

where an independent noise term, η_t , has been added to account for the error of discretization and/or stochastic forcing. Note that it is also reasonable (and probably more appropriate) to consider this error term to include model errors, in the sense that the original PDE should be made stochastic to reflect that it is only an approximation of the real process of interest.

As an example, consider a simple spatio-temporal process $Y(x; t)$ in one-dimensional space, $0 \leq x \leq L$ (i.e., $D_s = [0, L]$), and time t . We assume a *diffusion equation* to be an approximate representation of the true process:

$$\frac{\partial Y}{\partial t} = \frac{\partial}{\partial x} \left(b(x) \frac{\partial Y}{\partial x} \right),$$

where (for now) the diffusion coefficients, $b(x)$, the boundary conditions, $Y(0; t) = Y_0$ and $Y(L; t) = Y_L$, and the initial condition, $\{Y(x; 0) : 0 \leq x \leq L\}$, are known. This PDE is different from the diffusion-injection equation considered in the introduction to Chapter 6.

Now, as in traditional numerical analysis, we approximate the partial derivatives by finite differences. For example, we might use a centered difference,

$$\frac{\partial Y}{\partial x} \simeq \frac{Y(x + \Delta_x; t) - Y(x - \Delta_x; t)}{2\Delta_x},$$

along with a second-order difference given by

$$\frac{\partial^2 Y}{\partial x^2} \simeq \frac{Y(x + \Delta_x; t) - 2Y(x; t) + Y(x - \Delta_x; t)}{\Delta_x^2}.$$

For an approximation of the time derivative, we could use a forward difference:

$$\frac{\partial Y}{\partial t} \simeq \frac{Y(x; t + \Delta_t) - Y(x; t)}{\Delta_t}.$$

Thus, using a forward difference in time and centered differences in space, we obtain the finite-difference approximation of the one-dimensional diffusion equation:

$$\begin{aligned} \frac{Y(x; t + \Delta_t) - Y(x; t)}{\Delta_t} &\simeq b(x) \left(\frac{Y(x + \Delta_x; t) - 2Y(x; t) + Y(x - \Delta_x; t)}{\Delta_x^2} \right) \\ &\quad + \left(\frac{Y(x + \Delta_x; t) - Y(x - \Delta_x; t)}{2\Delta_x} \right) \\ &\quad \left(\frac{b(x + \Delta_x) - b(x - \Delta_x)}{2\Delta_x} \right), \end{aligned}$$

which can be rewritten as

$$\begin{aligned} Y(x; t + \Delta_t) &\simeq \left(1 - \frac{2b(x)\Delta_t}{\Delta_x^2} \right) Y(x; t) + Y(x + \Delta_x; t) \\ &\quad \left(\frac{\Delta_t}{\Delta_x^2} [b(x) + \{b(x + \Delta_x) - b(x - \Delta_x)\}/4] \right) \\ &\quad + Y(x - \Delta_x; t) \left(\frac{\Delta_t}{\Delta_x^2} [b(x) - \{b(x + \Delta_x) - b(x - \Delta_x)\}/4] \right). \end{aligned}$$

Collecting terms in the finite-difference representation, we obtain

$$Y(x; t + \Delta_t) \simeq \theta_1(x)Y(x; t) + \theta_2(x)Y(x + \Delta_x; t) + \theta_3(x)Y(x - \Delta_x; t),$$

where $\{\theta_i(x) : i = 1, 2, 3\}$ are functions of Δ_x , Δ_t , $b(x)$, $b(x - \Delta_x)$, and $b(x + \Delta_x)$. It is critical to note that $\theta_1(\cdot)$, $\theta_2(\cdot)$, and $\theta_3(\cdot)$ play the role of the m_{ij} from before. Thus, this is analogous to the “lagged-nearest-neighbor” structure

mentioned in Section 7.2.3. However, now the diffusion PDE suggests the parameterization. In particular, we see that if we know the spatially varying $b(x)$ and the discretization parameters, then we know the structure of the nearest-neighbor interactions. Then, just as in Section 6.3 for the diffusion-injection equation, we can write this in matrix form.

As a simple example, consider $n = 3$ equally spaced locations $s_1 = x_1$, $s_2 = x_2$, $s_3 = x_3$, and boundary points $s_0 = x_0 = 0$ and $s_4 = x_4 = L$. We can then write

$$\begin{bmatrix} Y(x_1; t + \Delta_t) \\ Y(x_2; t + \Delta_t) \\ Y(x_3; t + \Delta_t) \end{bmatrix} \simeq \begin{bmatrix} \theta_1(x_1)Y(x_1; t) + \theta_2(x_1)Y(x_2; t) + \theta_3(x_1)Y(x_0; t) \\ \theta_1(x_2)Y(x_2; t) + \theta_2(x_2)Y(x_3; t) + \theta_3(x_2)Y(x_1; t) \\ \theta_1(x_3)Y(x_3; t) + \theta_2(x_3)Y(x_4; t) + \theta_3(x_3)Y(x_2; t) \end{bmatrix},$$

which can be written as

$$\begin{bmatrix} Y(x_1; t + \Delta_t) \\ Y(x_2; t + \Delta_t) \\ Y(x_3; t + \Delta_t) \end{bmatrix} \simeq \begin{bmatrix} \theta_1(x_1) & \theta_2(x_2) & 0 \\ \theta_3(x_2) & \theta_1(x_2) & \theta_2(x_2) \\ 0 & \theta_3(x_3) & \theta_1(x_3) \end{bmatrix} \begin{bmatrix} Y(x_1; t) \\ Y(x_2; t) \\ Y(x_3; t) \end{bmatrix} + \begin{bmatrix} \theta_3(x_1) & 0 \\ 0 & 0 \\ 0 & \theta_2(x_3) \end{bmatrix} \begin{bmatrix} Y_0 \\ Y_L \end{bmatrix}.$$

In general matrix form, this is

$$\mathbf{Y}_{t+\Delta_t} \simeq \mathbf{M}(\theta)\mathbf{Y}_t + \mathbf{M}^{(b)}(\theta)\mathbf{Y}_t^{(b)},$$

where $\mathbf{M}^{(b)}(\theta)$ is the boundary propagator matrix, and $\mathbf{Y}_t^{(b)} \equiv (Y_0, Y_L)'$ is the boundary condition. In the HM framework, a prior distribution can be placed on the boundary process, which can simplify computation tremendously (see Wikle, Berliner, and Milliff, 2003).

In the one-dimensional case with general n , $\mathbf{M}(\theta)$ is tridiagonal with elements on the three main diagonals, and zeros elsewhere. This sparseness is of great benefit, both computationally and with regard to parameter estimation, due to the reduction in dimensionality of the parameter space. In higher spatial dimensions, the matrix is still relatively sparse, with just a few nonzero diagonals [e.g., for two-dimensional space, the diffusion equation has five diagonals corresponding to the five nearest neighbors; see Wikle (2003b)].

Given an initial condition, the discretized PDE can then be “integrated” forward by iteration from 0 to Δ_t , from Δ_t to $2\Delta_t$, and so on. The initial condition, $\{Y(x; 0) : x \in D_s\}$, is assumed to be known, but again we could put a prior distribution on it.

The model above is now made stochastic to account for discretization error and uncertainty concerning model misspecification. This is most easily done by adding an error term:

$$\mathbf{Y}_{t+\Delta_t} = \mathbf{M}(\theta)\mathbf{Y}_t + \mathbf{M}^{(b)}(\theta)\mathbf{Y}_t^{(b)} + \boldsymbol{\eta}_{t+\Delta_t},$$

where typically, $\boldsymbol{\eta}_t \sim iid \text{Gau}(\mathbf{0}, \mathbf{Q}_\eta)$.

In most problems, we know neither the diffusion coefficients, $\mathbf{b} \equiv (b(x_1), \dots, b(x_n))'$, nor the covariance parameters, $\mathbf{Q}_\eta$, that control the noise. These must be estimated or modeled. This is not easy in general, but it is simplified greatly by a hierarchical representation where these parameters are given distributions at the next level of the hierarchy. For pure numerical integration, various conditions on the parameters guarantee the stability of the integration; see also Section 6.3. For example, in the simple diffusion equation above with $b(x) \equiv b$, the condition

$$\left| \frac{2\Delta_t b}{\Delta_x^2} \right| < 1$$

must be satisfied or the solution will be unstable [grow without bound; e.g., see Haberman (1987)]. Now, such stability issues are less critical in the presence of random effects and data, as the data act to stabilize the integration for the posterior distribution. However, in that case, the diffusion coefficients adjust to fit the data, and they may lose their interpretability.

Of course, the simple diffusion equation presented above is just one of a very large number of linear PDEs that could be used to motivate a dynamical spatio-temporal process. For example, if we were interested in a process in which the quantity of concern (e.g., a pollutant) is propagating due to the influence of some velocity field (e.g., wind), then a linear advection-diffusion equation would be an appropriate PDE to motivate the process model. For example, in two spatial dimensions (say, $\mathbf{s} = (x, y) \in D_s \subset \mathbb{R}^2$), this would be

$$\frac{\partial Y}{\partial t} = c_1(x, y) \frac{\partial Y}{\partial x} + c_2(x, y) \frac{\partial Y}{\partial y} + \frac{\partial}{\partial x} \left(b(x, y) \frac{\partial Y}{\partial x} \right) + \frac{\partial}{\partial y} \left(b(x, y) \frac{\partial Y}{\partial y} \right), \quad (7.50)$$

where the first two and last two terms on the right-hand side correspond to the advection and diffusion portions of the process, respectively, and it is understood that Y is a function of (x, y) and t . As an illustration, consider the numerical simulations of the advection-diffusion process shown in Figure 7.1. The left panels in this plot show a simulation of the process with advection coefficients that are zero (i.e., $c_i(x, y) \equiv 0$, for $i = 1, 2$). One can see that, from an initial state in the center, the process spreads (“diffuses”) with time, consequently lowering its magnitude. The center panels of this figure show the advection-diffusion simulation with an advection term that is a (nonzero) constant, such that the process is advected to the lower right corner of the image through time. Observe that, in addition to obvious movement with time, there is a spread with time as well. Finally, the right panels in this figure show the simulated advection-diffusion process with advection parameters that are varying with space as shown in Figure 7.2. In this case, as the object propagates, it is affected by the spatially varying advection in a way that makes it propagate at different velocities, thereby changing over time the propagation direction, shape, and orientation of the object.

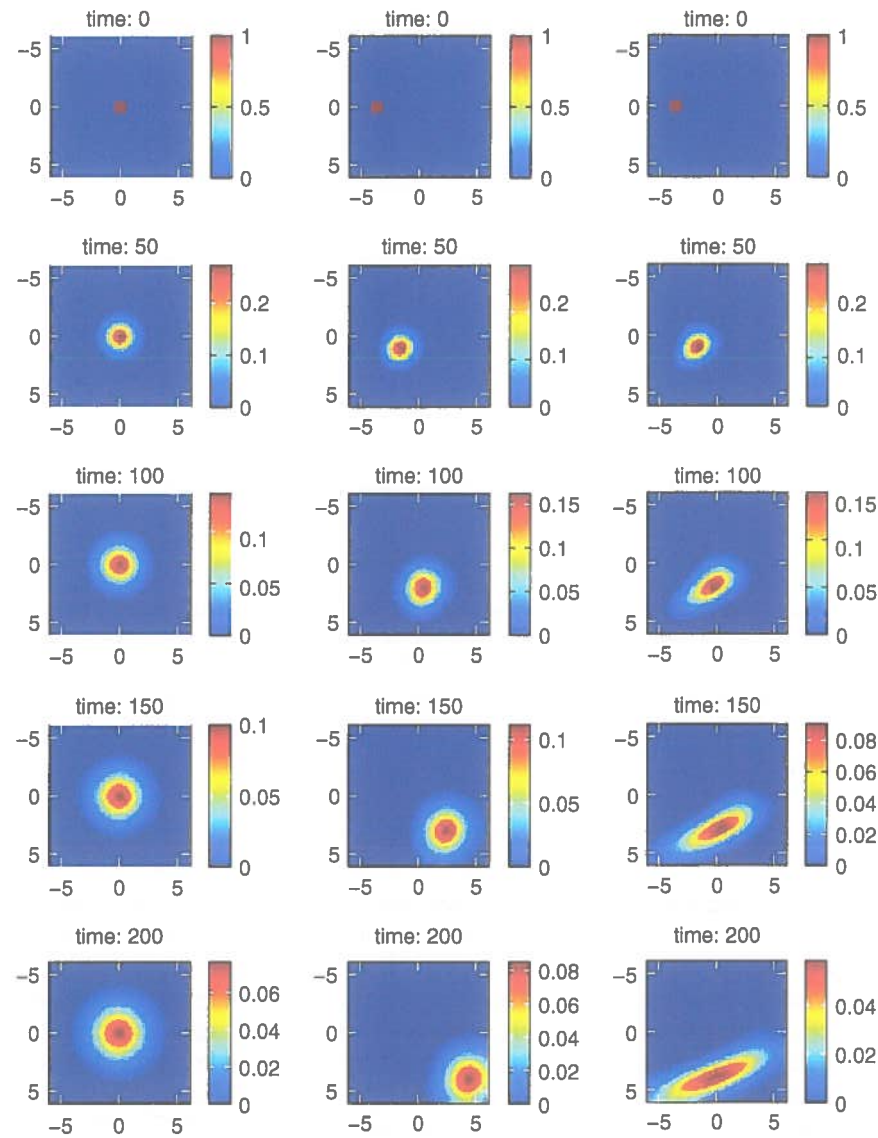


Figure 7.1 Simulations of the evolution of a two-dimensional advection-diffusion process. The left column shows the evolving process with the advection parameter set to zero across the spatial domain (i.e., a pure diffusion process). The center column shows the evolving process with a constant (in space) advection parameter that advects the process toward the lower-right corner. The right column shows an evolving process with spatially varying advection coefficients as given in Figure 7.2.

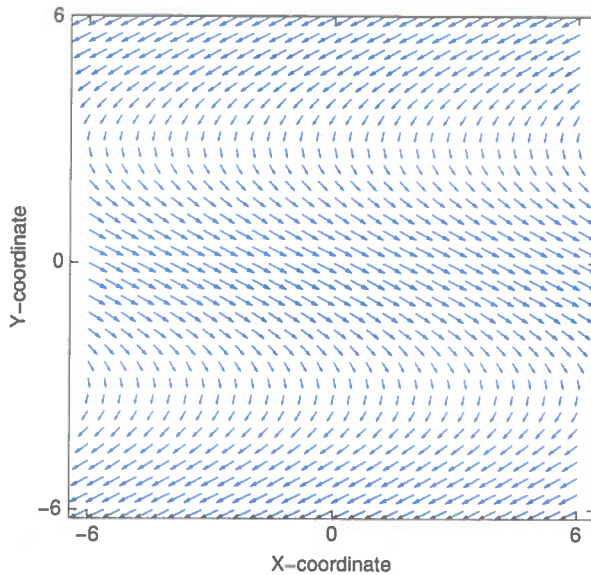


Figure 7.2 Graphical representation of spatially varying advection coefficients associated with the simulation of the two-dimensional advection-diffusion equation shown in the right column of Figure 7.1. The process is advected in the direction shown by the arrows in this plot.

The form of the PDE might also suggest a higher-order Markovian structure in the discretized dynamics. For example, consider the hyperbolic wave equation,

$$\frac{\partial^2 Y}{\partial t^2} = \frac{\partial}{\partial x} \left(d(x) \frac{\partial Y}{\partial x} \right),$$

where it is understood that Y is a function of x and t . Approximating this equation with centered finite differences suggests a second-order Markov (e.g., VAR(2)) statistical model due to the second-order time derivative. Note also that higher-order differencing schemes can imply higher-order discretized model forms as well. Indeed, it is certainly the case that other discretization schemes can be used, including higher-order finite-difference methods, finite-element methods, and Runge–Kutta methods. These might be explicit or implicit formulations (e.g., see Haberman, 1987). The choice is largely dependent on the type of equations considered and the inferential goals at hand. It is important to note that each of these discretization schemes implies a different statistical-model parameterization.

7.2.5 IDE-Based Dynamics

To capture dynamical interactions in space–time that are realistic for many processes (especially those in ecology), the propagator matrix $\mathbf{M}$ must contain nonzero off-diagonal elements beyond just the nearest neighbors. That is, long-range dependence can sometimes be important in the dynamics. This

might suggest using a general form of $\mathbf{M}$, which we have already discounted as impractical when the number of process locations n is large. However, a general non-sparse $\mathbf{M}$ can be parameterized so that the true number of parameters necessary to describe $\mathbf{M}$ is much smaller than n^2 . One way to consider such a parameterization again comes from a theoretical deterministic framework, in particular integro-difference equations (IDEs).

Consider the deterministic IDE model, used in population ecology (e.g., Kot, Lewis, and van den Driessche, 1996) when time is discrete and space is continuous. In the case of linear dynamics, the IDE model is simply

$$Y_t(\mathbf{s}) = \int_{D_s} m(\mathbf{s}, \mathbf{x}; \theta) Y_{t-1}(\mathbf{x}) d\mathbf{x}, \quad \mathbf{s} \in D_s, \quad (7.51)$$

where $m(\mathbf{s}, \mathbf{x}; \theta)$ is the redistribution kernel, which depends on parameters θ . One could think of this process as simply redistributing in space the process from the previous time, where the redistribution is controlled by the kernel m . In particular, $m(\mathbf{s}, \mathbf{x}; \theta)$ gives the redistribution weight for the process at location $\mathbf{x}$ at the previous time, relative to the location $\mathbf{s}$ at the current time. The simple addition of a random noise term, independent of $Y_{t-1}(\cdot)$, then gives a stochastic IDE model:

$$Y_t(\mathbf{s}) = \int_{D_s} m(\mathbf{s}, \mathbf{x}; \theta) Y_{t-1}(\mathbf{x}) d\mathbf{x} + \eta_t(\mathbf{s}), \quad \mathbf{s} \in D_s, \quad (7.52)$$

where the error process $\{\eta_t(\cdot)\}$ is independent in time but may be correlated in space.

It has been shown that this redistribution kernel can model diffusive wave fronts (e.g., Kot, Lewis, and van den Driessche, 1996), where the shape and speed of diffusion depends on kernel width and tail behavior (i.e., kernel dilation). For example, Figure 7.3a and Figure 7.4a show simulations of one-dimensional spatial deterministic IDE processes based on Gaussian kernels $m(\|\mathbf{s} - \mathbf{x}\|; \sigma^2)$ with standard deviations $\sigma = 1$ and $\sigma = 3$, respectively. Note how much more quickly the initial disturbance disperses when the variance (width) of the kernel is larger. Figures 7.3b and 7.4b shows simulations from the respective stochastic IDE processes with an additive Gaussian white-noise error process $\eta_t(\cdot)$, in which $\text{var}(\eta_t(\mathbf{s})) = 1$.

In addition, so-called nondiffusive propagation (e.g., advection) can be modeled via the relative displacement of the kernel (i.e., translation from its initial location $\mathbf{s}$), which is manifested as skewness; see Wikle (2002a). Figure 7.5a and Figure 7.6a show simulated one-dimensional spatial, deterministic IDE processes with a “skewed-Gaussian” kernel with skewness (i.e., displacement) parameter equal to 0.2 and -0.5 , respectively. Note that, in the case of a positively skewed kernel (Figure 7.5), the propagation through time is to the right; when the kernel is negatively skewed (Figure 7.6), the propagation is to the left. Figure 7.5b and Figure 7.6b show the simulations from the respective stochastic IDE with an additive Gaussian white-noise (in space and time) error

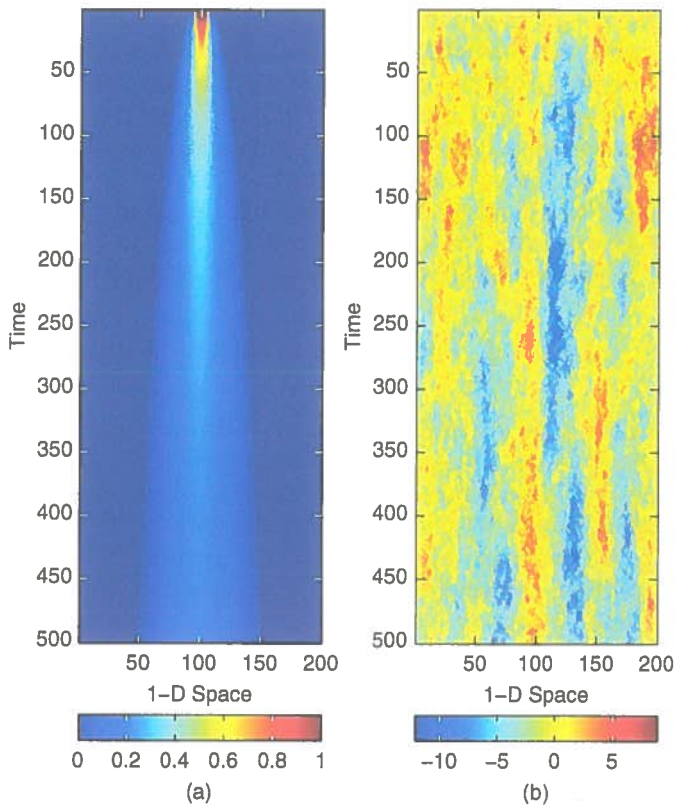


Figure 7.3 Simulation of a one-dimensional IDE process with a symmetric Gaussian transition kernel that has standard deviation, 1. (a) The deterministic IDE given in equation (7.51). (b) The stochastic IDE given in equation (7.52), with white noise (in space and time) variance, 1.

process $\eta_t(\cdot)$, in which $\text{var}(\eta_t(s)) = 1$. This methodology is even more powerful when the kernel translation and dilation parameters are allowed to vary with spatial location (e.g., Wikle, 2002a; Xu, Wikle, and Fox, 2005). Also, nonlinear forms of IDE models are important for spatio-temporal growth processes (e.g., Lewis, 2000; Wikle and Holan, 2011).

The advantage of an IDE dynamical model is that it can accommodate complicated spatio-temporal dynamics with relatively few parameters. For example, discretizing the integral in (7.52) gives

$$Y_t(s_i) = \sum_{j=1}^n m_{ij}(\theta) Y_{t-1}(s_j) + \eta_t(s_i). \quad (7.53)$$

Thus, this discretized version suggests potential parameterizations of $\mathbf{M}$ in terms of relatively few kernel parameters θ . Such parameterizations include nonzero off-diagonal elements, and typically the propagator matrix is not symmetric (i.e., $m_{ij}(\theta) \neq m_{ji}(\theta)$). This is most useful when knowledge of

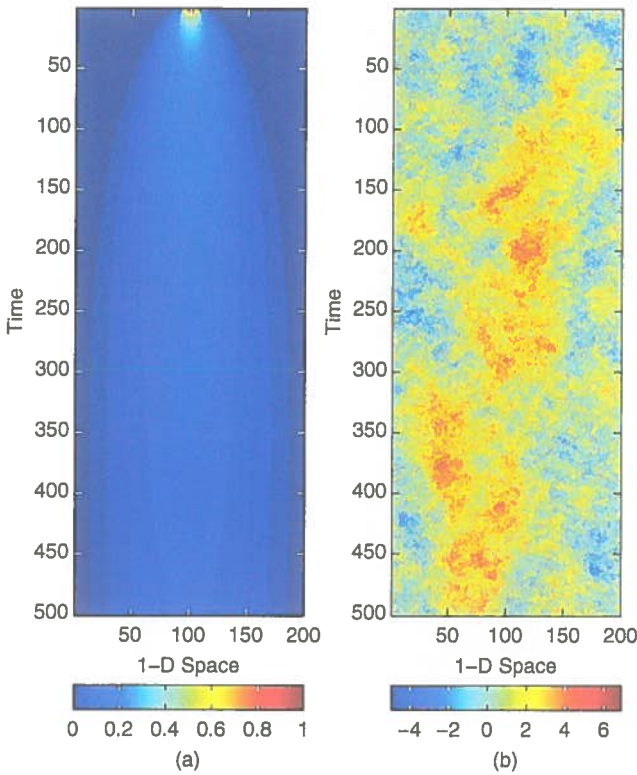


Figure 7.4 Simulation of a one-dimensional IDE process with a symmetric Gaussian transition kernel that has standard deviation, 3. (a) The deterministic IDE given in equation (7.51). (b) The stochastic IDE given in equation (7.52), with white noise (in space and time) variance, 1.

qualitative features in the dynamics (e.g., diffusion, advection) is presumed, rather than explicit forms.

Disadvantages of this approach have to do with implementation. For example, even if θ is known, the dimensionality of $\mathbf{M}$ can still be problematic from a computational perspective, if n is large. This is compounded by the non-sparse structure of $\mathbf{M}$. Furthermore, estimation of θ can be difficult in general, although the hierarchical Bayesian paradigm simplifies this greatly. That is, one can put a prior on θ at the next stage of the hierarchy. In addition, the choice of the appropriate kernel is not always clear. Hopefully, some *a priori* information regarding the qualitative aspects of the dynamics are available to help with this. Alternatively, mixtures of kernels can allow more complicated long-range dependencies. In addition, it is certainly possible to let the kernel parameters vary with space and time, as suggested above. This can accommodate complicated dynamics but, again, care must be taken not to replace one complicated spatio-temporal model by several complicated spatio-temporal models elsewhere in the hierarchy!

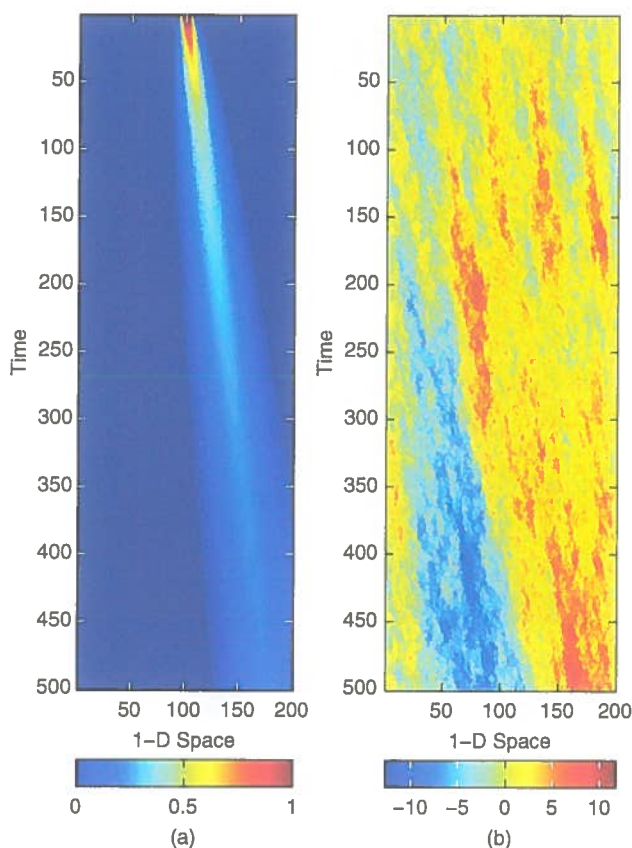


Figure 7.5 Simulation of a one-dimensional IDE process with a skewed-Gaussian transition kernel with standard deviation, 1, and skewness parameter, 0.2, implying propagation to the right. (a) The deterministic IDE given in equation (7.51). (b) The stochastic IDE given in equation (7.52) with white noise variance (in space and time), 1.

We note that an alternative implementation, which can be more efficient, consists of writing the IDE model in its spectral form. This will be illustrated in Section 7.2.6.

7.2.6 Spectral Parameterizations

As mentioned above, it is often convenient to consider spectral representations for dimension reduction and/or to facilitate scientific interpretation. Such transformations can simplify the parameterization of high-dimensional spatio-temporal processes in different ways. Several of these approaches are discussed here.

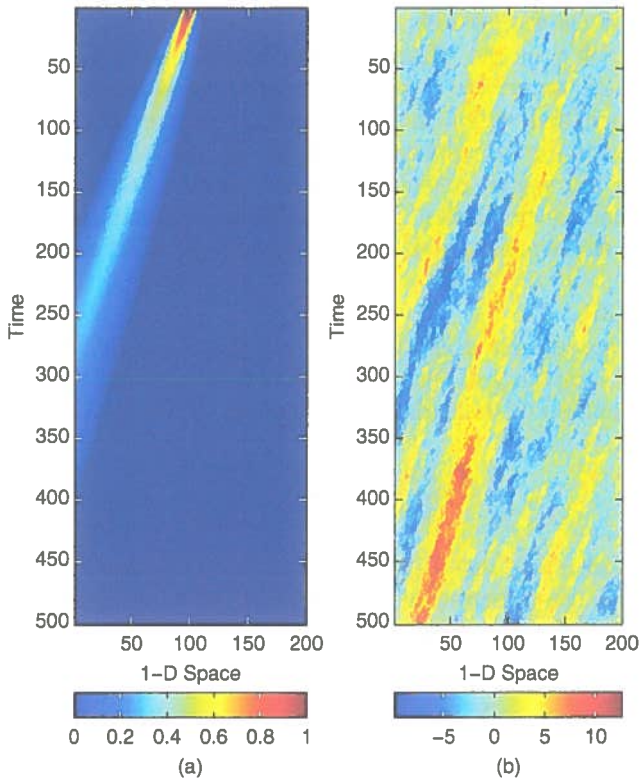


Figure 7.6 Simulation of a one-dimensional IDE process with a skewed-Gaussian transition kernel with standard deviation, 1, and skewness parameter, -0.5 , implying propagation to the left. (a) The deterministic IDE given in equation (7.51). (b) The stochastic IDE given in equation (7.52) with white noise (in space and time) variance, 1.

Consider the general linear DSTM:

$$\mathbf{Y}_t = \mathbf{M}\mathbf{Y}_{t-1} + \boldsymbol{\eta}_t.$$

Now, expanding $\mathbf{Y}_t$ in terms of some complete class of spatial spectral basis vectors, given by the columns of $\boldsymbol{\Phi}$, we obtain

$$\mathbf{Y}_t = \boldsymbol{\Phi}\boldsymbol{\alpha}_t,$$

where $\boldsymbol{\alpha}_t$ are the associated spectral coefficients (Section 7.1.3). Note that at this point we are not assuming any truncation, but rather a simple linear transformation of the spatial process at time t into the spectral domain. One can show (e.g., Wikle and Cressie, 1999) that this suggests the equivalent dynamical relationship:

$$\boldsymbol{\alpha}_t = \mathbf{M}_\alpha \boldsymbol{\alpha}_{t-1} + \boldsymbol{\gamma}_t, \quad (7.54)$$

where $\mathbf{M}_\alpha \equiv (\Phi' \Phi)^{-1} \Phi' \mathbf{M} \Phi$ is the spectral propagator matrix, and $\gamma_t = (\Phi' \Phi)^{-1} \Phi' \eta_t$ is the spectrally transformed noise-forcing process. Note that if the basis vectors are orthogonal, then (7.54) is simplified, since we can choose Φ such that $\Phi' \Phi = \mathbf{I}$. The advantage of the spectral representation is that $\mathbf{M}_\alpha$ and the covariance matrix for γ_t are often simpler than their analogues in the spatial domain, since the spectral transform typically acts as a decorrelator (see Section 3.5). In addition, spectral approaches can also facilitate the implementation of science-based (e.g., PDE and IDE) models.

A Simple PDE

Suppose we are interested in a problem in which a simple advection-diffusion equation (for simplicity of illustration, in one-dimensional space) is a reasonable approximation of the underlying dynamics. This equation is given by

$$\frac{\partial Y}{\partial t} = c \frac{\partial Y}{\partial x} + b \frac{\partial^2 Y}{\partial x^2},$$

where it is understood that Y is a function of $x \in D_s$ and $t \in D_t$. Notice that this is just the one-dimensional version of (7.50). Now, assume solutions are a superposition of wave modes, namely,

$$Y_t(x) = \sum_{j=1}^J [\alpha_{1j}(t) \cos(\omega_j x) + \alpha_{2j}(t) \sin(\omega_j x)],$$

where $\omega_j = 2\pi j / |D_s|$ is the spatial frequency of a sinusoid over spatial domain D_s with spatial wave number $j = 1, \dots, J$ (e.g., see Pedlosky, 1987 for similar derivations). Thus, for all spatial locations of interest, we have $\mathbf{Y}_t = \Phi \alpha_t$, where Φ is made up of the Fourier basis functions, and $\{\alpha_t\}$ is the collection of all spectral coefficients.

The deterministic solution gives formulas for $\alpha_{1j}(t)$ and $\alpha_{2j}(t)$, which are exponentially decaying sinusoids in time. For $j = 1, \dots, J$, we have

$$\alpha_{1j}(t) = \exp(-b\omega_j^2 t) \sin(c\omega_j t),$$

$$\alpha_{2j}(t) = \exp(-b\omega_j^2 t) \cos(c\omega_j t).$$

Deterministic linear wave theory suggests

$$\alpha_j(t + \Delta_t) = \mathbf{G}_j \alpha_j(t), \quad j = 1, \dots, J,$$

where $\alpha_j(t) \equiv (\alpha_{1j}(t) \ \alpha_{2j}(t))'$. For this example, we have

$$\begin{bmatrix} e^{-b\omega_j^2(t+\Delta_t)} \sin\{\omega_j(t+\Delta_t)\} \\ e^{-b\omega_j^2(t+\Delta_t)} \cos\{\omega_j(t+\Delta_t)\} \end{bmatrix} = \mathbf{G}_j \begin{bmatrix} e^{-b\omega_j^2 t} \sin\{\omega_j t\} \\ e^{-b\omega_j^2 t} \cos\{\omega_j t\} \end{bmatrix},$$

where

$$\mathbf{G}_j = \begin{bmatrix} e^{-b\omega_j^2\Delta_t} \cos\{\omega_j\Delta_t\} & e^{-b\omega_j^2\Delta_t} \sin\{\omega_j\Delta_t\} \\ -e^{-b\omega_j^2\Delta_t} \sin\{\omega_j\Delta_t\} & e^{-b\omega_j^2\Delta_t} \cos\{\omega_j\Delta_t\} \end{bmatrix}.$$

Now, we do not expect the true process to behave *exactly* as the linear wave theory suggests. Rather, we let this be the motivation for the DSTM. In particular, we let $\alpha_{1j}(t)$ and $\alpha_{2j}(t)$ be *stochastic* and add the noise term $\gamma_j(t)$ to account for uncertainty. Finally, we let the propagator matrix be $\mathbf{M}_j$ with *prior mean* $\mathbf{G}_j$. That is, conditional on $\mathbf{M}_j$, we obtain

$$\alpha_j(t + \Delta_t) = \mathbf{M}_j \alpha_j(t) + \gamma_j(t + \Delta_t), \quad j = 1, \dots, J.$$

Collecting these equations together, we obtain

$$\alpha_{t+1} = \mathbf{M} \alpha_t + \gamma_{t+1},$$

where $\alpha_t \equiv (\alpha'_1(t) \dots \alpha'_J(t))'$; $\gamma_t \equiv (\gamma'_1(t), \dots, \gamma'_J(t))'$; $\mathbf{M}$ is block diagonal with blocks $\{\mathbf{M}_j : j = 1, \dots, J\}$, and we have assumed that $\Delta_t = 1$. This idea of using spectral basis functions motivated by the deterministic PDE is a powerful modeling framework, and it can be applied to much more complicated PDE dynamical systems (Wikle et al., 2001).

Reduced Dimensionality in the Spectral Domain

The effective dimensionality of α_t is often much less than that of $\mathbf{Y}_t$, since much of the variability of $\mathbf{Y}_t$ can be captured by relatively few spectral modes (see Chapter 3). This is often critical for estimation and implementation. In the simplest cases, the dimensionality of α_t is so reduced ($p_\alpha \ll n$) that it is possible to estimate the propagator (in linear systems) directly if $p_\alpha \ll n < T$. This proceeds as with traditional state-space models (see Section 8.3).

There are also cases where dimension reduction is helpful in the spectral domain, when motivated by science-based deterministic models. In particular this has been applied to the PDE and IDE models, as described in Wikle and Cressie (1999), Wikle (2002a), Wikle et al. (2001), and Xu, Wikle, and Fox (2005). This approach is sketched out below.

Assume the process $\{Y_t(\mathbf{s}) : \mathbf{s} \in D_s, t \in D_t\}$ can be decomposed as

$$Y_t(\mathbf{s}) = Y_t^P(\mathbf{s}) + v_t(\mathbf{s}),$$

where $\{Y_t^P(\mathbf{s})\}$ is assumed to be dynamical, and $v_t(\mathbf{s})$ is a nondynamical process (i.e., it does not exhibit Markovian dependence). Furthermore, assume that $\{Y_t^P(\mathbf{s})\}$ follows a stochastic IDE model, such as (7.52):

$$Y_t^P(\mathbf{s}) = \int_{D_s} m(\mathbf{s}, \mathbf{x}) Y_{t-1}^P(\mathbf{x}) d\mathbf{x} + \xi_t(\mathbf{s}), \quad (7.55)$$

where $m(\mathbf{s}, \mathbf{x})$ is a transition kernel that maps the process $Y_{t-1}^P(\cdot)$ at the previous time, over the domain D_s , to the process at the current time at location $\mathbf{s}$. For notational simplicity, we do not show the parameters used to specify $m(\cdot, \cdot)$.

Now, assume that the dynamical process can be expanded in terms of p_α basis functions $\{\phi_i(\mathbf{s}): i = 1, \dots, p_\alpha\}$:

$$Y_t^P(\mathbf{s}) = \sum_{i=1}^{p_\alpha} \alpha_t(i) \phi_i(\mathbf{s}). \quad (7.56)$$

In addition, expand the transition kernel in terms of the basis set $\{\psi_j(\mathbf{x}): j = 1, \dots, n_J\}$:

$$m(\mathbf{s}, \mathbf{x}) = \sum_{j=1}^{n_J} b_j(\mathbf{s}) \psi_j(\mathbf{x}), \quad (7.57)$$

where the basis functions $\phi_i(\cdot)$ and $\psi_i(\cdot)$ are often the same. Substituting (7.56) and (7.57) into (7.55) and assuming that the basis functions are biorthonormal (for simplicity), namely

$$\int_{D_s} \phi_i(\mathbf{x}) \psi_j(\mathbf{x}) d\mathbf{x} = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} \quad (7.58)$$

we obtain

$$\begin{aligned} Y_t^P(\mathbf{s}) &= \sum_{i=1}^{p_\alpha} b_i(\mathbf{s}) \alpha_t(i) + \xi_t(\mathbf{s}) \\ &= \mathbf{b}(\mathbf{s})' \boldsymbol{\alpha}_t + \xi_t(\mathbf{s}), \end{aligned} \quad (7.59)$$

where $p_\alpha \leq n_J$, $\mathbf{b}(\mathbf{s}) \equiv (b_1(\mathbf{s}), \dots, b_{p_\alpha}(\mathbf{s}))'$ and $\boldsymbol{\alpha}_t \equiv (\alpha_t(1), \dots, \alpha_t(p_\alpha))'$. Note that $\mathbf{Y}_t^P = \boldsymbol{\Phi} \boldsymbol{\alpha}_t$, where the $n \times p_\alpha$ basis function matrix is

$$\boldsymbol{\Phi} \equiv \begin{pmatrix} \phi(\mathbf{s}_1)' \\ \vdots \\ \phi(\mathbf{s}_n)' \end{pmatrix},$$

and $\phi(\mathbf{s}_i) \equiv (\phi_1(\mathbf{s}_i), \dots, \phi_{p_\alpha}(\mathbf{s}_i))'$; $i = 1, \dots, n$. In addition, define the $n \times p_\alpha$ matrix

$$\mathbf{B} \equiv \begin{pmatrix} \mathbf{b}(\mathbf{s}_1)' \\ \vdots \\ \mathbf{b}(\mathbf{s}_n)' \end{pmatrix}$$

and the n -dimensional vector, $\boldsymbol{\xi}_t \equiv (\xi_t(\mathbf{s}_1), \dots, \xi_t(\mathbf{s}_n))'$. Then, after combining all n process locations, we can write

$$\boldsymbol{\alpha}_t = (\boldsymbol{\Phi}' \boldsymbol{\Phi})^{-1} \boldsymbol{\Phi}' \mathbf{B} \boldsymbol{\alpha}_{t-1} + (\boldsymbol{\Phi}' \boldsymbol{\Phi})^{-1} \boldsymbol{\Phi}' \boldsymbol{\xi}_t, \quad (7.60)$$

$$\equiv \mathbf{M}_\alpha \boldsymbol{\alpha}_{t-1} + \boldsymbol{\eta}_t, \quad (7.61)$$

where the p_α -dimensional error vector is given by $\eta_t \equiv (\Phi' \Phi)^{-1} \Phi' \xi_t$, and the $p_\alpha \times p_\alpha$ propagator matrix is given by $\mathbf{M}_\alpha \equiv (\Phi' \Phi)^{-1} \Phi' \mathbf{B}$. Finally, we note the obvious simplification in $\mathbf{M}_\alpha$ and η_t if the basis functions Φ are orthogonal (i.e., $\Phi' \Phi = \mathbf{I}$), in which case $\mathbf{M}_\alpha = \Phi' \mathbf{B}$ and $\eta_t = \Phi' \xi_t$.

Clearly, the spectral truncation (7.56) leads to a reduced-dimension dynamical process (choose $p_\alpha \ll n$). One would still have to estimate the $n \times p_\alpha$ kernel-transition matrix $\mathbf{B}$ and the $p_\alpha \times p_\alpha$ covariance matrix associated with η_t , in order to proceed. One distinct advantage of the IDE formulation is that the kernel, $m(\mathbf{s}, \mathbf{x})$, can be parameterized explicitly in terms of just a few critical parameters, namely, $m(\mathbf{s}, \mathbf{x}; \theta)$. For example, as discussed in Section 7.2.5, θ might control the width (diffusion) and skewness (advection) of the kernel. In this case, one might even be able to select appropriate basis functions $\{\psi_j(\mathbf{s}) : j = 1, \dots, J\}$, so that the expansion coefficients for the kernel are also given (analytically) in terms of the parameters θ . An example is the Fourier basis set, where the Fourier transform of the kernel gives the characteristic function (i.e., Wikle, 2002a; Xu, Wikle, and Fox, 2005). The advantage of this approach is seen when the parameters are allowed to vary with spatial location (i.e., $\theta(\mathbf{s})$ replaces θ), and these spatially varying parameters are then modeled as random fields at the next level of the hierarchy. In this case, the matrix $\mathbf{B}(\theta_s)$ is given in terms of the parameter vector $\theta(\mathbf{s})$, and thus $\mathbf{M}_\alpha$ is completely determined by $\theta(\mathbf{s})$ (since the basis functions Φ are assumed to be known).

In the PDE and IDE examples presented above, the obvious choices for basis functions were Fourier modes. This is not always the case, as discussed above in Section 7.1. Thus, one has to put considerable care into the choice of basis functions, whether they be Fourier, wavelet, EOF, bisquare, spline, smoothing kernel, and so on. Furthermore, one must decide if the expansion should be truncated (implying dimension reduction of the state process), or whether the parameter space is sufficiently reduced by a complete expansion. One must also be concerned with boundary (edge) effects, as well as implementation on nonregular grids.

At this point, it is instructive to discuss the relationship between the IDE framework and the smoothing-kernel framework presented in Section 7.1.3. The smoothing-kernel formulation given in (7.34) is similar to the IDE formulation given in (7.52), but clearly the main difference is that in (7.34) interest is in smoothing a latent process (i.e., $\{\alpha_t(\cdot)\}$) that is distinct from the process of interest (i.e., $\{Y_t(\cdot)\}$). In contrast, the IDE kernel operates on past values of the Y -process. This distinction is also evident in the discretized version of both formulations given in (7.35) and (7.53). However, the differences blur when we consider the spectral formulation of the IDE model. In this case, the IDE-based HM includes a process model such as given by (7.24), which is similar to the vector form of the kernel smoothing data model given in (7.36). The distinction is then that this spectral matrix Φ is parameterized in terms of the kernel parameters in (7.36), but in the IDE context this matrix is assumed *known* (e.g., Fourier basis vectors). More critically, the IDE model then allows

the kernels to play a role in the dynamical propagator on the process model for $\{\alpha_t\}$, whereas the kernel-smoothing approach typically assumes a simple structure on the dynamics of $\{\alpha_t\}$ (to satisfy identifiability concerns). Thus, the IDE framework is designed to give the modeler much more flexibility in modeling the *dynamics* of the process, while the kernel-smoothing framework places the modeling emphasis on representing the spatial structure of the process at various times, with much less regard for the underlying dynamics. As we have emphasized repeatedly in this book, we believe that, when possible, the modeling emphasis should be placed on the dynamics, since it is closer to the etiology of the real-world spatio-temporal phenomenon being modeled. That being said, there are situations where one simply does not know much about the underlying process dynamics, and it can be useful to take a kernel-smoothing approach.

7.2.7 Multiresolution DSTMs

In many respects, the spectral-basis-function expansion suggested in Section 7.2.6 does, in fact, lead to a multiresolution DSTM. This is because many basis-function expansions (e.g., EOF, Fourier, wavelet, bisquare) are inherently “multiresolutional” in the sense that the basis functions vary by spatial scale. Alternatively, one can consider a tree-structured approach to capture the multiresolution behavior explicitly.

One approach to modeling tree-structured multiresolution DSTMs was given by Johannesson, Cressie, and Huang (2007), in which they extend a multiresolution spatial model (e.g., Chou, Willsky, and Nikoukhah, 1994; Huang and Cressie, 2001; Huang, Cressie, and Gabrosek, 2002) to the dynamic spatio-temporal setting by allowing temporal dynamics on the coarsest scale. Since all scales are linked, the marginal-dependence structure of the process includes dynamics. To illustrate this in more detail, recall from Section 4.1.6 that levels of resolution are defined by partitioning the spatial domain, say D_s , into subdomains at the r th resolution:

$$\{D(i, r) : i = 1, \dots, n_r\},$$

where $r = 0, \dots, R$ are the spatial resolutions of D_s (with $r = 0$ the coarsest resolution), and n_r corresponds to the number of cells at the r th level of resolution. We require $\{D(i, r)\}$ to be disjoint, $D_s = \bigcup_i D(i, r)$, and $D(i, r) = \bigcup_{j=1}^{m_r} D(ch_j(i, r))$, where $ch_j(i, r)$ is the index for the j th child node (cell) of the cell at (i, r) ; $j = 1, \dots, m_r$.

Now, consider the spatio-temporal process,

$$\{Y_t(\mathbf{s}) : \mathbf{s} \in D_s, t = 1, 2, \dots\},$$

and define the *aggregated* spatio-temporal process as

$$Y_t(i, r) \equiv \frac{1}{|D(i, r)|} \int_{D(i, r)} Y_t(\mathbf{s}) d\mathbf{s}.$$

The process model is then defined as follows:

$$\mathbf{Y}_t(ch(i, r)) = \mathbf{1} Y_t(i, r) + \omega_t(ch(i, r)),$$

where

$$\begin{aligned}\mathbf{Y}_t(ch(i, r)) &\equiv (Y_t(ch_1(i, r)), \dots, Y_t(ch_{m_r}(i, r)))', \\ \omega_t(ch(i, r)) &\equiv (\omega_t(ch_1(i, r)), \dots, \omega_t(ch_{m_r}(i, r)))',\end{aligned}$$

and $\omega_t(ch(i, r)) \sim ind. Gau(\mathbf{0}, \sigma^2 \mathbf{W}(ch(i, r)))$. Critically, the temporal dynamics are induced through a dynamical linear model on the coarsest scale:

$$\mathbf{Y}_t(0) = \mathbf{M}_t \mathbf{Y}_{t-1}(0) + \eta_t,$$

where $\mathbf{Y}_t(0) \equiv (Y_t(1, 0), \dots, Y_t(n_0, 0))'$, $\{\mathbf{M}_t\}$ are the propagator matrices, and $\eta_t \sim ind. Gau(\mathbf{0}, \sigma^2 \mathbf{Q}_t)$.

The implementation of this model (given a suitable data model) makes use of the *change-of-resolution Kalman filter* (see Chou, Willsky, and Nikoukhah, 1994; Fieguth et al., 1995; Huang and Cressie, 2001). An advantage of this approach, beyond computational efficiency, is that one can capture spatial nonstationarity through the covariance matrices, $\{\mathbf{W}(ch(i, r))\}$, although these must be specified, estimated, or given a prior distribution, depending on the implementation at hand. In addition, the propagator and covariance matrices associated with the coarse-resolution dynamics must be specified or estimated. In principle, any of the parameterization approaches for the dynamics described above could be considered here as well. It should be noted that, typically, one wants to constrain the propagator matrices in multiresolution models to satisfy conditions of *nonnegativity*, *mass balance*, and *constant mean* (e.g., Gelpke and Künsch, 2001). Finally, extending the dynamical model to finer scales and interactions between them, results in a breakdown of computational efficiency, although the model may be more appropriate for the phenomenon under study.

7.3 PROCESS MODELS FOR THE DSTM: NONLINEAR MODELS

Linear dynamics, as discussed in Section 7.2, can be quite useful in applications. However, many processes in nature are governed by nonlinear spatio-temporal interactions. For example, in ecological population dynamics, processes are encountered that show state-dependent (i.e., “density-dependent”) growth as given, for example, by

$$\frac{\partial Y}{\partial t} = Yg(Y; \theta),$$

for some nonlinear growth function g and where it is understood that Y is a function of $s \in D_s$ and $t \in D_t$. Similarly, in geophysics one encounters

processes that exhibit what is often referred to as *nonlinear advection*. For example, in the one-dimensional spatial setting,

$$\frac{\partial Y}{\partial t} = Y \frac{\partial Y}{\partial x},$$

where again it is understood that Y is a function of $x \in D_s$ and $t \in D_t$. Nonlinear advection comes from fluid dynamics, where Y might be one of the velocity components that make up the flow. Thus, there is a nonlinear interaction between the velocity component and how that component of the flow is advected downstream. In the case of spatio-temporal hierarchical statistical modeling, we must have process models that can accommodate these types of processes, as well as many others. Just like with linear dynamical models, the key is that the nonlinear behavior in spatio-temporal processes generally comes from *interactions* across a variety of spatio-temporal scales of variability and even across multiple processes. Hence, nonlinear DSTMs should reflect this.

As we have noted, the issues of dimensionality and efficient parameterization present the most significant challenges for statistical modeling of linear DSTMs. Perhaps not surprisingly, these issues are even more critical for nonlinearly evolving spatio-temporal dynamical processes. Indeed, we live in a nonlinear world, and sooner or later the nonlinearity has to be modeled in order to capture the essence of the true dynamics. In addition, for forecasting, we know from chaos theory that seemingly insignificant events can make a huge difference in the time evolution of a process (see Section 3.2.4). That is, uncertainties in initial and boundary conditions can affect dramatically our ability to predict such processes. For these reasons, from a parametric perspective, the modeler must have an even stronger sense of the basic nonlinear dynamical form in order to perform scientific inference and/or prediction.

Currently, the Statistics literature that considers nonlinear spatio-temporal processes is somewhat sparse. However, by borrowing information from nonlinear statistical time series analysis and other disciplines (e.g., atmospheric science, ecology, etc.) we can begin to formulate classes of nonlinear DSTMs. We note the distinction between nonlinearity in the state variables and in the parameters. In hierarchical nonlinear DSTMs, both parameter and process nonlinearities are possible. It is often the case in spatio-temporal dynamics that one is interested in predicting (or forecasting) at a finite set of points in space. Thus, we focus our discussion on the vector-valued process, $\{\mathbf{Y}_t : t = 1, 2, \dots\}$, as defined previously. This discussion follows closely that in Wikle and Hooten (2010).

A general class of nonlinear dynamical models is the *nonlinear autoregressive model*:

$$\mathbf{Y}_t = \mathcal{M}(\mathbf{Y}_{t-1}, \boldsymbol{\eta}_t; \boldsymbol{\theta}_t); \quad t = 1, 2, \dots, \quad (7.62)$$

where $\mathcal{M}$ is a nonlinear function operating on elements of the processes $\{\mathbf{Y}_t\}$ and $\{\boldsymbol{\eta}_t\}$, and $\{\boldsymbol{\theta}_t\}$ are parameters that control the process' evolution. Although

this model is conceptually quite simple, without a specified form for the operator $\mathcal{M}$, (7.62) is not particularly useful for statistical modeling, other than to indicate that there is first-order Markov dependence. We should mention that higher-order temporal lag terms in the process could be considered as well, but for simplicity we focus here on first-order lag dependence. (The curse of dimensionality can make the higher-order case much more difficult to implement in practice.) In the remainder of this section, we consider several specific formulations of (7.62); however, these are not an exhaustive set of possible models that could be considered.

7.3.1 Local Linear Approximations

A long-standing approach for modeling nonlinear systems is simply to linearize the process. That is, the nonlinear evolution operator is approximated by a locally linear function. Specifically, we can expand the process in a Taylor series about some value of the state process. This value is typically an estimate of the current state or its mean. For example, consider the nonlinear first-order AR model:

$$\mathbf{Y}_t = \mathcal{M}(\mathbf{Y}_{t-1}) + \boldsymbol{\eta}_t, \quad (7.63)$$

where $\mathcal{M}(\cdot)$ is some known nonlinear function, $\mathcal{M}(\cdot) \equiv (m_1(\cdot), \dots, m_n(\cdot))'$, with $m_i(\cdot)$ a scalar-valued nonlinear function. Now, let $\hat{\mathbf{Y}}_{t-1}$ be an estimate of the mean of $\mathbf{Y}_{t-1}$, and consider the Taylor series expansion:

$$\mathcal{M}(\mathbf{Y}_t) = \mathcal{M}(\hat{\mathbf{Y}}_{t-1}) + \mathbf{M}_t \cdot (\mathbf{Y}_{t-1} - \hat{\mathbf{Y}}_{t-1}) + \dots, \quad (7.64)$$

where we use the $\cdot$ notation in this section to emphasize the product operator and avoid ambiguity with the functional argument. The ellipses in (7.64) refer to higher-order terms in $(\mathbf{Y}_{t-1} - \hat{\mathbf{Y}}_{t-1})$, and $\mathbf{M}_t$ is an $n \times n$ matrix of first partial derivatives of $\mathcal{M}(\cdot)$, evaluated at $\mathbf{Y}_{t-1} = \hat{\mathbf{Y}}_{t-1}$. That is, the (i, j) th element of $\mathbf{M}_t$ is

$$(\mathbf{M}_t)_{i,j} = \left. \frac{\partial m_i(\mathbf{Y}_t)}{\partial Y_t(s_j)} \right|_{\mathbf{Y}_t = \hat{\mathbf{Y}}_{t-1}}. \quad (7.65)$$

Thus, for $\mathbf{M}_t$ to be defined, we require that $m_i(\cdot)$ be at least once-differentiable with respect to its vector argument, $\mathbf{Y}_{t-1}$, for each $i = 1, \dots, n$. Typically, we assume that the higher-order terms are negligible or can be absorbed into $\boldsymbol{\eta}_t$ (however, see Section 7.3.3 for the quadratic case), and thus

$$\mathbf{Y}_t = \mathbf{c}_t + \mathbf{M}_t \mathbf{Y}_{t-1} + \boldsymbol{\eta}_t, \quad (7.66)$$

where

$$\mathbf{c}_t \equiv \mathcal{M}(\hat{\mathbf{Y}}_{t-1}) - \mathbf{M}_t \hat{\mathbf{Y}}_{t-1}. \quad (7.67)$$

This setting is analogous to the linear DSTM and, thus one can use similar estimation and inference approaches for the process $\{\mathbf{Y}_t\}$. Note also that we might wish to expand the Taylor series around some quantity other than $\hat{\mathbf{Y}}_{t-1}$. For example, we might use a forecasted value (from this model, or from a deterministic model), historical values (e.g., climatology), $E(\mathbf{Y}_{t-1})$, or some other state vector. Such methods have a long history in the Kalman-filtering literature and are known as *extended Kalman filters* (e.g., see Grewal and Andrews, 1993). Note that we have assumed that we know the functional form, $\mathcal{M}(\cdot)$. However, in spatio-temporal statistical problems it is often the case that we do not know the form of this operator, or it would be too complicated to work with directly (e.g., numerical weather-forecasting models). In this case, $\mathbf{c}_t$ and $\mathbf{M}_t$ would have to be estimated, and the same difficulties described for linear DSTMs would apply, with the additional complication of the time-varying, state-dependent nature of these quantities.

7.3.2 State-Dependent Processes

One way to simplify the general nonlinear first-order AR process model (7.62) is by considering the *nonlinear state-dependent model*:

$$\mathbf{Y}_t = \mathbf{M}(\mathbf{Y}_{t-1}; \theta_t^M) \cdot \mathbf{Y}_{t-1} + \eta_t, \quad (7.68)$$

where the propagator matrix $\mathbf{M}(\cdot; \cdot)$ depends on the state process at the previous time, in addition to (possibly) time-varying parameters $\{\theta_t^M\}$. This model is certainly not new in the time-series literature [e.g., see Section 3.4.8 and overviews in Priestley (1988), Tong (1990), and Fan and Yao (2005)], but it is somewhat less common in the spatio-temporal statistics literature.

Such state-dependent models are powerful and include many nonlinear models as special cases. For example, one might consider exponential autoregressive models, bilinear models, and threshold autoregressive models from this perspective (e.g., Tong, 1990). As an example, consider the threshold AR model:

$$\mathbf{Y}_t = \mathbf{M}_t(\mathbf{Y}_{t-1}) \cdot \mathbf{Y}_{t-1} + \eta_t \equiv \begin{cases} \mathbf{M}_1 \mathbf{Y}_{t-1} + \eta_{1,t}, & \text{if } f_1(\gamma_t) \in c_1 \\ \vdots & \vdots \\ \mathbf{M}_K \mathbf{Y}_{t-1} + \eta_{K,t}, & \text{if } f_K(\gamma_t) \in c_K, \end{cases} \quad (7.69)$$

where $f_k(\gamma_t)$ is a function of a time-varying parameter γ_t , and $f_k(\gamma_t) \in c_k$; $k = 1, \dots, K$, is the condition under which the k th equation is to be followed. Although written as a scalar here, this process $\{\gamma_t\}$ could be multivariate, and it might depend on the state process $\mathbf{Y}_{t-1}$ or other processes and parameters. In summary, through dependence on the value of the (typically hidden) process γ_t , the model uses a different linear evolution operator and, potentially, a different error process. This is related to the state-dependent model (7.68) in the situation

where γ_t is a function of $\mathbf{Y}_{t-1}$. An example of such a model is given by Berliner, Wikle, and Cressie (2000), with regard to long-lead forecasting of tropical Pacific sea surface temperature; see Chapter 5 and Section 8.1.

As is often the case in spatio-temporal problems, the state dimension n can be quite large, leading to difficulties in parsimonious parameter specifications [e.g., the matrices $\{\mathbf{M}_k\}$ in (7.69)]. Furthermore, there are often specific types of process behavior, known *a priori*, that one wishes to include in the process model (e.g., advection, diffusion, repulsion, density-dependent growth, etc.). In this case, parameterizing the state-dependent transition matrices, so that they are consistent with these processes (as with the PDE and IDE examples shown previously for the linear case), leads to more efficient inferences. As is always the case in the hierarchical context, these parameterizations can often be simplified by modeling complicated dependence at lower levels of the HM hierarchy.

We thus specify the following, rather general, nonlinear DSTM:

$$\mathbf{Y}_t = \mathbf{M}(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_t^M) \cdot \mathbf{Y}_{t-1} + \mathbf{F}(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_t^F) \cdot \boldsymbol{\eta}_t, \quad (7.70)$$

where $\mathbf{M}(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_t^M)$ is an evolution matrix depending on the state $\mathbf{Y}_{t-1}$ and some (possibly time-varying) parameters $\boldsymbol{\theta}_t^M$, and the “noise forcing matrix” $\mathbf{F}(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_t^F)$ likewise may depend on the state process and some forcing parameters $\boldsymbol{\theta}_t^F$. Here, $\{\boldsymbol{\theta}_t^M\}$ and $\{\boldsymbol{\theta}_t^F\}$ are components of the process-model parameters, $\boldsymbol{\theta}_P$. Our focus here is on parameterization of $\mathbf{M}(\cdot; \cdot)$. We write

$$\mathbf{M}(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_t^M) = \mathbf{A}(\boldsymbol{\theta}_t^A) + \mathbf{B}(\boldsymbol{\theta}_t^B) \cdot \mathbf{G}(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_t^G) \cdot \mathbf{C}(\boldsymbol{\theta}_t^C), \quad (7.71)$$

where $\boldsymbol{\theta}_t^M \equiv \{\boldsymbol{\theta}_t^A, \boldsymbol{\theta}_t^B, \boldsymbol{\theta}_t^C, \boldsymbol{\theta}_t^G\}$. The form of this matrix is quite general and can model realistically the behavior of many nonlinear dynamical processes (Wikle and Hooten, 2010). Examples of special cases include:

- $\mathbf{M}(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_t^M) = \mathbf{A}(\boldsymbol{\theta}_t^A)$: In this case, the model is simply the basic first-order linear DSTM. An example is a system of linear difference equations, or a linear advection-diffusion PDE discretized by finite differences. In the latter case, $\mathbf{A}$ is quite sparse as shown in Section 7.2.4 (with a diagonal band corresponding to the discretization “stencil”; e.g., there are five nonzero diagonals for the typical two-dimensional finite-difference case).
- $\mathbf{M}(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_t^M) = \mathbf{B}(\boldsymbol{\theta}_t^B) \cdot \mathbf{G}(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_t^G)$: In this case, the model reduces to the spatio-temporal “matrix model” (Caswell, 2001; Hooten et al., 2007; and Section 9.3) or the discretized nonlinear IDE model. Here, $\mathbf{G}$ accounts for growth, and $\mathbf{B}$ accounts for diffusion or spread. Typically, $\mathbf{B}$ is parameterized in terms of a kernel function (e.g., Hooten et al., 2007), and the functional form of the the growth process that defines $\mathbf{G}$ is known and typically nonlinear.
- $\mathbf{M}(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_t^M) = \mathbf{A}(\boldsymbol{\theta}_t^A) + \mathbf{G}(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_t^G)$: In this case, the model accommodates discretized linear PDE dynamics with a nonlinear growth term; typically, $\mathbf{G}(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_t^G)$ is diagonal.

- $\mathbf{M}(\mathbf{Y}_{t-1}; \theta_t^M) = \mathbf{A}(\theta^A) + \mathbf{G}(\mathbf{Y}_{t-1}; \theta^G) \cdot \mathbf{C}(\theta^C)$: In this case, the model accomodates a discretized PDE with nonlinear advection.

7.3.3 General Quadratic Nonlinearity (GQN)

We focus on a special case of (7.70) and (7.71), characterized by what Wikle and Hooten (2010) refer to as *general quadratic nonlinearity* (GQN):

$$Y_t(\mathbf{s}_i) = \sum_{j=1}^n a_{ij} Y_{t-1}(\mathbf{s}_j) + \sum_{k=1}^n \sum_{l=1}^n b_{i,kl} Y_{t-1}(\mathbf{s}_k) g(Y_{t-1}(\mathbf{s}_l); \theta^G) + \eta_t(\mathbf{s}_i), \quad (7.72)$$

for $i = 1, \dots, n$, where there are linear terms and quadratic interactions. We note that, although (7.72) is not written that way, the coefficients could also depend on t . The first term on the right-hand side of (7.72) contains a linear combination of process values at the previous time, and the second term on the right-hand side includes interactions between different locations of the process at the previous time. The term “general” is due to the function $g(Y_{t-1}(\mathbf{s}_l); \theta^G)$, which is included in the quadratic interaction. There are $O(n^3)$ parameters in this model, which could be quite large for the typical number of process locations n that one might see in applications.

There are several different ways this model could be written in matrix notation. We consider two forms that will be useful below. First, define the $n \times n$ matrix,

$$\mathbf{B}_i \equiv (b_{i,kl}), \quad (7.73)$$

where $b_{i,kl}$ is defined in (7.72) and is put in the (k, l) th entry of the matrix, $\mathbf{B}_i$, for $i = 1, \dots, n$. Then,

$$\mathbf{Y}_t = \mathbf{A} \mathbf{Y}_{t-1} + (\mathbf{I}_n \otimes \mathcal{G}(\mathbf{Y}_{t-1}; \theta^G))' \mathbf{B} \mathbf{Y}_{t-1} + \boldsymbol{\eta}_t, \quad (7.74)$$

where $\otimes$ denotes the Kronecker product, $\mathcal{G}(\mathbf{Y}_{t-1}; \theta^G) \equiv (g(Y_{t-1}(\mathbf{s}_1); \theta^G), \dots, g(Y_{t-1}(\mathbf{s}_n); \theta^G))'$, and the $n^2 \times n$ matrix $\mathbf{B}$ is given by

$$\mathbf{B} \equiv \begin{pmatrix} \mathbf{B}_1 \\ \mathbf{B}_2 \\ \vdots \\ \mathbf{B}_n \end{pmatrix}. \quad (7.75)$$

In the context of the general state-dependent model presented in (7.68), we have

$$\mathbf{M}(\mathbf{Y}_{t-1}; \theta^M) = \mathbf{A} + (\mathbf{I}_n \otimes \mathcal{G}(\mathbf{Y}_{t-1}; \theta^G))' \mathbf{B}, \quad (7.76)$$

where the parameters are $\theta^M \equiv \{\mathbf{A}, \mathbf{B}, \theta^G\}$. Although it is possible to allow these parameters to vary with time, we suppress that notation here for simplicity.

Alternatively, we can also write the model as

$$\mathbf{Y}_t = \mathbf{A}\mathbf{Y}_{t-1} + \mathbf{B}_v \cdot (\mathbf{Y}_{t-1} \otimes \mathcal{G}(\mathbf{Y}_{t-1}; \theta^G)) + \eta_t, \quad (7.77)$$

where the $n \times n^2$ matrix $\mathbf{B}_v$ is given by

$$\mathbf{B}_v = \begin{pmatrix} \text{vec}(\mathbf{B}_1)' \\ \text{vec}(\mathbf{B}_2)' \\ \vdots \\ \text{vec}(\mathbf{B}_n)' \end{pmatrix}, \quad (7.78)$$

and the $n^2 \times 1$ vector $\mathbf{Y}_{t-1} \otimes \mathcal{G}(\mathbf{Y}_{t-1}; \theta^G)$ is given by

$$\begin{aligned} (\mathbf{Y}_{t-1} \otimes \mathcal{G}(\mathbf{Y}_{t-1}; \theta^G))' = & (Y_{t-1}(s_1)g(Y_{t-1}(s_1); \theta^G), \dots, \\ & Y_{t-1}(s_1)g(Y_{t-1}(s_n); \theta^G), Y_{t-1}(s_2)g(Y_{t-1}(s_1); \theta^G), \dots, \\ & Y_{t-1}(s_n)g(Y_{t-1}(s_n); \theta^G)), \end{aligned}$$

which contains all of the pair-wise interaction terms. We refer to these as *dyadic* interactions.

It is not surprising that for many spatio-temporal processes, $\mathbf{A}$ and $\mathbf{B}$ have too many parameters to permit reliable estimation; however, these matrices could be parameterized in terms of θ^A and θ^B , namely $\mathbf{A}(\theta^A)$ and $\mathbf{B}(\theta^B)$, respectively. In the case of the linear portion of the propagator, $\mathbf{A}(\theta^A)$, the types of parameterizations discussed previously are still relevant. What about $\mathbf{B}(\theta^B)$? First, note that if $\mathcal{G}(\mathbf{Y}_{t-1}; \theta^G) = \mathbf{Y}_{t-1}$, then we have only $n(n+1)/2$ unique dyadic interactions. Although this greatly simplifies the model, it may limit the appropriateness of the parameterization. We consider some special cases and motivating models in the examples below.

Example: Second-Order Taylor Expansion

If the form of the nonlinearity is known (i.e., the propagator is known), then a common approach to approximating the nonlinearity is to construct a local-linear approximation as discussed in Section 7.3.1. To improve accuracy, the associated Taylor expansion can be extended to second-order terms (e.g., Jazwinski, 1970; West and Harrison, 1997). Such an expansion is a special case of the GQN model presented here.

Recall the deterministic vector-valued nonlinear function:

$$\mathbf{Y}_t = \mathcal{M}(\mathbf{Y}_{t-1}), \quad (7.79)$$

where $\mathcal{M}(\mathbf{Y}_{t-1}) \equiv (m_1(\mathbf{Y}_{t-1}), \dots, m_n(\mathbf{Y}_{t-1}))'$, with $\{m_i\}$ being scalar-valued (nonlinear) functions. A second-order Taylor expansion about some specified vector $\hat{\mathbf{Y}}_t$ can then be written:

$$\begin{aligned} \mathcal{M}(\mathbf{Y}_t) &= \mathcal{M}(\hat{\mathbf{Y}}_t) + \mathbf{M}_t \cdot (\mathbf{Y}_t - \hat{\mathbf{Y}}_t) \\ &\quad + \frac{1}{2} (\mathbf{I}_n \otimes (\mathbf{Y}_t - \hat{\mathbf{Y}}_t)') \cdot \mathbf{H}_t \cdot (\mathbf{Y}_t - \hat{\mathbf{Y}}_t) + \dots, \end{aligned} \quad (7.80)$$

where recall that we use the $\cdot$ notation to emphasize the product operator; $\mathbf{M}_t$ is the matrix of first partial derivatives of $\mathcal{M}(\mathbf{Y}_t)$ evaluated at a given $\mathbf{Y}_t = \hat{\mathbf{Y}}_t$, as given in equation (7.65) in Section 7.3.1; and

$$\mathbf{H}_t \equiv \mathbf{H}_t(\hat{\mathbf{Y}}_t) = \begin{pmatrix} \mathbf{H}_{1t}(\mathbf{Y}_t) \\ \vdots \\ \mathbf{H}_{nt}(\mathbf{Y}_t) \end{pmatrix} \bigg|_{\mathbf{Y}_t = \hat{\mathbf{Y}}_t}, \quad (7.81)$$

where $\{\mathbf{H}_{it}(\mathbf{Y}_t) : i = 1, \dots, n\}$ are the Hessian matrices (i.e., matrices of second-partial derivatives) of $\mathcal{M}(\cdot)$, evaluated at a given $\mathbf{Y}_t = \hat{\mathbf{Y}}_t$. For $i = 1, \dots, n$, the (k, l) th element of $\mathbf{H}_{it}(\hat{\mathbf{Y}}_t)$ is

$$(\mathbf{H}_{it}(\hat{\mathbf{Y}}_t))_{kl} = \frac{\partial^2 m_i(\mathbf{Y}_t)}{\partial Y_t(s_k) \partial Y_t(s_l)} \bigg|_{\mathbf{Y}_t = \hat{\mathbf{Y}}_t}, \quad k, l = 1, \dots, n. \quad (7.82)$$

This equation fits into the GQN form (with respect to $\mathbf{Y}_t - \hat{\mathbf{Y}}_t$), where $\mathbf{M}_t$ and $\mathbf{H}_t$ correspond to $\mathbf{A}$ and $\mathbf{B}$, respectively. As mentioned previously, such an implementation requires complete knowledge of the nonlinear functions (i.e., $m_i(\mathbf{Y}_t)$), which is not typically available for most spatio-temporal processes that require statistical modeling.

Example: Reaction-Diffusion (Growth) Models

Many processes in population ecology are motivated by a reaction-diffusion equation. For example, invasive species tend to spread (diffuse) and grow proportional to the current abundance, so long as the environmental carrying capacity has not been obtained. Such processes can be modeled by a variety of approaches, depending on whether time and space are continuous or discrete. In the context of continuous space and continuous time, a reaction-diffusion PDE provides a good description of the potential spread of the invasive species. For example, consider the PDE:

$$\begin{aligned} \frac{\partial Y}{\partial t} &= \frac{\partial}{\partial x} \left(\delta(x, y) \frac{\partial Y}{\partial x} \right) + \frac{\partial}{\partial y} \left(\delta(x, y) \frac{\partial Y}{\partial y} \right) \\ &\quad + \gamma_0(x, y) Y \exp \left(1 - \frac{Y}{\gamma_1(x, y)} \right), \end{aligned} \quad (7.83)$$

where $\mathbf{s} \equiv (x, y)'$; and the diffusion coefficient δ , growth parameter γ_0 , and carrying capacity γ_1 are, in general, spatially varying. The growth (i.e., reaction) process in (7.83) follows a logistic formulation. A simple finite-difference discretization of this PDE suggests the following multivariate difference equation:

$$\mathbf{Y}_t \simeq \mathbf{A}(\delta)\mathbf{Y}_{t-1} + \mathbf{B}_v(\mathbf{Y}_{t-1} \otimes \mathcal{G}(\mathbf{Y}_{t-1}; \boldsymbol{\gamma}_0, \boldsymbol{\gamma}_1)), \quad (7.84)$$

where $\mathbf{A}(\delta)$ is a sparse $n \times n$ matrix containing five non-zero diagonals with elements from δ (representing the diffusion component of the model, as discussed in Section 7.2.4 for the linear DSTM); $\mathbf{B}_v$ is a very sparse matrix in which the submatrices $\mathbf{B}_i$ each contain zeros except for a 1 in the (i, i) th location; and the i th element of $\mathcal{G}(\cdot)$ is $\gamma_0(\mathbf{s}_i) \exp\{1 - Y_{t-1}(\mathbf{s}_i)/\gamma_1(\mathbf{s}_i)\}$.

As an example, Figure 7.7 shows a simulation of this deterministic two-dimensional reaction-diffusion process with logistic growth; see (7.83). The left panels show five snapshots of the simulation assuming a constant diffusion coefficient ($\delta = 4$) across the domain, as well as a constant growth rate ($\gamma_0 = .55$) and constant carrying capacity ($\gamma_1 = 2$). The right panels show an analogous simulation, but with a nonhomogeneous diffusion rate ($\delta = .1$ on the left half of the domain, and $\delta = 4$ on the right half of the domain), a nonhomogeneous growth rate ($\gamma_0 = .2$ on the lower half of the domain, and $\gamma_0 = .55$ on the upper half of the domain), and constant carrying capacity ($\gamma_1 = 2$). In both simulations, the process initially decays in its intensity, as with a diffusion process (see Figure 7.1), then increases in its intensity, and it is starting to reach its carrying capacity by the last time shown (400 time steps). The nonhomogeneous diffusion rate in the right panels clearly suggests much stronger spread to the right than to the left. Similarly, the upper-lower variation in the growth rate gives much greater intensities in the upper portion of the domain than in the lower portion. One could imagine these parameters varying spatially for a real-world invasive species, perhaps according to habitat preference.

In our context, (7.84) could serve as “motivation” for a *statistical* model that would typically have an additive or multiplicative error process and random parameters $\{\delta, \boldsymbol{\gamma}_0, \boldsymbol{\gamma}_1\}$, whose distributions (e.g., spatial random fields) would be given at the next level of the HM. An example of this hierarchical approach applied to the prediction of the spread of the invasive Eurasian Collared Dove is given in Wikle and Hooten (2006) and Hooten and Wikle (2008). Of course, other growth functions could be used in this context (e.g., see Wikle and Hooten, 2010).

Example: Nonlinear Process in Geophysics

Consider the case of a dynamical process motivated by a one-dimensional nonlinear advection equation:

$$\frac{\partial Y}{\partial t} = Y \frac{\partial Y}{\partial x}, \quad (7.85)$$

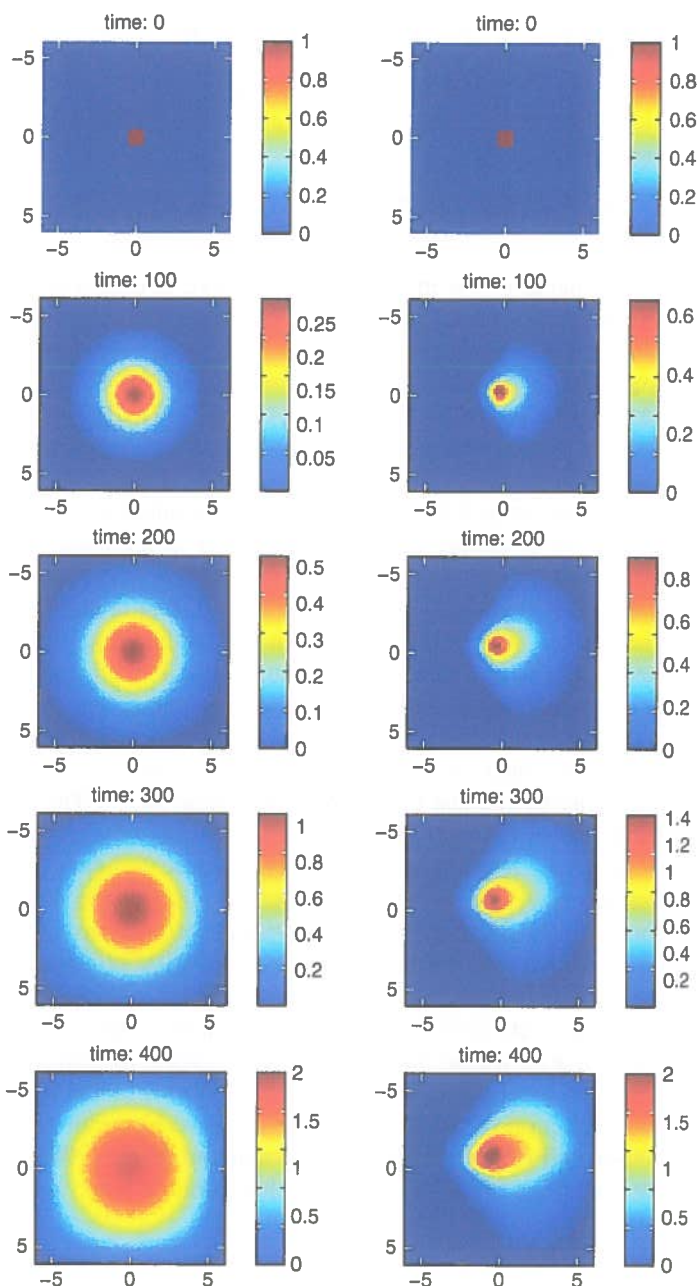


Figure 7.7 Simulation of a two-dimensional reaction-diffusion process with logistic growth as in (7.83). The left panels show the simulated process at five times ($t = 0, 100, 200, 300, 400$) assuming a constant diffusion coefficient ($\delta = 4$) across the domain, as well as a constant growth rate ($\gamma_0 = .55$) and constant carrying capacity ($\gamma_1 = 2$). The right panels show an analogous simulation but with nonhomogeneous diffusion rate ($\delta = .1$ on the left half of the domain, and $\delta = 4$ on the right half of the domain), nonhomogeneous growth rate ($\gamma_0 = .2$ on the lower half of the domain, and $\gamma_0 = .55$ on the upper half of the domain), and constant carrying capacity ($\gamma_1 = 2$).

such as occurs in geophysical fluid dynamics. A basic finite-difference discretization of (7.85) gives

$$Y_t(x) \simeq Y_{t-\Delta_t}(x) + \frac{\Delta_t}{2\Delta_x} Y_{t-\Delta_t}(x) Y_{t-\Delta_t}(x + \Delta_x) - \frac{\Delta_t}{2\Delta_x} Y_{t-\Delta_t}(x) Y_{t-\Delta_t}(x - \Delta_x). \quad (7.86)$$

For simplicity of presentation, assume that $\Delta_t = 1$, and also that we have regularly spaced grid points along x given by $x_0, x_1, x_2, \dots, x_n, x_{n+1}$. Furthermore, let $b \equiv \Delta_t/2\Delta_x$, and assume that boundary values $Y_t(x_0)$ and $Y_t(x_{n+1})$ are zero in this simple example. We can then write (7.86) as the vector equation,

$$\mathbf{Y}_t \simeq \mathbf{A}\mathbf{Y}_{t-1} + (\mathbf{I}_n \otimes \mathbf{Y}'_{t-1})\mathbf{B}\mathbf{Y}_{t-1}, \quad (7.87)$$

where $\mathbf{A} = \mathbf{I}_n$, and $\mathbf{B}$ is given by (7.75). Here, $\mathbf{B}_1$ has b as the (row, column) element (2, 1) and zeros elsewhere; $\{\mathbf{B}_i : i = 2, \dots, n-1\}$ are zero matrices with the exception of having b as the element $(i+1, i)$ and $-b$ as the element $(i-1, i)$; and $\mathbf{B}_n$ has $-b$ in matrix location $(n, n-1)$ and zeros elsewhere.

Note that by writing (7.86) as (7.87), we obtain a very sparse matrix $\mathbf{B}$. Thus, the vector form of (7.86) can be written more directly as

$$\mathbf{Y}_t \simeq (\mathbf{I} + \text{diag}(\mathbf{Y}_{t-1})\tilde{\mathbf{B}})\mathbf{Y}_{t-1}, \quad (7.88)$$

where $\tilde{\mathbf{B}}$ is a matrix of zeros except for b down the first diagonal above the main diagonal and $-b$ down the first diagonal below the main diagonal.

Of course, using this as motivation for a statistical model, an additive error term in (7.88) would be included, and the parameters would be modeled at a lower level of the HM (BHM) or would be estimated directly (EHM). Obviously, more complicated models for geophysical processes could be considered. For example, a very complex process model for so-called *quasi-geostrophic* flow in the ocean (see Section 7.4) is given in Berliner, Milliff, and Wikle (2003), and this is shown by Wikle and Hooten (2010) to fit into the GQN framework.

Example: Reduced-Dimension State Process

Certainly, there are many situations where the underlying dynamics of a process are not known well enough to motivate it by specific deterministic forms. Or,

it may be that the underlying dynamics are known to some degree, but they are just too complicated to be used as the foundation for a process model in an HM. In addition, it may be the case that in a spectral setting we have no clear prior scientific understanding of how best to truncate the expansion. In such cases, one might consider dimension reduction of the state process itself. Recall the dimension reduced form given in (7.24), where α_t is p_α -dimensional and $p_\alpha \ll n$. In this case, (7.72) for $\{\alpha_t\}$ yields

$$\alpha_t(i) = \sum_{j=1}^{p_\alpha} a_{ij} \alpha_{t-1}(j) + \sum_{k=1}^{p_\alpha} \sum_{l=1}^{p_\alpha} b_{i,kl} \alpha_{t-1}(k) g(\alpha_{t-1}(l); \theta_g) + \eta_t(i), \quad (7.89)$$

for $i = 1, \dots, p_\alpha$. Recall that there are $O(p_\alpha^3)$ parameters in this formulation. If p_α is very small, we can proceed by estimating all of the parameters directly. However, in most cases, p_α is not small enough for the amount of data available. Thus, we have to make additional simplifying assumptions and/or perform model selection to choose particular interactions that are important [e.g., see de Luna and Genton (2005) for an example in the linear case and Wikle and Holan (2011) for a GQN example]. Another alternative is to simplify using a dimension-reduction technique called *scale analysis*, where the consequence of recognizing an additional scale is analyzed.

First, assume it is reasonable to decompose the process as

$$\mathbf{Y}_t = \Phi^{(1)} \alpha_t^{(1)} + \Phi^{(2)} \alpha_t^{(2)} + \mathbf{v}_t, \quad (7.90)$$

where $\alpha_t^{(i)}$ is $p_{\alpha,i}$ -dimensional and where $p_{\alpha,i} \ll n$; $i = 1, 2$. Typically, the spectral coefficients $\alpha_t^{(1)}$ correspond to “large-scale” basis functions in some sense, and $\alpha_t^{(2)}$ correspond to “small-scale” basis functions. Let $\alpha_t = (\alpha_t^{(1)'} , \alpha_t^{(2)'})'$ satisfy the GQN model (7.89). Motivated loosely by the concept of “Reynolds averaging” in turbulence theory (e.g., see Holten, 2004, Chapter 5), we want the pairwise interactions between components of $\alpha_t^{(1)}$ to be explicit and described by $\alpha_t^{(1)}$ substituted for α_t in (7.89), but we want those among the “small-scale” components $\alpha_t^{(2)}$ to be in some sense “noise,” and the interactions between the components of $\alpha_t^{(1)}$ and $\alpha_t^{(2)}$ to suggest *random* coefficients in a linear combination of the large-scale basis functions (Wikle and Hooten, 2010).

As a simple example, consider a situation where $g(x) = x$ is the *identity function*, $\alpha_t^{(1)} \equiv (\alpha_t^{(1)}(1), \alpha_t^{(1)}(2))'$, and $\alpha_t^{(2)} \equiv (\alpha_t^{(2)}(1), \alpha_t^{(2)}(2), \alpha_t^{(2)}(3))'$. In this case, there are three “resolved” dyadic interactions corresponding to the three quadratic combinations of $\alpha_t^{(1)}(i)$, for $i = 1, 2$. In addition, the six unique pairwise interactions between the large-scale $\alpha_t^{(1)}(i)$ and the small-scale $\alpha_t^{(2)}(j)$; $i = 1, 2$, $j = 1, 2, 3$, coefficients imply linear terms in the large-scale components, with the small-scale components giving rise to unknown but *random* coefficients (i.e., $\{a_{ij}\}$ in (7.89)). Finally, the six dyadic interactions

between the small scales $\{\alpha_t^{(2)}(j) : j = 1, 2, 3\}$, are assumed to correspond to the noise process, and because they are quadratic combinations, it suggests that they can be absorbed into the additive error process in the GQN (7.89), resulting in a dependent error process.

More generally (still assuming the function $g(\cdot)$ is the identity function), the GQN model for $\alpha_t^{(1)}$ can be written as

$$\alpha_t^{(1)} = \mathbf{A}\alpha_{t-1}^{(1)} + \left(\mathbf{I}_{p_1} \otimes \alpha_{t-1}^{(1)'}\right) \mathbf{B}\alpha_{t-1}^{(1)} + \eta_t; \quad \eta_t \sim iid \text{Gau}(\mathbf{0}, \mathbf{Q}). \quad (7.91)$$

These models have not been explored extensively in the spatio-temporal statistics literature, but they have also been considered in other disciplines (see the Bibliographic Notes in Section 7.8). The challenge in this setting is still the choice of meaningful, science-based parameterizations. In addition, we could certainly consider higher-order (e.g., triad) interactions (e.g., Wikle and Holan, 2011), and we could motivate truncations in a similar way to that considered in the turbulence literature (e.g., Lesieur, 2008). However, the curse of dimensionality is so dominant in this case, that the hierarchical statistical modeler must really have a strong science-based approach for parameterization and truncation.

7.3.4 Agent-Based Models

A relatively new approach to statistical modeling of nonlinear and non-Gaussian spatio-temporal dynamical processes comes from *agent-based* models. Such models have been used in a variety of disciplines to mimic the behavior of natural processes. Agent-based procedures model complicated large-scale behavior by focusing on intuitive rules for the process at the smallest scales. In this way, agent-based models can be equated to *cellular automata* (CA) algorithms (e.g., Wolfram, 1984). Basically, an automata can be thought of as a collection of entities or agents whose states are determined by a simple set of rules. Automata are most often defined in a deterministic paradigm, in which the state of the “neighborhood” of an agent determines the future state of the agent. This can be generalized to allow specified probabilities to determine the future state of the agent. Traditionally, such models are used in a simulation context and statistical implementations are less common. However, as shown in Hooten and Wikle (2010), agent-based models can be formulated in a spatio-temporal hierarchical statistical context to characterize processes such as the spread of an epidemic. They show that such a model can be based on an intuitive parameterization of the system dynamics, and it can accommodate features such as long-distance and directionally varying dispersal, as well as spatial heterogeneity.

Consider a deterministic CA that is defined, in general, as

$$\mathbf{Y}_t = \mathcal{M}(\mathbf{Y}_{t-1}, \mathcal{R}, \mathcal{N}), \quad (7.92)$$

where $\mathcal{M}$ is an evolution function that depends on a set of neighborhoods, $\mathcal{N} \equiv \{\mathcal{N}_i : i = 1, \dots, n\}$, and a set of rules $\mathcal{R}$. Specifically, each element $Y_{i,t}$ (the automata) of the state vector $\mathbf{Y}_t \equiv (Y_{1,t}, \dots, Y_{n,t})'$ is a result of the function $\mathcal{M}$ and the previous set of neighboring automata, $\mathbf{Y}_{\mathcal{N}_i,t-1} \equiv (Y_{j,t-1} : j \in \mathcal{N}_i)'$, where it is possible here that $i \in \mathcal{N}_i$ (unlike for Markov random fields, described in Section 4.2.2). At first glance, this does not seem all that different from the neighbor-based vector autoregressive models (linear or nonlinear) described previously. The difference is in how the function $\mathcal{M}$ is defined, which is now made up of a set of rules, defining the map from all possible states of the set $\mathbf{Y}_{\mathcal{N}_i,t-1}$ to the possible states of $Y_{i,t}$. A CA defined in this simple way can exhibit very complex dynamical behavior (e.g., Wolfram, 1984).

The rules that govern the CA system are critical. However, the space of potential rules can be quite large, even for relatively small neighborhoods. For example, in a simple binary-state automaton (i.e., $Y_{i,t} \in \{0, 1\}$), a mapping function $\mathcal{M}$ over a simple first-order neighborhood in one dimension has $2^3 = 8$ possible neighborhood states, which leads to $2^8 = 256$ possible rules for the binary automaton $Y_{i,t}$. In general, for such a binary-state model, with $|\mathcal{N}_i|$ neighbors (including i), there will be $2^{2^{|\mathcal{N}_i|}}$ possible rules; the rules may or may not be automaton-dependent. As an example of how cumbersome this could be, consider a binary CA on a two-dimensional regular grid of cells and a “Queen’s neighborhood” (i.e., the eight nearest neighbors in a cell to which the Queen could move one space in a game of chess), for which $|\mathcal{N}_i| \equiv 9$. In this case, the number of possible rules is 1.34×10^{154} . Thus, if the rules are not known *a priori*, it would not be feasible to do an exhaustive search in order to infer which is governing the observed binary data. Instead, we could consider a stochastic or probabilistic CA.

In essence, a stochastic or probabilistic CA consists of random variables for the cells’ states that are conditioned on the neighborhood’s variables at the previous time. In this context, the usual vector autoregressive models for time series are stochastic CAs. The conditional distribution describing the probabilistic CA can be written as

$$Y_{i,t} | \mathbf{Y}_{\mathcal{N}_i,t-1} \sim [Y_{i,t} | \mathcal{M}(\mathbf{Y}_{\mathcal{N}_i,t-1})], \quad i = 1, \dots, n. \quad (7.93)$$

As an example, for binary automata, the distribution in (7.93) could be a Bernoulli distribution, in which the Bernoulli probabilities are a function of the neighborhood at the previous time, along with some parameters. An obvious advantage of this probabilistic specification over the deterministic one, is that, from a statistical perspective, it is easier to search the space of probability rules. However, it might be viewed that a disadvantage of the probabilistic approach is that the evolution from the previous state to the current state is no longer completely predictable. Note that there is a strong similarity between these models and so-called auto-logistic models (e.g., Zhu, Huang, and Wu, 2005; Royle and Dorazio, 2008).

Example: Modeling a Raccoon Rabies Epidemic

As a simple illustration of how one might apply a stochastic CA, consider the model presented in Hooten and Wikle (2010). The authors were interested in modeling the spread of the raccoon rabies epidemic on the east coast of the United States during the early 1990s [for more discussion of such epidemics, see, e.g., Nettles et al. (1979) and Smith et al. (2002)]. The process in this case is binary ($Y_{i,t} = 1$, corresponding to the presence of rabies in the i th grid box at a time t , and $Y_{i,t} = 0$, corresponding to no rabies present). It is assumed that data and process here are equivalent, and hence the HM consists of the following process model:

$$Y_{i,t} | \mathbf{Y}_{t-1} \sim \text{Ber}(\theta_{i,t}), \quad (7.94)$$

where the Bernoulli probability, $\theta_{i,t}$, that the disease is present in the i th grid cell at time t is modeled as

$$\theta_{i,t} = \phi Y_{i,t-1} + (1 - Y_{i,t-1})(I_{\mathcal{N}_{i,t-1}} \bar{p}_{i,t} + (1 - I_{\mathcal{N}_{i,t-1}}) \psi), \quad (7.95)$$

for all i and t in the spatio-temporal support of the data. In (7.95),

$$I_{\mathcal{N}_{i,t-1}} \equiv \begin{cases} 1, & \sum_{j \in \mathcal{N}_i} Y_{j,t-1} > 0, \\ 0, & \sum_{j \in \mathcal{N}_i} Y_{j,t-1} = 0, \end{cases} \quad (7.96)$$

and $\mathcal{N}_i$ is the spatial neighborhood of grid cell i (assumed to be a Queen's neighborhood, made up of the eight neighbors immediately adjacent to cell i , as well as cell i).

There are three disjoint situations represented on the right-hand side of (7.95):

- $Y_{i,t-1}$: Cell i has rabies present (i.e., is “occupied” at time $t - 1$),
- $(1 - Y_{i,t-1})(I_{\mathcal{N}_{i,t-1}})$: At least one of the neighbors of the i th cell is occupied at time $t - 1$, but cell i is not occupied; and
- $1 - I_{\mathcal{N}_{i,t-1}}$: No cell in $\mathcal{N}_i$ (including cell i) is occupied at time $t - 1$.

Then the parameter ϕ in (7.95) is the probability that, once the phenomenon is present in cell i , it is present in that cell at the next time. Correspondingly, ψ is the probability of “out-of-neighborhood dispersal,” meaning that ψ is the probability that the phenomenon will jump to cell i without occupying any neighborhood cell in $\mathcal{N}_i$. The neighborhood dispersal in this model is handled by the second term on the right-hand side of (7.95), which is controlled by the probability parameter $\bar{p}_{i,t}$. There are many possible specifications for this probability, depending on the spatial domain, level of realistic detail desired, and ease of estimation. Hooten and Wikle (2010) consider

$$\bar{p}_{i,t} = 1 - \exp((\mathbf{Y}_{\mathcal{N}_i,t-1})' \log(\mathbf{1} - \mathbf{p})), \quad (7.97)$$

where $\mathbf{Y}_{\mathcal{N}_i, t-1}$ consists of ones and zeros corresponding to the state of the neighborhood of the i th cell, $\mathbf{1}$ is a vector whose entries are all 1s, and $\mathbf{p} \equiv (p_1, p_2, \dots, p_{|\mathcal{N}_i|})'$ is a vector of transition probabilities to cell i from each of its $|\mathcal{N}_i|$ neighbors, which they assumed did not depend on i . As shown in Hooten and Wikle (2010), this specification considers the probability of presence at time t , for a cell that has been previously unoccupied to be the union of transition probabilities from occupied neighbors. To be clear, the model (7.95)–(7.97) assumes that the neighborhood-based dispersal is dynamically anisotropic (unequal $\{p_i\}$) but homogeneous ($\{p_i\}$ do not depend on t). This model has been extended to the case of heterogeneous transition probabilities, dependent on some environmental conditions, by Hooten and Wikle (2010).

In a BHM, the parameters ϕ , ψ , and $\mathbf{p}$ would be given distributions at the next level of the hierarchy. For example, ϕ and ψ could each be given a beta prior distribution, and $\mathbf{p}$ could be given a Dirichlet prior distribution, possibly dependent on hyperparameters (Hooten and Wikle, 2010).

7.4 PROCESS MODELS FOR THE DSTM: MULTIVARIATE MODELS

Just as most real-world processes are not linear, most processes do not exist in isolation from other processes. For example, although we could consider *surface ozone* as a spatio-temporal process, a model for its behavior would surely be better if its relationship with *ambient air temperature* were included. One might just think of temperature as a covariate in this case, since typically surface ozone responds to temperature, and not so much the other way around (although this is not the case in the upper atmosphere). What about temperature and pressure? These are physically related, and one may think about their (statistical) dependence jointly, or in a conditional-probability setting, depending on the circumstances. Obviously, modeling joint statistical dependence could be extended to more meteorological processes (e.g., wind, pressure, temperature, density, water vapor, etc.). In this section, we consider both joint and conditional formulations for multivariate DSTM process models.

7.4.1 Augmenting the State Process (Multivariate DSTMs)

From a dynamical perspective, all of the approaches for modeling DSTMs are still applicable in the multivariate context. For example, if we were modeling a system of several processes in which first-order linear dynamics were appropriate, we could simply augment the state variable (by concatenating the various state vectors) and use the models described above: Say we have processes $\mathbf{Y}_t^{(k)}$ of dimension n_k ; $k = 1, \dots, K$. The augmented state variable is $\mathbf{Y}_t = (\mathbf{Y}_t^{(1)'} \dots \mathbf{Y}_t^{(K)'})'$, and hence we can use a dynamical model to capture its spatio-temporal variability, such as the linear DSTM:

$$\mathbf{Y}_t = \mathbf{M}\mathbf{Y}_{t-1} + \boldsymbol{\eta}_t; \quad \boldsymbol{\eta}_t \sim \text{iid Gau}(\mathbf{0}, \mathbf{Q}_\eta), \quad (7.98)$$

which is of dimension $n_+ \equiv \sum_k n_k$. Of course, the nonlinear evolution model discussed in Section 7.3 could also be considered from a multivariate perspective, but we focus here on the linear case.

The propagator matrix $\mathbf{M}$ allows each process at the previous time to have an impact on itself and the others at the current time. Similarly, the error covariance matrix $\mathbf{Q}_\eta$ is in block form, corresponding to the covariance matrix of each process (the diagonal blocks) and the cross-covariance matrices (the off-diagonal blocks). Note that higher-order time lags could be used, and the propagator matrices could vary over time. However, the curse of dimensionality would limit such models, except in very highly parameterized settings. In what follows, we shall focus on the linear, first-order lag structure and non-time-varying propagator matrices, since they are sufficient to illustrate general multivariate-modeling issues.

Obviously, the potential difficulty now concerns dimensionality. If there was concern about dimensionality with just one process, the problems will only be compounded by the extra dimensions coming from the augmentation of the state with more vector-valued processes. Any of the strategies to reduce dimensionality considered previously in Section 7.1 (e.g., neighbor-based parameterization, truncation of spectral representations, etc.) could be used here as well. For example, the discretization (e.g., using finite-difference, finite-element, or spectral methods) of a system of PDEs provides a framework to motivate a model such as (7.98) (or the nonlinear equivalents of it, as suggested in Section 7.3.3). That is, the propagator matrix should be relatively sparse, and the associated parameters could either be estimated directly (EHM) or modeled probabilistically (BHM) at a lower level of the HM.

As an example of how (7.98) can be simplified, consider the spectral decomposition of the process vectors: $\mathbf{Y}_t^{(k)} = \Phi^{(k)} \alpha_t^{(k)}$, where the $\Phi^{(k)}$ are $n_k \times n_k$ matrices of spatial basis functions. Then, (7.98) can be written as

$$\alpha_t = \mathbf{M}_\alpha \alpha_{t-1} + \eta_{\alpha,t}, \quad (7.99)$$

where $\alpha_t \equiv (\alpha_t^{(1)'}, \dots, \alpha_t^{(K)'})'$, $\mathbf{M}_\alpha \equiv (\mathbf{W}'\mathbf{W})^{-1} \mathbf{W}'\mathbf{M}\mathbf{W}$, $\eta_{\alpha,t} \equiv (\mathbf{W}'\mathbf{W})^{-1} \mathbf{W}'\eta_t$ (assuming $(\mathbf{W}'\mathbf{W})^{-1}$ exists), and $\mathbf{W} \equiv (\Phi^{(1)'}, \dots, \Phi^{(K)'})'$. As is the case for univariate spectral representations, one would expect that the dependence structure in $\eta_{\alpha,t}$ would be simpler than that of η_t in (7.98), given the decorrelating effects of spectral transformations.

More practically, one would most likely perform dimension reduction by writing $\mathbf{Y}_t^{(k)} = \Phi^{(k)} \alpha_t^{(k)} + \nu_t^{(k)}$, where $\alpha_t^{(k)}$ is $p_{\alpha,k}$ -dimensional ($p_{\alpha,k} \ll n_k$), and thus $\Phi^{(k)}$ has dimension $n_k \times p_{\alpha,k}$; $k = 1, \dots, K$. Additional computational savings are possible if the basis functions are mutually orthogonal (or, at least, orthogonal for each process, k). Furthermore, one might simply choose to estimate the $p_{\alpha,+} \times p_{\alpha,+}$ propagator matrix $\mathbf{M}_\alpha$ directly, if $p_{\alpha,+} \equiv \sum_k p_{\alpha,k}$ is sufficiently small.

The joint-modeling approach can also be considered in the context of an integro-difference equation (IDE). For example, say we have K spatio-temporal

processes in continuous space and discrete time, $\{Y_t^{(k)}(\cdot) : k = 1, \dots, K\}$. The multivariate version of the linear stochastic IDE process model given in Section 7.2.5 can then be written as

$$Y_t^{(k)}(\mathbf{s}) = \int_{D_s} \sum_{j=1}^K m^{(kj)}(\mathbf{s}, \mathbf{x}; \boldsymbol{\theta}) Y_{t-1}^{(j)}(\mathbf{x}) d\mathbf{x} + \eta_t^{(k)}(\mathbf{s}), \quad (7.100)$$

for $k = 1, \dots, K$, and where the kernels $m^{(kj)}(\mathbf{s}, \mathbf{x}; \boldsymbol{\theta})$ are assumed to depend on some parameters $\boldsymbol{\theta}$ (that, in principle, could also be spatially and/or temporally varying). We can write (7.100) in matrix notation as

$$\mathbf{Y}_t(\mathbf{s}) = \int_{D_s} \mathbf{M}(\mathbf{s}, \mathbf{x}; \boldsymbol{\theta}) \mathbf{Y}_{t-1}(\mathbf{x}) d\mathbf{x} + \boldsymbol{\eta}_t(\mathbf{s}), \quad (7.101)$$

where $\mathbf{Y}_t(\mathbf{s}) \equiv (Y_t^{(1)}(\mathbf{s}), \dots, Y_t^{(K)}(\mathbf{s}))'$, $\mathbf{M}(\mathbf{s}, \mathbf{x}; \boldsymbol{\theta})$ is a $K \times K$ matrix with (k, j) th element given by $m^{(kj)}(\mathbf{s}, \mathbf{x}; \boldsymbol{\theta})$, and $\boldsymbol{\eta}_t(\mathbf{s}) \equiv (\eta_t^{(1)}(\mathbf{s}), \dots, \eta_t^{(K)}(\mathbf{s}))'$. Notice that $\mathbf{M}$ in (7.101) does not depend on t .

Discretization of (7.101) in space would imply a form such as (7.98), and the usual issues related to the curse of dimensionality could play a role. However, as described in Section 7.2.5, an advantage of the IDE formulation defined by kernels in the dynamical process is that the kernels can be parameterized efficiently to capture specific (qualitative) dynamical behavior (e.g., diffusion, advection, interaction, etc.). In addition, as described in Section 7.2.6, one might expand both the process and the kernels in terms of spectral basis functions and then consider the evolution of the reduced-dimensional spectral coefficients in the dynamical model, analogous to the formulation given in (7.99).

7.4.2 Dependence on Common Processes

An obvious way to build spatio-temporal dependence between processes is to represent these processes in terms of a common process. As with the univariate spectral decomposition discussed in Section 7.2.6, this hidden process is dynamical, but may or may not be spatial, depending on the implementation. For example, as with the multivariate IDE model given in (7.101), consider K spatio-temporal processes, $\{Y_t^{(k)}(\mathbf{s}) : k = 1, \dots, K\}$. In the spirit of Ver Hoef and Barry (1998), consider the construction of a multivariate spatio-temporal process given by

$$Y_t^{(k)}(\mathbf{s}) = \int_{D_s} \sum_{j=1}^J h^{(kj)}(\mathbf{s}, \mathbf{x}; \boldsymbol{\theta}) \alpha_t^{(j)}(\mathbf{x}) d\mathbf{x} + \gamma_t^{(k)}(\mathbf{s}), \quad \mathbf{s} \in D_s, \quad (7.102)$$

for $k = 1, \dots, K$, and where the kernels $h^{(kj)}(\mathbf{s}, \mathbf{x}; \boldsymbol{\theta})$ are assumed to depend on some parameters $\boldsymbol{\theta}$. Notice that these parameters do not (in general) depend

on t and hence do not contribute to the dynamics. In matrix notation, we can write (7.102) as

$$\mathbf{Y}_t(\mathbf{s}) = \int_{D_s} \mathbf{H}(\mathbf{s}, \mathbf{x}; \boldsymbol{\theta}) \boldsymbol{\alpha}_t(\mathbf{x}) d\mathbf{x} + \boldsymbol{\gamma}_t(\mathbf{s}), \quad \mathbf{s} \in D_s, \quad (7.103)$$

where $\mathbf{Y}_t(\mathbf{s}) \equiv (Y_t^{(1)}(\mathbf{s}), \dots, Y_t^{(K)}(\mathbf{s}))'$, $\mathbf{H}(\mathbf{s}, \mathbf{x}; \boldsymbol{\theta})$ is a $K \times J$ matrix with (k, j) th element given by $h^{(kj)}(\mathbf{s}, \mathbf{x}; \boldsymbol{\theta})$, $\boldsymbol{\alpha}_t(\mathbf{x}) \equiv (\alpha_t^{(1)}(\mathbf{x}), \dots, \alpha_t^{(J)}(\mathbf{x}))'$, and $\boldsymbol{\gamma}_t(\mathbf{s}) \equiv (\gamma_t^{(1)}(\mathbf{s}), \dots, \gamma_t^{(K)}(\mathbf{s}))'$. That is, a K -dimensional process $\mathbf{Y}_t(\cdot)$ is expressed in terms of a J -dimensional process $\boldsymbol{\alpha}_t(\cdot)$. This has benefit if the hidden dynamical process, $\{\alpha_t^{(j)}(\cdot) : j = 1, \dots, J\}$, has simplifying structure (e.g., they are a set of J independent AR(1) processes with unit variances). The α -processes induce dependence between the various Y -processes, defined by the smoothing induced from the kernels. In order for this framework for building DSTMs to induce realistic dynamics, one would typically need nontrivial parameterizations of some combination of the propagator and error covariance matrices. Indeed, as described in Section 7.1.3 for the univariate DSTM, without strong assumptions about the structure of the kernels and the dynamical parameters associated with the hidden processes, such models are not typically identifiable. Furthermore, this model can suffer from a serious curse of dimensionality. For example, if we approximate the integrals in (7.102) and (7.103) with summations, then for the k th process we have

$$\int_{D_s} \sum_{j=1}^J h^{(kj)}(\mathbf{s}, \mathbf{x}; \boldsymbol{\theta}) \alpha_t^{(j)}(\mathbf{x}) d\mathbf{x} \simeq \sum_{i=1}^{p_\alpha} \sum_{j=1}^J h^{(kj)}(\mathbf{s}, \mathbf{x}_i; \boldsymbol{\theta}) \alpha_t^{(j)}(\mathbf{x}_i), \quad (7.104)$$

where $\{\mathbf{x}_i : i = 1, \dots, p_\alpha\}$ correspond to a discrete set of locations in D_s . Thus, in matrix notation, a discretized model for (7.103) is given by

$$\mathbf{Y}_t(\mathbf{s}) = \mathbf{H}(\mathbf{s}) \boldsymbol{\alpha}_t + \boldsymbol{\gamma}_t(\mathbf{s}), \quad \mathbf{s} \in D_s, \quad (7.105)$$

where $\mathbf{Y}_t(\mathbf{s})$ and $\boldsymbol{\gamma}_t(\mathbf{s})$ are defined above in (7.103); $\boldsymbol{\alpha}_t \equiv (\alpha_t^{(1)'}, \dots, \alpha_t^{(J)'})'$ with $\alpha_t^{(j)} \equiv (\alpha_t^{(j)}(\mathbf{x}_1), \dots, \alpha_t^{(j)}(\mathbf{x}_{p_\alpha}))'$; and $\mathbf{H}(\mathbf{s})$ is a $K \times Jp_\alpha$ matrix whose k th row is made up of Jp_α -dimensional vectors, $\{(h^{(kj)}(\mathbf{s}, \mathbf{x}_i; \boldsymbol{\theta}) : i = 1, \dots, p_\alpha) : j = 1, \dots, J\}$.

Now, suppose that in (7.105), we assume $p_\alpha = 1$ and let $\mathbf{x}_1 = \mathbf{s}$. Then (7.105) becomes

$$\mathbf{Y}_t(\mathbf{s}) = \mathbf{H} \tilde{\boldsymbol{\alpha}}_t(\mathbf{s}) + \boldsymbol{\gamma}_t(\mathbf{s}); \quad \mathbf{s} \in D_s, \quad (7.106)$$

where $\tilde{\boldsymbol{\alpha}}_t(\mathbf{s}) \equiv (\alpha_t^{(1)}(\mathbf{s}), \dots, \alpha_t^{(J)}(\mathbf{s}))'$ is J -dimensional, and the $K \times J$ matrix $\mathbf{H}$ has (k, j) th entry $h_{kj} \equiv h^{(kj)}(\mathbf{s}, \mathbf{s}; \boldsymbol{\theta})$. Fix t , and suppose the spatial processes, $\alpha_t^{(1)}(\cdot), \dots, \alpha_t^{(J)}(\cdot)$, are independent. Then (7.106) gives the linear

model of coregionalization, with an extra, additive noise term $\gamma_t(\cdot)$ (e.g., Gelfand et al., 2004; Gelfand, Banerjee and Gamerman, 2005; Sansó, Schmidt, and Nobre, 2008; see also Section 4.1.5, for discussion of this model in the spatial context).

Now consider the K processes at common multiple locations, $\{s_1, s_2, \dots, s_n\}$; that is, $n_1 = \dots = n_K = n$. Then write $\mathbf{Y}_t^{(k)} \equiv (Y_t^{(k)}(s_1), \dots, Y_t^{(k)}(s_n))'$, which is an n -dimensional vector. (Note that this could be generalized to the case where locations for each process are different, but we do not do so for ease of presentation.) Then define the augmented (Kn) -dimensional spatio-temporal processes, $\mathbf{Y}_t \equiv (\mathbf{Y}_t^{(1)'}, \dots, \mathbf{Y}_t^{(K)'})'$. Hence, for multiple locations, the process model (7.105) can be rewritten as

$$\mathbf{Y}_t = \tilde{\mathbf{H}}\tilde{\boldsymbol{\alpha}}_t + \boldsymbol{\gamma}_t, \quad (7.107)$$

where $\tilde{\boldsymbol{\alpha}}_t$ is a (Jn) -dimensional vector given by $\tilde{\boldsymbol{\alpha}}_t \equiv (\tilde{\boldsymbol{\alpha}}_t^{(1)'}, \dots, \tilde{\boldsymbol{\alpha}}_t^{(J)'})'$, with $\tilde{\boldsymbol{\alpha}}_t^{(j)} \equiv (\alpha_t^{(j)}(s_1), \dots, \alpha_t^{(j)}(s_n))'$; $\tilde{\mathbf{H}}$ is a $(Kn) \times (Jn)$ matrix (which we could allow to be time-varying, in general); and $\boldsymbol{\gamma}_t$ is a (Kn) -dimensional vector of error processes, $\{\gamma_t^{(k)} : k = 1, \dots, K\}$.

Now, if $\{\tilde{\boldsymbol{\alpha}}_t\}$ has relatively interesting dynamics, then spatio-temporal dependence will be induced in $\{\mathbf{Y}_t^{(k)} : k = 1, \dots, K\}$ through the temporal and between-component dependence in $\{\tilde{\boldsymbol{\alpha}}_t\}$ and, critically, through the matrix $\tilde{\mathbf{H}}$. As discussed above, even in the case where this matrix is not time-varying, there is an identifiability issue with estimating the parameters of $\tilde{\mathbf{H}}$ along with any proposed non-trivial dynamical propagator for $\{\tilde{\boldsymbol{\alpha}}_t\}$.

Another special case of (7.105) for multiple locations is given by Calder (2007):

$$\mathbf{Y}_t = (\mathbf{K} \otimes \mathbf{F})\mathbf{A}_t + \boldsymbol{\gamma}_t, \quad (7.108)$$

where $\mathbf{K}$ is an $n \times p_\alpha$ matrix; $\mathbf{F}$ is a $K \times J$ matrix; and $\boldsymbol{\gamma}_t = (\gamma_t^{(1)'}, \dots, \gamma_t^{(K)'})'$, with $\gamma_t^{(k)} \sim \text{ind. Gau}(\mathbf{0}, \lambda_\gamma^{(k)}\mathbf{I})$; $k = 1, \dots, K$. This can be rewritten as

$$\mathcal{Y}_t = \mathbf{K}\mathbf{A}_t\mathbf{F}' + \mathbf{G}_t, \quad (7.109)$$

where $\mathcal{Y}_t \equiv (\mathbf{Y}_t^{(1)}, \dots, \mathbf{Y}_t^{(K)})$ is an $n \times K$ matrix of process vectors, $\mathbf{A}_t \equiv (\boldsymbol{\alpha}_t^{(1)}, \dots, \boldsymbol{\alpha}_t^{(J)})$ is a $p_\alpha \times J$ matrix of hidden processes, $\mathbf{G}_t \equiv (\gamma_t^{(1)}, \dots, \gamma_t^{(K)})$ is an $n \times K$ matrix of error processes, and $\mathbf{K}$ and $\mathbf{F}$ are as defined above. In this form, the model is related to dynamical factor models for multiple time series (e.g., Aguilar and West, 2000). Here there are non-trivial dynamics on the hidden processes that make up $\mathbf{A}_t$, and there are identifiability issues to be taken into account. Calder (2007) uses a discrete-kernel convolution matrix for $\mathbf{K}$ and proposes identifiability restrictions on $\mathbf{F}$ based on those given in Aguilar and West (2000). However, the introduction of parameters in the evolution processes $\{\boldsymbol{\alpha}_t^{(j)} : j = 1, \dots, J\}$ may require further identifiability restrictions.

In certain scientific applications, the hidden process $\{\tilde{\alpha}_t\}$ in (7.107) might correspond to a real-world process, and there might be known relationships between $\tilde{\alpha}_t$ and $\mathbf{Y}_t$ that suggest efficient parameterizations of $\tilde{\mathbf{H}}$. For example, Royle et al. (1999) considered spatial processes given by the east–west and north–south wind components and conditioned them on a hidden spatial process given by atmospheric pressure. Theoretical relationships (e.g., *geostrophy*) suggest that the wind components are proportional to the gradient of pressure, to a first-order approximation, in the mid-latitudes. This, in turn, suggests a low-dimensional parameterization for the matrix $\tilde{\mathbf{H}}$. For more details of such an example in the spatio-temporal context, see Berliner, Milliff, and Wikle (2003) and Milliff et al. (2010). We expand on this notion of scientific-process conditioning in the next section.

7.4.3 Conditional Formulation

Once again, conditioning can be used to help with the reduction of dimensionality, to ensure that valid models are being fitted, and (in many cases) to facilitate the incorporation of scientific information. For example, consider the case where we have two spatio-temporal processes $\{\mathbf{Y}_t^{(1)}\}$ and $\{\mathbf{Y}_t^{(2)}\}$. From the first-order dynamical perspective, we have $[\mathbf{Y}_t^{(1)}, \mathbf{Y}_t^{(2)} | \mathbf{Y}_{t-1}^{(1)}, \mathbf{Y}_{t-1}^{(2)}]$ (where dependence on parameters in the conditional distribution is suppressed for notational clarity). Now, we could make use of conditional probabilities and write

$$[\mathbf{Y}_t^{(1)}, \mathbf{Y}_t^{(2)} | \mathbf{Y}_{t-1}^{(1)}, \mathbf{Y}_{t-1}^{(2)}] = [\mathbf{Y}_t^{(2)} | \mathbf{Y}_t^{(1)}, \mathbf{Y}_{t-1}^{(1)}, \mathbf{Y}_{t-1}^{(2)}][\mathbf{Y}_t^{(1)} | \mathbf{Y}_{t-1}^{(1)}, \mathbf{Y}_{t-1}^{(2)}]. \quad (7.110)$$

Hopefully, there would be a good scientific reason for conditioning $\mathbf{Y}_t^{(2)}$ on $\mathbf{Y}_t^{(1)}$; in the example of near-surface ozone and temperature mentioned at the beginning of this section, it would make sense to consider ozone ($\mathbf{Y}_t^{(2)}$, say) conditioned on temperature ($\mathbf{Y}_t^{(1)}$, say), since the production of surface ozone is partially driven by ambient air temperature. Now, it might be the case that there is further scientific insight that allows sensible assumptions to be made about the conditional models on the right-hand side of (7.110). For example, say we believe that (on the given temporal scale) it is reasonable to allow $\mathbf{Y}_t^{(2)}$ to be independent of $\mathbf{Y}_{t-1}^{(1)}$ given $\mathbf{Y}_t^{(1)}$ (often a reasonable assumption), and perhaps we believe that $\mathbf{Y}_t^{(1)}$ is independent of $\mathbf{Y}_{t-1}^{(2)}$ given $\mathbf{Y}_{t-1}^{(1)}$. Then, (7.110) becomes

$$[\mathbf{Y}_t^{(1)}, \mathbf{Y}_t^{(2)} | \mathbf{Y}_{t-1}^{(1)}, \mathbf{Y}_{t-1}^{(2)}] = [\mathbf{Y}_t^{(2)} | \mathbf{Y}_t^{(1)}, \mathbf{Y}_{t-1}^{(2)}][\mathbf{Y}_t^{(1)} | \mathbf{Y}_{t-1}^{(1)}]. \quad (7.111)$$

Of course, this is just one example, and the type of assumptions that would be valid for a given system are certainly problem-dependent and should be justified. The point is that (7.111), which was derived from (7.110), has simplified the model through a conditional-probability specification, for which more scientifically meaningful propagator and covariate parameterizations might be

suggested. While this will reduce the number of parameters that need to be estimated/modeled, the curse of dimensionality still needs to be dealt with.

Example: Bayesian Hierarchical Modeling of Air–Sea Interaction

Berliner, Milliff, and Wikle (2003) consider a model for air–sea interaction that is multivariate and includes process models motivated by relatively complex PDE dynamics. The scientific problem considered in this study was to understand the interaction between atmospheric wind forcing and the associated ocean response in the Labrador Sea. The Labrador Sea is a locus of deep convection in the ocean, with processes occurring on relatively short time scales with small spatial footprints. It is an important region, since the deep convection provides a source of deep-water formation and transport, a driving factor in climate-scale ocean circulation. The primary interest in Berliner, Milliff, and Wikle (2003) was in the so-called *preconditioning* phase of the deep convection, in which the coincidence of momentum concentration from the atmosphere and weakened upper-ocean stratification leads to regional surface-buoyancy-forcing events. The study was focused on determining whether a hierarchical model of coupled ocean and atmospheric processes, with relatively sparsely sampled winds and sea-surface heights from satellite-based scatterometry and altimetry, could capture this preconditioning. Our focus here is on the coupled air-sea portion of the process model, although we present some aspects of the full HM to facilitate the motivation.

First, it is helpful to consider the full hierarchical model from a schematic perspective. Let D_a and D_o denote the atmosphere (scatterometer) and ocean (altimeter) data, respectively. Then, the hierarchical model can be written as

$$\begin{aligned} \text{Data model:} & \quad [D_a, D_o | \text{Atm, Ocean}, \theta_a, \theta_o] \\ \text{Process model:} & \quad [\text{Atm, Ocean} | \eta_a, \eta_{o|a}] \\ \text{Parameter model:} & \quad [\theta_a, \theta_o, \eta_a, \eta_{o|a}] \end{aligned}$$

where “Atm” and “Ocean” represent the atmosphere and ocean processes, respectively, and the associated data and process parameters are θ_a , θ_o , η_a , and $\eta_{o|a}$. Efficient modeling in this case is achieved by assuming conditional independence of the atmosphere and ocean observations, as well as a conditional-probability specification of the process model. Thus, the data and process models become

$$\begin{aligned} \text{Data model:} & \quad [D_a, D_o | \text{Atm, Ocean}, \theta_a, \theta_o] = [D_a | \text{Atm}, \theta_a][D_o | \text{Ocean}, \theta_o] \\ \text{Process model:} & \quad [\text{Atm, Ocean} | \eta_a, \eta_{o|a}] = [\text{Ocean} | \text{Atm}, \eta_{o|a}][\text{Atm} | \eta_a] \end{aligned}$$

Hence, the air–sea model is a stochastic atmosphere model coupled to a stochastic ocean-conditional-on-atmosphere model. The posterior distribution is then

$$[\text{Atm, Ocean}, \theta_a, \theta_o, \eta_a, \eta_{o|a} | D_a, D_o],$$

which represents a complete coupling between the atmosphere and ocean processes.

Specifically, the ocean-process model was motivated by *quasigeostrophy* (e.g., see Pedlosky, 1987). That is, if ψ is the ocean streamfunction (a scalar function of two-dimensional fluid motion such that lines of constant streamfunction are everywhere tangent to the local flow), then the quasigeostrophic equation is given by

$$\left(\nabla^2 - \frac{1}{r^2}\right) \frac{\partial \psi}{\partial t} = -J(\psi, \nabla^2 \psi) - \beta \frac{\partial \psi}{\partial x} + \frac{1}{\rho H} \text{curl}_z \tau - \gamma \nabla^2 \psi - a_H \nabla^4 \psi, \quad (7.112)$$

where $J(a, b) \equiv \partial a / \partial x \cdot \partial b / \partial y - \partial b / \partial x \cdot \partial a / \partial y$ is the Jacobian (nonlinear in ψ); $\text{curl}_z \tau$ is the “wind stress curl” given by $\partial \tau_y / \partial x - \partial \tau_x / \partial y$ (where the wind-stress components τ_y and τ_x are related, respectively, to the integral of the u - and v -wind components through a portion of the atmosphere boundary layer); and $r, \beta, \rho, H, \gamma, a_H$ are parameters related to the flow. Using standard finite-difference approximations to temporal derivatives and spatial derivatives in (7.112), Berliner, Milliff, and Wikle (2003) obtain a *vector* representation of this ocean-conditional-on-atmosphere process model with additive errors. Wikle and Hooten (2010) show that this model, discretized, is a special case of the GQN models discussed in Section 7.3.3.

The atmosphere process model in this HM was similar in spirit to the so-called “stochastic geostrophy” model first outlined in Royle et al. (1999). In that case, the spatio-temporal component-wind processes were conditioned on the spatio-temporal pressure field, and fairly simple data models for the scatterometer and altimeter were considered.

Obviously, the posterior distribution is analytically intractable and must be approximated numerically (e.g., by Monte Carlo methods). Berliner, Milliff, and Wikle (2003) exploit the special conditional nature of this posterior form, and they show that a hybrid-sampling approach using an importance sampler and a Gibbs sampler leads to efficient sampling.

7.5 DSTM PARAMETER MODELS

We begin by reiterating that the distinction between “process” and “parameters” is often quite arbitrary. Indeed, one person’s process may be another person’s parameters. However, at least in science-based, spatio-temporal hierarchical statistical modeling, at a given level of the hierarchy it is usually possible to distinguish between parameters and processes. A strength of the hierarchical approach is that it allows parameters from one level to be modeled as a stochastic process at the next level. This ability to construct models by exploiting conditional dependence can lead to very flexible joint probability distributions. The spatial, temporal, and spatio-temporal models used to build these conditional distributions have been discussed in Chapters 3, 4, and 6

and this chapter, respectively. This being said, there are usually components of models (e.g., variance–covariance parameters and parameters that control propagation) that most would agree are parameters. We focus on those in this section.

As we have seen, a key consideration when constructing hierarchical spatio-temporal models is efficient parameterization at both the data-model and the process-model levels. In some cases, these levels are sufficiently simple or low-dimensional, so that it is reasonable to estimate the parameters directly (EHM), or to assign them relatively noninformative prior distributions (BHM). We shall explore this in more detail in Chapter 8, and we give examples in Chapter 9.

The more typical case, at least for complex data models and process models, is to model the parameters explicitly in the parameter stage of the hierarchical model. There are myriad choices for such models, with specific choices depending on the data model and the process model employed. Nevertheless, there are some general issues and approaches that are often of importance. We discuss some of these below.

7.5.1 Data-Model Parameters

It is often the case that some prior information is available regarding the measurement-error variance, allowing a fairly informative parameter model to be specified. For example, if we are assuming conditionally independent measurement errors that are *iid* $Gau(0, \sigma_\epsilon^2)$, then we should specify an informative prior for σ_ϵ^2 . Say we were interested in ambient air temperature, and we saw that the instrument manufacturer's specifications indicated an "error" of $\pm 0.1^\circ\text{C}$. Assuming that this "error" corresponds to 2 standard deviations (an assumption that should be checked!), we might then specify σ_ϵ^2 to have a prior mean of $(0.1/2)^2 = 0.0025$. Because of the instrument manufacturer's specification, we would assume a distribution that had a clearly defined and fairly narrow peak at 0.0025 (e.g., inverse gamma). In fact, we could just fix σ_ϵ^2 at 0.0025; however, the data-model error may have other components of uncertainty too (Section 7.1). To avoid possible identifiability problems with process-model error, it is *very* important that modelers reduce the uncertainty in σ_ϵ^2 as much as the Science allows, including doing side studies designed to have replicated data.

Note that the *iid* error assumption in the data model is not always realistic. For example, when measuring ambient air temperature, it is certainly possible that the instrument error could be greater depending on the true ambient temperature (e.g., very cold or very warm temperatures). Because the data-model variances might change with state, with time, or in space, the data-model error might be modeled as a threshold process, a time series (stationary or nonstationary), or a spatial process. The most common form of dependence considered in DSTM models is spatial. Any of the spatial models discussed in Chapter 4 could be used in this case. Also, as discussed in Sections 7.2.6 and 7.1.3, in

the context of EOF expansions, we might be able to parameterize the (spatially dependent) data-model covariance in terms of a noncomplete set of orthogonal basis vectors:

$$\Sigma_\varepsilon = c\mathbf{I} + \sum_{k=1}^{n_\varepsilon} \lambda_k \boldsymbol{\phi}_k \boldsymbol{\phi}_k', \quad (7.113)$$

where c is an unknown constant, $\{\boldsymbol{\phi}_k\}$ are selected orthogonal basis functions, and $\{\lambda_k\}$ are their associated coefficients; these are selected to model small-scale to microscale spatial variations. Clearly, this suggests all manner of covariance-model parameterizations, depending on how c , n_ε , $\{\lambda_k\}$, and $\{\boldsymbol{\phi}_k\}$ are chosen (e.g., see Berliner, Wikle, and Cressie, 2000).

Although the discussion of data-model parameters has focused on variance (and covariance) parameters, the additive and multiplicative bias parameters discussed in Section 7.1 could also be modeled explicitly. One may have some *a priori* knowledge about these parameters or, if not, they may be given vague priors according to the specifics of the application. For example, say we expect the additive bias terms given in (7.10), $\{\mathbf{a}_t\}$, to be dependent on known covariates, $\mathbf{X}_t$. Then, we might assume $\mathbf{a}_t = \mathbf{X}_t \boldsymbol{\beta}$, where we assign a prior distribution to $\boldsymbol{\beta}$ (e.g., a multivariate Gaussian distribution) at this parameter stage. Alternatively, it might make more sense in some applications to let $\mathbf{a}_t$ follow a Gaussian random spatial process; for example, $\mathbf{a}_t \sim iid \text{Gau}(\boldsymbol{\mu}_a, \boldsymbol{\Sigma}_a)$, as in Chapter 4, or a random spatio-temporal process as in Chapter 6. Similar choices could be used for the multiplicative bias terms $\{\mathbf{h}_t\}$ in (7.10). In general, multiple levels of hierarchical specification could be applied, assuming there is relevant data and/or scientific information to inform such specifications. As shown in Wikle, Berliner, and Cressie (1998), considerable flexibility in a DSTM can be obtained by allowing the higher-level model parameters to vary spatially.

7.5.2 Process-Model Parameters

In this subsection, we consider some standard approaches to putting (prior) distributions on the process-model parameters.

Parameters of Process-Model Error

Recall that the covariance matrix of the process error $\{\eta_t\}$ is written as $\mathbf{Q}_\eta$. It is often the case that we assume homogeneous variance, σ_η^2 , in which case $\mathbf{Q}_\eta = \sigma_\eta^2 \mathbf{R}_\eta$, where $\mathbf{R}_\eta$ is a correlation matrix. Note that various prior distributions for the process variance σ_η^2 could be specified, although one typically does not have a great deal of *a priori* understanding about this variance. We also note that there is often some knowledge about how σ_η^2 and the data-model variance, σ_ε^2 , are related. Thus, in practice, the signal-to-noise ratio, $\xi = \sigma_\eta^2 / \sigma_\varepsilon^2$, and σ_ε^2 are often modeled, rather than the individual variances. In addition to a natural interpretation, the signal-to-noise ratio has the advantage of being scale-free.

As will be discussed in Section 8.3, an inverse-Wishart prior distribution is often applied to covariance matrices when implementing dynamical models in a Bayesian framework. Alternatively, in cases where $\mathbf{Q}_\eta$ represents a spatial covariance matrix, one could specify a relevant geostatistical or lattice covariance model (see Chapter 4) as the prior structure. The parameters that characterize those spatial models might also be allowed to vary with time, if there is sufficient justification. Yet another alternative is a covariance model based on a matrix decomposition. We saw a simple example of this in Section 7.5.1. One can also use a Cholesky decomposition,

$$\mathbf{T}\mathbf{Q}_\eta\mathbf{T}' = \mathbf{D}, \quad (7.114)$$

where $\mathbf{T}$ is a lower-triangular matrix, and $\mathbf{D}$ is a diagonal matrix with positive entries, facilitating interpretation and (MCMC) computation (e.g., Pourahmadi, 1999; Chen and Dunson, 2003).

There are also many cases where one might have scientific information about the prior covariance structure. For example, in Wikle et al. (2001), scientific considerations suggested that the tropical ocean winds should follow a turbulent power-law spectrum. This was incorporated by specifying a relationship between spectral scales in $\mathbf{Q}_\eta$.

Parameters of the Propagator Matrix

The parameter model should accommodate scientific understanding of the dynamical process. In the case of low-dimensional processes, the entire propagator matrix $\mathbf{M}$ in the DSTM might be modeled, as will be shown in Section 8.3. This could be done using a matrix prior (e.g., a matrix normal) or, equivalently, by modeling the vectorized form, $\text{vec}(\mathbf{M})$.

It is more common that the curse of dimensionality has led the modeler to parameterize $\mathbf{M}$ as $\mathbf{M}(\boldsymbol{\theta})$. Prior distributions for $\boldsymbol{\theta}$ would then depend on the particular type of parameterization. Some parameterizations assume that there is spatial dependence in at least some of the $\boldsymbol{\theta}$. For example, the basic spatial univariate AR model given in Section 7.2.2 assumes that $\mathbf{M} = \text{diag}(\mathbf{m})$. In this case, $\mathbf{m}$ might be assumed to follow a Gaussian spatial process, such as discussed in Chapter 4. This is quite powerful, in that the elements of $\mathbf{Y}_t$ might be assumed to be (conditionally) independent (i.e., $\mathbf{Q}_\eta = \sigma_\eta^2 \mathbf{I}$) but, if $\mathbf{M}$ exhibits spatial dependence, then the marginal distribution of $\mathbf{Y}_t$ would still show spatial dependence. Of course, these parameters could be allowed to vary with time as well, possibly being dependent on other covariates.

There might also be scientific information to help model $\mathbf{M}(\boldsymbol{\theta})$. For example, if we were modeling an ecological process that exhibited non-isotropic diffusion over time, it might be useful to model the process based on a discretized diffusion PDE, in which the diffusion parameters vary with space. As shown in Section 7.2.4, these diffusion coefficients are included in the off-diagonal entries of $\mathbf{M}(\boldsymbol{\theta})$. It would then be reasonable to model these coefficients as a spatial process, possibly related to habitat covariates (see Wikle, 2003b). In another example, one might have knowledge of advection dynamics, where the

relevant parameters θ could be related to wind fields (e.g., Xu, Wikle, and Fox, 2005), suggesting once again that the parameter model might involve spatial dependence.

Recall from our discussion of vector autoregression models in Section 3.4.7 that the propagator matrix $\mathbf{M}$ in a linear DSTM must have all of its eigenvalues less than one in modulus to ensure that the process is not explosive. In principle, it is not difficult to enforce such constraints in the implementation of a DSTM. However, depending on the parameterization of $\mathbf{M}$, this may or may not be easy to constrain in practice within a hierarchical context. In simple parameterizations it can be done efficiently (e.g., see Sansó, Schmidt, and Nobre, 2008). However, it is not always clear that such constraints should be enforced. For example, if one were using such a linear DSTM to characterize an environmental process, unconstrained estimation that suggested explosive behavior would probably indicate that the linear model is inappropriate and that the process is exhibiting some sort of nonlinear growth, which is reflected in the data driving the parameters into the explosive regime. In this case, one should probably be using a nonlinear process model. Alternatively, in a relatively-short-period forecasting situation (i.e., a *nowcast*), an explosive linear model that is suggested by the recent past might produce a realistic forecast of a process exhibiting growth (e.g., a thunderstorm growing in intensity). Thus, we should be careful employing restrictions on the parameter space, unless we are fairly certain that our statistical model is the “correct” model for the dynamics of the underlying spatio-temporal process.

7.5.3 Deterministic Process Models with Random Parameters

Recall from our discussion in Sections 7.2.4 and 7.3, that we can often motivate dynamical models with deterministic (e.g., PDE and IDE) models. For example, recall (7.62):

$$\mathbf{Y}_t = \mathcal{M}(\mathbf{Y}_{t-1}, \boldsymbol{\eta}_t; \boldsymbol{\theta}_t), \quad (7.115)$$

where $\mathcal{M}$ is some nonlinear dynamical operator depending on parameters $\boldsymbol{\theta}_t$, that may vary with t . If $\mathcal{M}$ is known and the parameters are allowed to be random, the behavior of the system may be unrealistic if it is not in some way constrained. In relatively simple systems, there may be enough understanding of the process parameters $\{\boldsymbol{\theta}_t\}$ to be able to specify reasonable constraints. Perhaps more importantly, the HM approach allows the data to help *control* the process’ behavior.

On the one hand, random parameters allow the process to have much more flexibility (even in the constrained case), but on the other hand these random parameters may not be interpretable in the context of the original process. This interpretability is further complicated when there is model error $\{\boldsymbol{\eta}_t\}$. It is often the case that one simply needs the error process because the deterministic model, even with random parameters, cannot accommodate real-world data. Clearly, there are now two sources trying to account for the unrepresentativeness of the underlying deterministic model, namely the random parameters $\{\boldsymbol{\theta}_t\}$

and the (additive) error process $\{\eta_t\}$. Although the trade-offs between data and process uncertainty, as they relate to system behavior, have long been considered in Control Theory (e.g., Aström, 1970), the interplay between these uncertainties *and* random parameters (in the context of hierarchical spatio-temporal statistical models) is presently under-developed and an active area of research.

7.6 DYNAMICAL DESIGN OF MONITORING NETWORKS

It is not usually possible to sample a spatio-temporal process completely. This section is concerned with choosing where (and when) to sample, through the design of a monitoring network. Typically, political, geographic, or economic factors govern the design, but choosing sampling locations without also taking into account statistical factors will lead to a sub-optimal design. There has been a great deal of work in the Statistics literature related to obtaining optimal designs for spatial processes [e.g., see the overviews in Müller (2000) and Le and Zidek (2006, Chapter 11)]. Indeed, many of these same methodologies are applicable to spatio-temporal processes as well. Most of the work done in this context has been related to finding a fixed design (over time) that considers the joint spatio-temporal dependence when determining the optimal design. However, as demonstrated in Wikle and Royle (1999), substantial efficiencies can be gained in the spatio-temporal context if the design is allowed to change with time (provided such a change is allowed by the monitoring protocol). This is sometimes referred to as targeted design or adaptive design. We prefer the term dynamical design, since it emphasizes the dynamical nature of the design changing with time and the fact that the underlying process is also evolving dynamically.

Wikle and Royle (1999) show that when considering Gaussian data and Gaussian dynamical processes, dynamical designs show greater efficiencies relative to static designs as the temporal correlation increases (their measure of efficiency is based on the average prediction variance across the spatial domain). Such an improvement is no surprise, in that if observations were made at location s at time t , then it might be redundant to take observations at this same location at time $t + 1$ when there is strong temporal dependence at location s . In addition, there will generally be large improvements in design efficiency if the spatial dependence is strong. Again, this is intuitive, since if there is strong dependence in space between two nearby locations at time t , then at $t + 1$, the monitor could be moved far away from both of these locations.

To illustrate a simple procedure for dynamic spatio-temporal design, consider Gaussian data and Gaussian process models:

$$\mathbf{Z}_t = \mathbf{H}_t \mathbf{Y}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \text{ind. Gau}(\mathbf{0}, \mathbf{R}_t), \quad (7.116)$$

$$\mathbf{Y}_t = \mathbf{M}_t \mathbf{Y}_{t-1} + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \text{ind. Gau}(\mathbf{0}, \mathbf{Q}_t), \quad (7.117)$$

where, as before, $\mathbf{Z}_t$ is an m_t -dimensional data vector, $\mathbf{Y}_t$ is an n -dimensional process vector, $\mathbf{H}_t$ is the $m_t \times n$ data-mapping matrix, $\mathbf{M}_t$ is the $n \times n$

propagator matrix, and $\mathbf{R}_t$ and $\mathbf{Q}_t$ are the data-error and process-error covariance matrices, respectively. Thus, the "design" at a given time is what relates the observations $\mathbf{Z}_t$ to the process $\mathbf{Y}_t$; this is precisely the data-mapping matrix $\mathbf{H}_t$. As discussed in Section 7.1, the form of this matrix can be quite general, so long as it is varying in time and can accommodate different potential observation networks (designs) at each time. For illustration, it is perhaps simplest to think of this as simply the incidence matrix of zeros and ones described in Section 7.1.2. In this case, the process is gridded and the observation locations correspond to a specific grid point. Thus, if a sampling design were proposed, we could easily specify the form of $\mathbf{H}_t$.

Given the measurement equation (7.116) (i.e., the data model) and state equation (7.117) (i.e., the process model), we can obtain the sequential designs quite simply. For now, assume we know (or have estimated) the parameter matrices $\mathbf{M}_t$, $\mathbf{Q}_t$, and $\mathbf{R}_t$. From the discussion of the Kalman filter recursions given in Section 8.2.1, we can obtain the following recursive equations for the prediction-error covariances:

$$\begin{aligned}\mathbf{P}_{t|t} &\equiv \text{var}(\mathbf{Y}_t | \mathbf{Z}_1, \dots, \mathbf{Z}_t) \\ &= (\mathbf{H}_t' \mathbf{R}_t^{-1} \mathbf{H}_t + \mathbf{P}_{t|t-1})^{-1} \\ &= \mathbf{P}_{t|t-1} - \mathbf{P}_{t|t-1} \mathbf{H}_t' (\mathbf{H}_t \mathbf{P}_{t|t-1} \mathbf{H}_t' + \mathbf{R}_t)^{-1} \mathbf{H}_t \mathbf{P}_{t|t-1},\end{aligned}\quad (7.118)$$

where

$$\begin{aligned}\mathbf{P}_{t|t-1} &\equiv \text{var}(\mathbf{Y}_t | \mathbf{Z}_1, \dots, \mathbf{Z}_{t-1}) \\ &= \mathbf{M}_t \mathbf{P}_{t-1|t-1} \mathbf{M}_t' + \mathbf{Q}_t.\end{aligned}\quad (7.119)$$

Now, given that $\mathbf{P}_{0|0}$ is specified, we could proceed with the recursion for $t = 1, 2, \dots$, by first obtaining $\mathbf{P}_{t|t-1}$ from (7.119) and then using it to obtain $\mathbf{P}_{t|t}$ in (7.118). In practice, we need to specify $\mathbf{P}_{0|0}$, $\mathbf{M}_t$, $\mathbf{R}_t$, and $\mathbf{Q}_t$ to implement these recursions. Data may have been collected in a pilot study, from which estimates could be obtained (see Section 8.3 for the case where the matrices are not varying in time).

The recursion in (7.118) and (7.119) provides the mechanism for the dynamic design of spatio-temporal monitoring networks. If, as is typically the case, the parameter matrices are not time-varying, then the only temporal dependence in $\mathbf{P}_{t|t}$ comes from the design matrix $\mathbf{H}_t$. That is, it changes through time only as a function of the observation locations at each time. Thus, by allowing the monitoring locations to change with t , we should be able to find more efficient designs by optimizing some function of $\mathbf{P}_{t|t}$ (the *design criterion*). That is, we can search the space of potential designs to find the "optimal" design. For example, a common design criterion is the spatial average of the prediction error variance, which one would minimize with respect to $\mathbf{H}_t$, for $t = 1, 2, \dots$.

Typically, it is not possible to search exhaustively the complete design space but we can make use of algorithms that can find a "nearly optimal" or, at least, a "good" design. For example, one can use an exchange algorithm that

evaluates the design criterion successively for different designs, and the design is updated by exchanging “bad” locations for “better” locations. Such algorithms are common in practice, and there are many different variants (e.g., see Cook and Nachtsheim, 1980; Nychka, Yang, and Royle, 1997).

Notice that the procedure described above does not take into account the uncertainty associated with the estimation of the model parameters; in that sense, it is EHM-based. In addition, this approach is greatly simplified by the Gaussian assumptions, since the prediction covariances do not depend on the data, only on the data locations and parameters. If one wants to take into account the parameter estimation and/or account for non-Gaussian or nonlinear structures, then one might consider a Bayesian approach to design [e.g., see Chaloner and Verdinelli (1995) for an overview; also see Diggle and Lophaven (2006)]. For example, the approach suggested by Müller (1999) relies on simulation and is useful for such problems. In essence, a Bayesian-based design seeks to maximize the expected value of some utility function of interest. Finding the design that maximizes this expected utility is typically intractable analytically, and one is forced either to simplify the utility greatly or to approximate the expected-utility maximization by simulation. Such approaches still require very computationally efficient optimization algorithms. Müller (1999) shows that this can be facilitated when augmenting the model by including the design as a random quantity. As in any relatively high-dimensional optimization problem, there is no guarantee to obtain *the* optimal design with this procedure, but it does give “good” designs (which is a huge help in complicated design problems). Royle and Wikle (2005) demonstrate this in the context of dynamical spatio-temporal design with a non-Gaussian data model and a DSTM, where good designs were chosen for the sampling locations associated with annual July brood-count surveys of North American waterfowl.

7.7 SWITCHING THE EMPHASIS OF TIME AND SPACE

This chapter has focused on an HM where a data model relates a vector of spatial observations at a given time either to a true spatial process evolving over time, or to a spectral representation of such a spatial process evolving over time. This perspective is motivated at least in part by our preference that the more natural (i.e., physical, biological, etc.) way to present a spatio-temporal process is for a spatial field to evolve over time. However, from a modeling perspective, it may be more efficient to consider the alternative perspective of a time series varying in space, though this is done at the expense of a “dynamical” interpretation.

Consider a situation in which we have many more time observations than spatial observations (i.e., $m \ll T$). We can define $\mathbf{Z}_i \equiv (Z(s_i; 1), \dots, Z(s_i; T))'$ to be the T -dimensional vector of observations at location s_i , for times $t = 1, \dots, T$ and $i = 1, \dots, m$. We could then characterize the temporal behavior functionally in terms of a dimension-reduced (say,

$q < m < T$) set of basis functions:

$$\mathbf{Z}_i = \Psi \boldsymbol{\beta}_i + \boldsymbol{\varepsilon}_i, \quad i = 1, \dots, m,$$

where Ψ is a $T \times q$ matrix of basis-function coefficients, $\boldsymbol{\beta}_i \equiv (\beta_i(1), \dots, \beta_i(q))'$ is a q -dimensional vector of coefficients for the i th spatial location, and $\{\boldsymbol{\varepsilon}_i\}$ is a T -dimensional error process. In this case, each basis function (column of Ψ) is a function of time (i.e., a time series). These might be chosen deterministically (e.g., Fourier, wavelets, bisquares) or from the data (e.g., EOFs, splines), just as with the spatially referenced basis functions discussed in Section 7.1.3. Now, it is natural to re-orient the coefficients $\{\boldsymbol{\beta}_i : i = 1, \dots, m\}$ into spatial processes. For example, fix the j th coefficient and let i vary over $\{1, 2, \dots, m\}$. Define

$$\boldsymbol{\beta}(j) \equiv (\beta_1(j), \dots, \beta_m(j))', \quad j = 1, \dots, q,$$

which corresponds to an m -dimensional spatial field for the j th basis-function coefficients. In an HM context, we could then specify a spatial process model for each of $\boldsymbol{\beta}(j)$; $j = 1, \dots, q$ (see Chapter 4). It may even be the case that an additional spectral expansion/dimension reduction could be considered on these spatial fields. For example, write

$$\boldsymbol{\beta}(j) = \Phi \boldsymbol{\alpha}(j) + \boldsymbol{\nu}(j),$$

where Φ is an $m \times p_\alpha$ matrix of spatial-basis-function coefficients, $\boldsymbol{\alpha}(j)$ is the associated p_α -dimensional vector of spectral coefficients, and $\boldsymbol{\nu}(j)$ is an error process that accounts for the variation remaining after the dimension reduction. See Wikle and Anderson (2003) for an implementation of such an approach applied to modeling spatio-temporal distributions of tornado counts given climatological time series. In addition, Sansó, Schmidt, and Nobre (2008) give a discussion of such models from the perspective of the linear model of coregionalization.

To sum up, although switching the modeling emphasis of a spatio-temporal process from time to space can be a very useful modeling strategy, it is not dynamical in the sense of the earlier sections of this chapter. Consequently, it may be difficult to imagine real-world processes whose etiologies are consistent with such a formulation.

7.8 BIBLIOGRAPHIC NOTES

General Hierarchical Formulation

As discussed in the Bibliographic Notes of Chapter 3 (Section 3.7), the motivation for hierarchical modeling has been around in Statistics for quite some time. Almost any modern book on Bayesian Statistics (e.g., Carlin and Louis, 2000; Gelman et al., 2003) cover this topic in detail. Furthermore, modern approaches to time series analysis (e.g., West and Harrison, 1997; Shumway and Stoffer, 2006; Prado and West, 2010) give significant coverage to hierarchical representations. This is also the case with modern treatments of spatial

statistics (e.g., Banerjee, Carlin, and Gelfand 2004; Diggle and Ribeiro, 2007). Specific discussion of dynamical linear models in the spatial context can be found in Gamerman and Lopes (2006). Our view of hierarchical modeling is based on the important paper of Berliner (1996), which discusses hierarchical modeling of environmental time series from the perspective of a series of conditional models for “data given process and parameters,” “process given parameters,” and “parameters.” This approach is further developed by Wikle, Berliner, and Cressie (1998), Berliner, Wikle, and Cressie (2000), Wikle et al. (2001), Berliner, Milliff, and Wikle (2003), Wikle (2003a,b), and Wikle and Hooten (2010), among many others discussed here and in other chapters.

Data Models—Incidence Matrices and Change-of-Support

Incidence matrices are essentially design matrices for mapping a spatial process vector to a spatial data vector by using ones and zeros. This perspective is illustrated in Wikle, Berliner, and Cressie (1998), Royle et al. (1999), Wikle and Royle (1999), and Wikle et al. (2001).

Spatial change of support is outlined in Section 4.1.3. An excellent review of this problem and traditional modeling approaches addressing it is given in Gotway and Young (2002). The hierarchical perspective is given in Muggin, Carlin, and Gelfand (2000) and Gelfand, Zhu, and Carlin (2001). Our perspective, which implies a specific form of the data-mapping matrix in the hierarchical DSTM context, is motivated by Wikle and Berliner (2005). Recent examples of hierarchical change-of-support for spatio-temporal data are given in Berrocal, Gelfand, and Holland (2010) and Sahu, Gelfand, and Holland (2009).

Data Models—Spectral Decomposition

We use the term “spectral decomposition” quite broadly in this book. That is, representation of the spatial process vector for a given time, as the product of a basis function matrix and a random vector (which need not be spatially referenced), is a “spectral” expansion. In this respect, there is a very large literature in which such decompositions are used in DSTMs, with the primary difference being which basis set is being used, and, to a lesser extent, the type of estimation being employed. Examples include empirical orthogonal functions, Fourier series, bisquares, polynomials, splines, and multiresolution wavelets (e.g., Wikle, 1996; Mardia et al., 1998; Wikle and Cressie, 1999; Berliner, Wikle, and Cressie, 2000; Stroud, Müller, and Sansó, 2001; Wikle et al., 2001; Cressie and Wikle, 2002; Xu, Wikle, and Fox, 2005; Johannesson, Cressie, and Huang, 2007; Malmberg et al., 2008; Kang and Cressie, 2011; Cressie, Shi, and Kang, 2010; Kang, Cressie, and Shi, 2010). In these formulations, the spectral coefficients are not typically spatially referenced, but correspond to some hidden process that is transformed into physical space by multiplying by the basis functions. In addition, the basis functions in these settings are typically assumed known. For an overview of these methods, see Wikle (2010b).

Data Models—Parameterized Low-Rank Expansions

An alternative perspective is to allow the random process to correspond to a finite number of “reference” spatial locations and then to build dependence by smoothing these. Indeed, this is the primary motivation behind so-called “process convolution” methods in spatial statistics (e.g., Barry and Ver Hoef, 1996; Ver Hoef and Barry, 1998; Higdon, 1998). In the discrete-convolution setting and the linear model of coregionalization, one can then apply a dynamical linear spatio-temporal process model on the process at the reference locations (e.g., Calder, Holloman, and Higdon, 2002; Sansó, Schmidt, and Nobre, 2008; Lemos and Sansó, 2009). Alternatively, one can consider kernels derived from kriging-based support points, as done for so-called “predictive processes” (Banerjee et al., 2008; Finley et al., 2009). The kernels depend on parameters, which limits the size of datasets that the predictive-process approach can handle.

Rather than fix the basis functions or parameterize the kernels, one could seek to estimate the transformation matrix directly, as with factor analysis in multivariate statistics (e.g., Wang and Wall, 2003; Forni et al., 2005). An example of this in the DSTM setting is given in Calder (2007), Lopes, Salazar, and Gamerman (2008), and Sáfiadi and Peña (2008). In this case, as with factor analysis, there are fundamental identifiability issues associated with estimating both the data-mapping matrix and the transition matrix in the dynamical process model; see also Strickland et al. (2011). More generally, this is just the “blind source separation” problem in Engineering (e.g., Jutten, Herault, and Guerin, 1988), and there are many algorithmic approaches to dealing with the problem [e.g., independent component analysis; see Hyvärinen, Karhunen, and Oja (2001)]. Knuth (1999, 2005) considers Bayesian approaches for source separation. For an overview of blind source separation and independent component analysis, see Choi et al. (2005).

Data Models—Alternative System Reduction Approaches

There is a long history in the Control Engineering literature of seeking optimal dimension reduction for dynamical systems (e.g., Kailath, 1980; Moore, 1981). Although not specific to spatio-temporal systems, *per se*, these methods are applicable to the kinds of spatio-temporal processes one encounters in the environmental sciences. For example, Farrell and Ioannou (2001) discuss stochastic optimals, EOFs, and balanced truncation in the context of continuous-time spatio-temporal processes in Atmospheric Science. Crommelin and Majda (2004) compare various optimal basis methodologies for geophysical processes, including principal interaction patterns (PIPs) and optimal persistence patterns (OPPs). More generally, Antoulas (2005) provides an overview of methods that have been used to approximate large-scale dynamical systems.

Data Models—Nonlinear and Non-Gaussian Models

Nonlinear data models are typically specified by the scientific problem at hand. Alternatively, one might include a transformation of the process at this level of the hierarchy (e.g., Sansó and Guenni, 1999). More common is the case of non-Gaussian data. Such models were specified in the spatial statistics literature by Diggle, Tawn, and Moyeed (1998) and have been considered by

numerous authors in the context of state-space models (e.g., Kitagawa, 1987; Fahrmeir and Kaufmann, 1991; Carlin, Polson, and Stoffer, 1992; Fahrmeir, 1992; Gamerman, 1998). The use of such data models in spatio-temporal processes is also common (e.g., Brix and Diggle, 2001; Wikle, 2003b; Wikle and Hooten, 2006; Hooten et al., 2007; Hooten and Wikle, 2008). In each of these contexts, one assumes that the data are from an exponential family and some transformation of the true spatio-temporal dynamical process is Gaussian. There are situations with non-Gaussian data where the usual exponential-family/canonical-link formulations may not be the most realistic, and one has to consider data models that are more appropriate for the process (e.g., Cangelosi and Hooten, 2009).

Basic Linear Process Model Specifications

Linear dynamical process evolution is certainly well characterized in the multivariate time series literature (see Chapter 3 for references). Bennett (1979) and Rouhani and Wackernagel (1990) considered time series at each spatial location and developed models accordingly. In general, these approaches can be included in the DSTM framework as in traditional state-space modeling. The challenges come as one must reduce the dimensionality of the parameter space or the process itself to avoid the curse of dimensionality. Huang and Cressie (1996) considered a simple dynamical parameterization in the DSTM, where the propagator was just a scalar multiplied by the identity matrix. Wikle, Berliner, and Cressie (1998) considered a more complicated hierarchical spatio-temporal model, which included seasonality, trends, and a dynamical process that was parameterized by a lagged-nearest-neighbor propagator. An original feature of this model was the use of spatial processes for parameters, which fits nicely into the hierarchical modeling framework.

PDE-Based Process Parameterizations

The use of PDEs to motivate statistical models is not new (e.g., see Hotelling, 1927). In fact, as outlined in Section 6.3, the connection between stochastic PDEs and spatial and spatio-temporal covariances is fundamental in spatial statistics (e.g., Heine, 1955; Whittle, 1954, 1962; Jones and Zhang, 1997). Mollison discusses diffusion and growth processes to motivate “spatial contact” models. More recently, the use of PDEs to motivate spatial and spatio-temporal statistical models was facilitated by the development of hierarchical Bayesian methods (e.g., Royle et al., 1999; Wikle et al., 2001; Berliner, Milliff, and Wikle, 2003; Wikle, 2003b; Wikle, Berliner, and Milliff, 2003; Hooten and Wikle, 2008; Malmberg et al., 2008; Cangelosi and Hooten, 2009; Stroud et al., 2010; Zheng and Aukema, 2011; Lindgren et al., 2011). Note that statistical fitting (e.g., parameter estimation), as distinct from model motivation, of PDE models in a scientific context is prevalent in several disciplines, including pharmacokinetics (e.g., Gelman, Bois, and Jiang, 1996), virology (e.g., Huang, Liu, and Wu, 2006), and ecology (Haario et al., 2006). A general approach in the context of functional data analysis is presented by Ramsay et al. (2007).

IDE-Based Process Parameterizations

Integro-difference equations (IDEs) have also proven to be a useful framework to motivate dynamical process models. Deterministic forms of IDE models have been useful in population ecology (e.g., Kot, 1992; Kot, Lewis, and van den Driessche, 1996). Spatio-temporal statistical formulations are presented in Wikle (1996), Wikle and Cressie (1999), Wikle (2002a), Huang and Hsu (2004), Xu, Wikle, and Fox (2005), Dewar, Scerri, and Kadirkamanathan (2009), and Scerri, Dewar, and Kadirkamanathan (2009). The implied covariance structure of these models is discussed in Brown et al. (2000) and Storvik, Frigessi, and Hirst (2002).

Reduced-Dimension Process Models

The spectral-representation, low-rank-parameterization, and IDE references listed above generally involve reduced dimensions. Most of them contain fairly simple multivariate-time-series process models, where dimension reduction follows from parameterizations of the process' propagator and covariance matrices. As discussed in Wikle and Cressie (1999), Xu and Wikle (2007), and Chapter 8, it is straight forward to perform estimation and prediction for such processes using Kalman filtering/smoothing, method-of-moments, EM algorithm, or fully Bayesian methodologies.

Multiresolution Process Models

The essence of the change-of-resolution multiresolution models presented here can be found in Chou, Willsky, and Nikoukhah (1994) and Fieguth et al. (1995). The models migrated into the Statistics literature through papers by Gelpke and Künsch (2001), Huang, Cressie, and Gabrosek (2002), Johannesson and Cressie (2004), and Johannesson, Cressie, and Huang (2007). Clearly, many of the spectral approaches described previously are also multiresolution in that the basis functions correspond to different spatial scales. For example, Nychka, Wikle, and Royle (2002) consider multiresolution wavelet basis functions to model spatially nonstationary processes. Such processes can be allowed to vary dynamically as well (Berliner, Wikle, and Milliff, 1999; Wikle et al., 2001). For a more general discussion of Bayesian multiscale modeling, see Ferreira and Lee (2007).

Nonlinear Process Models

Nonlinear statistical process models are most often considered in the time series literature (e.g., Priestley, 1988; Tong, 1990; Young, 2000; Chatfield, 2004; Fan and Yao, 2005). Historically, the papers by Jones (1965, 1978) were very insightful with regard to nonlinear prediction and nonlinear autoregressive processes. State-dependent models are common in spatio-temporal applications (e.g., Hughes and Guttorp, 1994; Berliner, Wikle, and Cressie, 2000). Relatively few examples exist for more general nonlinear spatio-temporal dynamical process models. Wendt, Irwin, and Cressie (2004) give a Bayesian waypoint analysis, where the state is the trajectory of an object moving according to

nonlinear equations of motion. Lu et al. (2009) describe a flexible, adaptively varying coefficient model that can accommodate nonlinear spatio-temporal interactions, as well as spatial nonstationarity and non-Gaussian data models. Wikle and Hooten (2010) show that general quadratic nonlinearity (GQN) is present in many real-world processes (e.g., Kondrashov et al., 2005), and this can be used to motivate statistical models (e.g., Wikle, Berliner, and Milliff, 2003; Wikle and Hooten, 2006; Hooten et al., 2007; Hooten and Wikle, 2008). This structure is also present in the multivariate functional coefficient autoregressive formulation for multivariate time series described in Harvill and Ray (2006). Functional polynomial regression models for time series are discussed in Yan and Müller (2010). Wikle and Holan (2011) consider a hierarchical formulation of spatio-temporal polynomial models in an IDE framework with stochastic search variable selection.

There are many fine discussions of nonlinear stochastic models in the geophysics literature, (e.g., Majda and Wang, 2006). For example, Majda, Abramov, and Grote (2005) consider such processes from an information theory perspective, and Majda and Franzke (2009) consider reduced dimensional EOF formulations of stochastic climate models.

Individual- or agent-based models (e.g., Lee et al., 1990; Grimm and Railsback, 2005) for spatio-temporal processes (e.g., Smith et al., 2002) are just now finding their way into the Statistics literature (e.g., Hooten and Wikle, 2010). There are strong connections between these models and models for non-Gaussian spatio-temporal processes (e.g., Zhu, Huang, and Wu, 2005; Royle and Kery, 2007). In a similar spirit, Fewster (2003) builds a spatio-temporal model for species spread, focusing on movement between individual sites.

Nonlinear effects of continuous covariates and temporal trends in spatio-temporal processes can be accommodated through nonparametric and semiparametric models. For example, Fahrmeir, Kneib, and Lang (2004) demonstrate such models using penalized structured additive regression in a Bayesian setting. Augustin et al. (2009) illustrate the use of generalized additive mixed models for spatio-temporal forest health data.

Multivariate Process Models

In many respects, statisticians are a bit “behind the curve” when it comes to formulating and implementing multivariate DSTMs. Note that such models are commonly (and operationally) implemented in the geophysical sciences in the context of data assimilation [e.g., see the overviews in Daley (1991), Ghil and Malanotte-Rizzoli (1991), Talagrand (1997), Bennett (2002), and Kalnay (2003)].

In Statistics, science-driven hierarchical multivariate spatial models were proposed by Royle and Berliner (1999). Although related to the geostatistical concepts of “linear models of coregionalization” [e.g., see the review in Wackernagel (2003); also see Schmidt and Gelfand (2003) and Gelfand et al. (2004)], it differs fundamentally in the emphasis on science-based conditional modeling (see Section 4.1.5). This idea was used in a dynamical process model

in Berliner, Milliff, and Wikle (2003), where high-dimensional spatio-temporal dynamical processes (atmosphere and ocean) were coupled. Motivated by the spatio-temporal dynamical model of Tonellato (2001), Shaddick and Wakefield (2002) consider a multivariate DSTM for multiple pollutants. The multivariate process-convolution approach, developed by Ver Hoef and Barry (1998), and the linear model of coregionalization can be used in the hierarchical framework to motivate DSTMs for multivariate processes (e.g., Gelfand, Banerjee, and Gamerman, 2005; Calder, 2007; Sahu and Challenor, 2008). Daniels, Zhou, and Zou (2006) consider the conditional specification of spatio-temporal models, more from a Markov-random-field perspective, although their methodology can be used for geostatistical processes of point or areal support.

Parameter Specifications

The essence of hierarchical modeling for spatio-temporal processes is the specification of parameter models at lower levels of the hierarchy. As illustrated in Wikle, Berliner, and Cressie (1998) and Gelfand et al. (2003), one can easily place spatial process priors on lower levels of the hierarchy for parameters of a dynamical spatio-temporal or purely spatial process. This is the obvious construction in hierarchical modeling and is one of the advantages of taking Berliner's (1996) view of hierarchical modeling of environmental processes.

Perhaps more critical are the distributions one might place on parameters at lower levels of the hierarchy, particularly the choice of hyperparameters. Overviews of these issues relative to hierarchical spatial processes are given in Banerjee, Carlin, and Gelfand (2004), Diggle and Ribeiro (2007), and Wikle (2010a).

Dynamical Design

The design of spatial monitoring networks is well covered in the literature [e.g., see reviews in Müller (2000); also see Le and Zidek (2006), Zimmerman (2006) and Diggle and Ribeiro (2007)]. In addition, ideas related to adaptive spatial designs were presented in Chao and Thompson (2001). Designs for dynamical processes were discussed by Titterton (1980). Bueso, Angulo, and Alonso (1998) use a state-space model to obtain an optimal static spatial design. The notion of choosing the design locations as a function of the underlying dynamical process was proposed in an adaptive-weather-observation context by Berliner, Lu, and Snyder (1999). In a series of papers, Wikle and Royle (1999, 2005) and Hooten et al. (2009) illustrate general dynamical design for monitoring networks for Gaussian and non-Gaussian data with underlying Gaussian dynamical processes.

Switching Time and Space

The formal development of semiparametric statistical models with emphasis on spatially varying time series is considered by Delicado et al. (2010) and Giraldo et al. (2010). This is summarized in the discussion of Mateu (2010), who describes how one might formulate an IDE model that is continuous in time yet discrete in space.

